

ÉCOLE CENTRALE DE NANTES  
LABORATOIRE ICONIC — CHU NANTES

# Récapitulatif par sujet

Débruitage EEG ART / ICA / ICLabel (Motor-Imagery)

*Comparaison CSP+LDA et MSE, sujet par sujet*

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29 juillet 2026

## Introduction

Nous avons identifié l'article « *Augmenting brain-computer interfaces with ART : An artifact removal transformer for reconstructing multichannel EEG signals* » de Chuang *et al.*, qui présente une nouvelle méthode de débruitage appelée **ART** (*Artifact Removal Transformer*) et montre de bons résultats sur des données d'imagerie motrice (*motor-imagery*). Dans leur étude, les auteurs entraînent ART sur la base *Eyes-Closed/ Eyes-Open Resting-State* (Chuang *et al.*, 2022). Nous avons identifié une limite à cette base pour l'entraînement : elle ne contient pas les artefacts habituels d'une séance EEG, comme les clignements des yeux (*blinks*).

C'est ce qui motive notre choix d'utiliser la base **Schalk *et al.* (2004), Motor Imagery (BCI2000)**, et d'y entraîner ART en prenant comme signaux propres ceux nettoyés par **ICLabel**. L'objectif est de tester si ART peut apprendre à nettoyer automatiquement les signaux aussi bien, voire mieux, qu'ICLabel.

Pour chaque sujet exclu (protocole *leave-one-subject-out*), ce document rapporte : le taux de classification gauche/droite (CSP+LDA) sur les différents signaux débruités disponibles (brut, ICA manuelle, ICLabel automatique, ART, ...); l'erreur quadratique moyenne (MSE) entre le signal nettoyé par ICA et celui nettoyé par ICLabel; ainsi que l'évolution du MSE (train/eval) pendant l'entraînement *leave-one-subject-out* d'ART.

**Mots-clés** : EEG, BCI, ART, ICA, ICLabel, CSP, MSE, motor-imagery.

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# 1 Méthodologie

## 1.1 Taux de classification (CSP+LDA)

Pour mesurer à quel point l'intention motrice (main gauche vs main droite) reste lisible dans un signal après débruitage, on applique une chaîne de classification standard en EEG : filtrage passe-bande dans la bande sensori-motrice (7–30 Hz), extraction d'une courte fenêtre temporelle après le stimulus, puis un filtrage spatial (CSP) qui recherche les directions de l'espace séparant le mieux les deux classes. La variance du signal filtré sert ensuite de caractéristique à un classifieur linéaire (LDA). La précision rapportée est la proportion d'essais correctement classés,

$$\text{Précision} = \frac{\text{essais bien classés}}{\text{essais totaux}},$$

moyennée sur plusieurs tirages aléatoires indépendants d'entraînement/test (80 %/20 %) pour lisser la variabilité d'un tirage à l'autre.

## 1.2 MSE ICA vs ICLabel

Pour comparer deux méthodes de débruitage entre elles, on calcule l'erreur quadratique moyenne (MSE) entre les signaux nettoyés correspondants, essai par essai :

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2,$$

où  $x_i$  et  $y_i$  sont les échantillons (tous canaux et instants confondus) des deux signaux nettoyés et  $N$  leur nombre total. Seuls les essais communs aux deux signaux (gauche/droite, repos exclu) sont comparés.

## 1.3 Évolution du MSE pendant l'entraînement (LOSO)

Le modèle de débruitage est évalué en *leave-one-subject-out* : pour chaque sujet, un modèle neuf est entraîné uniquement sur les autres sujets, puis évalué (même MSE que ci-dessus) sur le sujet exclu, jamais vu pendant l'entraînement. L'opération est répétée pour chaque sujet.

Pendant l'entraînement, on suit deux courbes : le MSE sur les données d'entraînement et le MSE sur le sujet exclu, après chaque passage complet sur les données (epoch). Un écart croissant entre les deux signale un sur-apprentissage.

## 2 Sujets

### 2.1 Sujet S001

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.63 \pm 0.21$ |
| ART         | $0.60 \pm 0.15$ |
| ICUNet      | $0.51 \pm 0.14$ |
| ICUNet++    | $0.42 \pm 0.16$ |
| ICUNet_attn | $0.49 \pm 0.17$ |
| DuoCL       | $0.60 \pm 0.11$ |
| GCTNet      | $0.62 \pm 0.07$ |
| ICLabel     | $0.67 \pm 0.13$ |

TABLE 1 – S001 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ART         | 64.30           | 84.91       |             |
| ICUNet      | 17.16           | 27.65       |             |
| ICUNet++    | 16.01           | 25.89       |             |
| ICUNet_attn | 19.43           | 23.01       | 26.5        |
| DuoCL       | 24.82           | 16.02       |             |
| GCTNet      | 24.27           | 12.54       |             |

TABLE 2 – S001 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

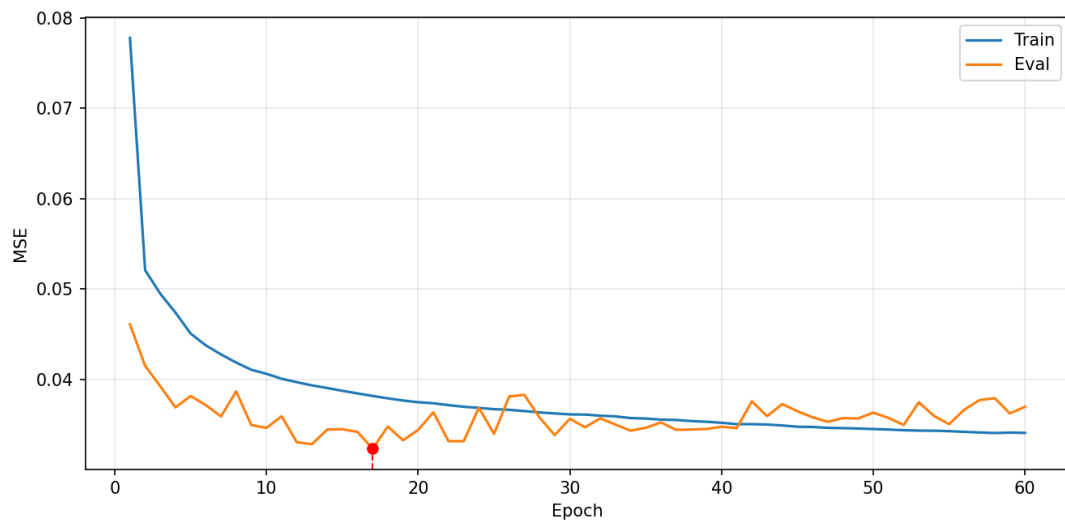


FIGURE 1 – S001 — Évolution du MSE.

Le MSE d'évaluation minimal (0.032) se présente à l'époch 17.

## Sujet S001 — ART par epoch

| Epoch | Accuracy                          | RMSE ( $\mu$ V, vs ICLabel) |
|-------|-----------------------------------|-----------------------------|
| 1     | $0.67 \pm 0.14$                   | 28.56                       |
| 2     | $0.68 \pm 0.14$                   | 28.63                       |
| 3     | $0.70 \pm 0.13$                   | 27.22                       |
| 4     | $0.68 \pm 0.18$                   | 28.52                       |
| 5     | $0.69 \pm 0.15$                   | 25.97                       |
| 6     | $0.64 \pm 0.12$                   | <b>25.65</b>                |
| 7     | $0.67 \pm 0.13$                   | 26.57                       |
| 8     | $0.69 \pm 0.18$                   | 25.88                       |
| 9     | $0.68 \pm 0.15$                   | 26.38                       |
| 10    | <b><math>0.72 \pm 0.19</math></b> | 26.88                       |
| 11    | $0.70 \pm 0.20$                   | 26.37                       |
| 12    | $0.68 \pm 0.18$                   | 26.94                       |
| 13    | $0.67 \pm 0.17$                   | 26.74                       |
| 14    | $0.67 \pm 0.12$                   | 27.38                       |
| 15    | $0.67 \pm 0.12$                   | 26.89                       |
| 16    | $0.68 \pm 0.20$                   | 27.00                       |
| 17    | $0.70 \pm 0.12$                   | 27.28                       |
| 18    | $0.70 \pm 0.14$                   | 26.10                       |
| 19    | $0.68 \pm 0.13$                   | 26.72                       |
| 20    | $0.69 \pm 0.15$                   | 27.66                       |
| 21    | $0.69 \pm 0.16$                   | 26.23                       |
| 22    | $0.68 \pm 0.12$                   | 27.78                       |
| 23    | $0.68 \pm 0.09$                   | 27.99                       |
| 24    | $0.71 \pm 0.12$                   | 28.32                       |
| 25    | $0.69 \pm 0.12$                   | 27.91                       |
| 26    | $0.71 \pm 0.09$                   | 28.43                       |
| 27    | $0.64 \pm 0.11$                   | 27.75                       |
| 28    | $0.66 \pm 0.13$                   | 27.98                       |
| 29    | $0.68 \pm 0.14$                   | 27.25                       |
| 30    | $0.70 \pm 0.12$                   | 27.23                       |
| 31    | $0.64 \pm 0.13$                   | 26.92                       |
| 32    | $0.67 \pm 0.10$                   | 27.60                       |
| 33    | $0.63 \pm 0.12$                   | 27.09                       |
| 34    | $0.62 \pm 0.16$                   | 27.12                       |
| 35    | $0.64 \pm 0.13$                   | 26.90                       |
| 36    | $0.70 \pm 0.12$                   | 27.51                       |
| 37    | $0.66 \pm 0.18$                   | 27.01                       |
| 38    | $0.63 \pm 0.16$                   | 27.34                       |
| 39    | $0.67 \pm 0.13$                   | 27.26                       |
| 40    | $0.59 \pm 0.16$                   | 27.44                       |
| 41    | $0.68 \pm 0.10$                   | 26.99                       |
| 42    | $0.71 \pm 0.12$                   | 27.18                       |
| 43    | $0.69 \pm 0.16$                   | 26.92                       |
| 44    | $0.68 \pm 0.14$                   | 28.11                       |
| 45    | $0.68 \pm 0.13$                   | 27.82                       |
| 46    | $0.67 \pm 0.14$                   | 27.72                       |
| 47    | $0.64 \pm 0.15$                   | 27.53                       |
| 48    | $0.64 \pm 0.13$                   | 27.74                       |
| 49    | $0.70 \pm 0.12$                   | 27.30                       |
| 50    | $0.68 \pm 0.08$                   | 27.19                       |
| 51    | $0.68 \pm 0.10$                   | 27.27                       |
| 52    | $0.68 \pm 0.14$                   | 27.48                       |
| 53    | $0.67 \pm 0.07$                   | 28.12                       |
| 54    | $0.64 \pm 0.13$                   | 28.40                       |
| 55    | $0.68 \pm 0.15$                   | 27.07                       |
| 56    | $0.71 \pm 0.17$                   | 28.42                       |
| 57    | $0.66 \pm 0.08$                   | 27.86                       |
| 58    | $0.70 \pm 0.14$                   | 26.88                       |
| 59    | $0.67 \pm 0.11$                   | 27.72                       |
| 60    | $0.69 \pm 0.08$                   | 27.71                       |

Le RMS de référence (signal nettoyé par ICLabel) pour S001 est de 26.5  $\mu$ V.

## Sujet S001 - ART : Evolution des critères d'évaluation

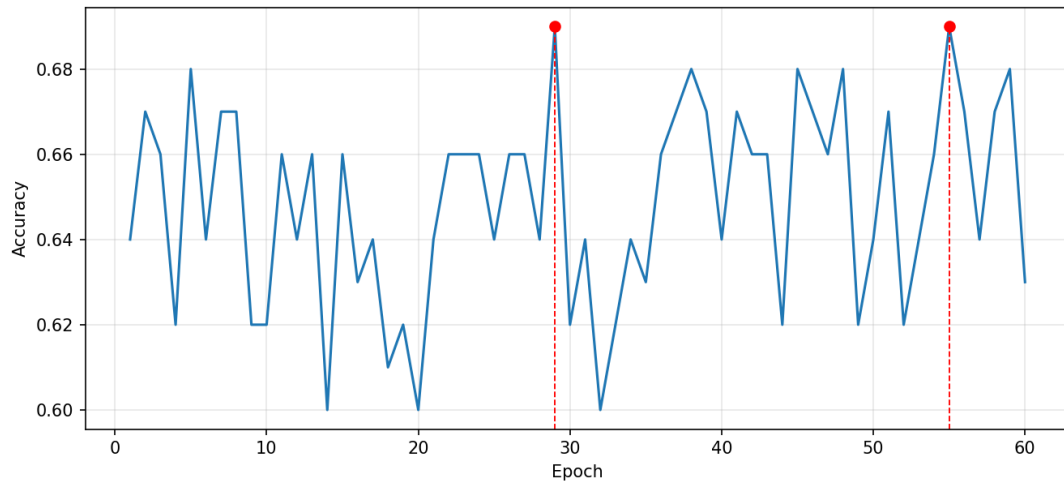


FIGURE 2 – S001 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.69) se présente à l'epoch 29 et à l'epoch 55.

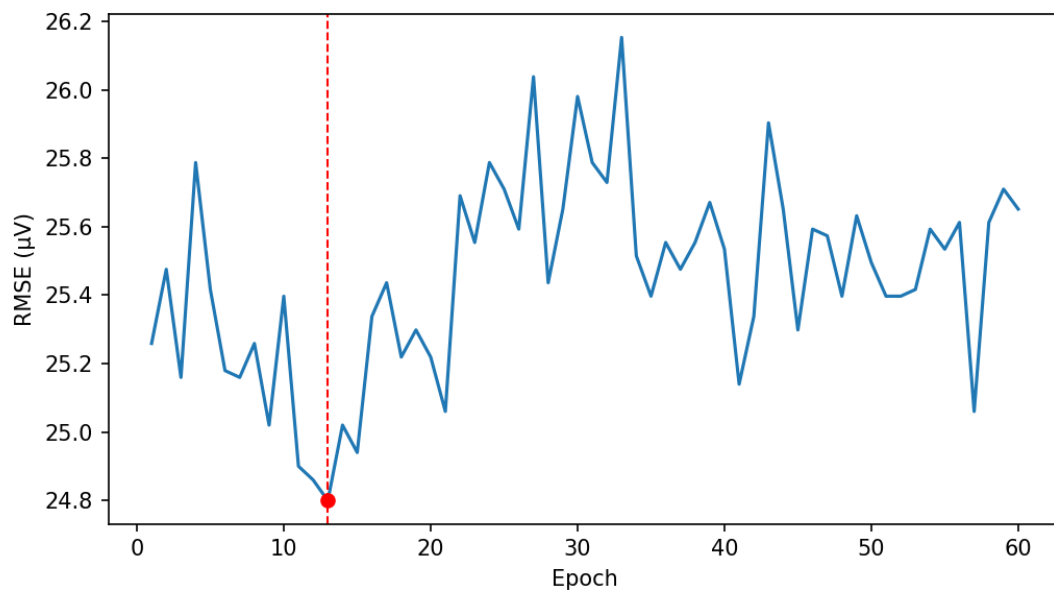


FIGURE 3 – S001 — Évolution du RMSE par epoch (ART vs ICLabel, checkpoints LOSO).

Le RMSE minimal (24.8  $\mu V$ ) se présente à l'epoch 13.

## 2.2 Sujet S002

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.66 \pm 0.18$ |
| ART         | $0.66 \pm 0.18$ |
| ICUNet      | $0.66 \pm 0.14$ |
| ICUNet++    | $0.54 \pm 0.20$ |
| ICUNet_attn | $0.64 \pm 0.13$ |
| DuoCL       | $0.67 \pm 0.12$ |
| GCTNet      | $0.68 \pm 0.14$ |
| ICLabel     | $0.70 \pm 0.19$ |

TABLE 3 – S002 — Accuracy de classification.

| Comparaison | RMSE ( $\mu V$ ) | RMS_Méthode | RMS_ICLABEL |
|-------------|------------------|-------------|-------------|
| ICUNet      | 7.14             | 11.77       |             |
| ICUNet++    | 6.68             | 12.03       |             |
| ICUNet_attn | 9.55             | 12.07       | 14.7        |
| DuoCL       | 12.12            | 9.17        |             |
| GCTNet      | 11.75            | 7.75        |             |

TABLE 4 – S002 — Erreur (RMSE,  $\mu V$ ) par rapport au signal nettoyé par ICLabel.

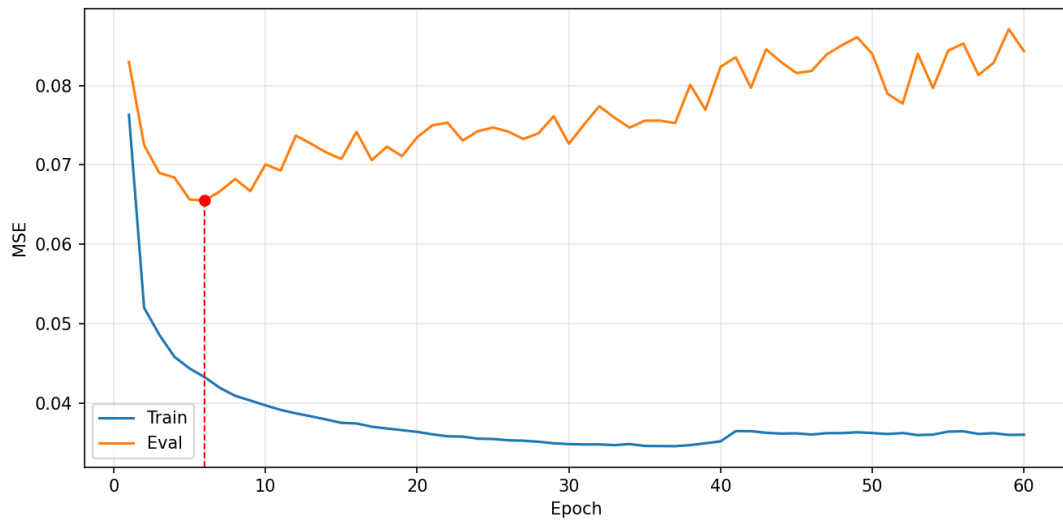


FIGURE 4 – S002 — Évolution du MSE.

Le MSE d'évaluation minimal (0.066) se présente à l'époch 6.

## Sujet S002 — ART par epoch

| Epoch | Accuracy                          | RMSE ( $\mu$ V, vs ICLabel) |
|-------|-----------------------------------|-----------------------------|
| 1     | $0.72 \pm 0.19$                   | 13.40                       |
| 2     | <b><math>0.77 \pm 0.18</math></b> | 13.15                       |
| 3     | $0.69 \pm 0.20$                   | 13.17                       |
| 4     | $0.69 \pm 0.20$                   | 13.10                       |
| 5     | $0.73 \pm 0.19$                   | 13.57                       |
| 6     | $0.69 \pm 0.13$                   | 13.32                       |
| 7     | $0.69 \pm 0.14$                   | 13.45                       |
| 8     | $0.71 \pm 0.15$                   | 13.32                       |
| 9     | $0.68 \pm 0.13$                   | 13.52                       |
| 10    | $0.70 \pm 0.17$                   | 13.26                       |
| 11    | $0.71 \pm 0.19$                   | 13.39                       |
| 12    | $0.72 \pm 0.19$                   | 13.30                       |
| 13    | $0.72 \pm 0.17$                   | 13.27                       |
| 14    | $0.68 \pm 0.17$                   | 13.11                       |
| 15    | $0.66 \pm 0.17$                   | 13.34                       |
| 16    | $0.64 \pm 0.18$                   | 12.86                       |
| 17    | $0.64 \pm 0.18$                   | 13.36                       |
| 18    | $0.68 \pm 0.20$                   | <b>12.42</b>                |
| 19    | $0.67 \pm 0.21$                   | 13.28                       |
| 20    | $0.64 \pm 0.21$                   | 13.33                       |
| 21    | $0.64 \pm 0.20$                   | 13.38                       |
| 22    | $0.70 \pm 0.19$                   | 13.22                       |
| 23    | $0.68 \pm 0.22$                   | 13.10                       |
| 24    | $0.68 \pm 0.20$                   | 12.99                       |
| 25    | $0.66 \pm 0.18$                   | 13.32                       |
| 26    | $0.69 \pm 0.21$                   | 13.04                       |
| 27    | $0.64 \pm 0.18$                   | 13.40                       |
| 28    | $0.70 \pm 0.21$                   | 13.35                       |
| 29    | $0.66 \pm 0.18$                   | 13.40                       |
| 30    | $0.71 \pm 0.21$                   | 13.30                       |
| 31    | $0.69 \pm 0.20$                   | 13.37                       |
| 32    | $0.64 \pm 0.17$                   | 13.01                       |
| 33    | $0.70 \pm 0.17$                   | 13.24                       |
| 34    | $0.66 \pm 0.18$                   | 13.24                       |
| 35    | $0.70 \pm 0.17$                   | 13.31                       |
| 36    | $0.73 \pm 0.22$                   | 13.41                       |
| 37    | $0.70 \pm 0.12$                   | 13.50                       |
| 38    | $0.70 \pm 0.18$                   | 13.17                       |
| 39    | $0.68 \pm 0.15$                   | 13.63                       |
| 40    | $0.70 \pm 0.20$                   | 13.13                       |
| 41    | $0.68 \pm 0.18$                   | 13.40                       |
| 42    | $0.71 \pm 0.17$                   | 13.16                       |
| 43    | $0.66 \pm 0.16$                   | 12.97                       |
| 44    | $0.69 \pm 0.21$                   | 13.65                       |
| 45    | $0.68 \pm 0.18$                   | 13.09                       |
| 46    | $0.71 \pm 0.13$                   | 13.03                       |
| 47    | $0.72 \pm 0.18$                   | 13.11                       |
| 48    | $0.63 \pm 0.17$                   | 13.30                       |
| 49    | $0.66 \pm 0.18$                   | 13.02                       |
| 50    | $0.68 \pm 0.19$                   | 13.34                       |
| 51    | $0.70 \pm 0.16$                   | 13.27                       |
| 52    | $0.69 \pm 0.18$                   | 13.33                       |
| 53    | $0.71 \pm 0.19$                   | 13.15                       |
| 54    | $0.69 \pm 0.15$                   | 13.33                       |
| 55    | $0.69 \pm 0.15$                   | 13.17                       |
| 56    | $0.68 \pm 0.16$                   | 12.99                       |
| 57    | $0.70 \pm 0.14$                   | 13.07                       |
| 58    | $0.70 \pm 0.14$                   | 13.30                       |
| 59    | $0.67 \pm 0.17$                   | 13.01                       |
| 60    | $0.69 \pm 0.14$                   | 13.00                       |

Le RMS de référence (signal nettoyé par ICLabel) pour S002 est de  $14.7 \mu$ V.

## Sujet S002 - ART : Evolution des critères d'évaluation

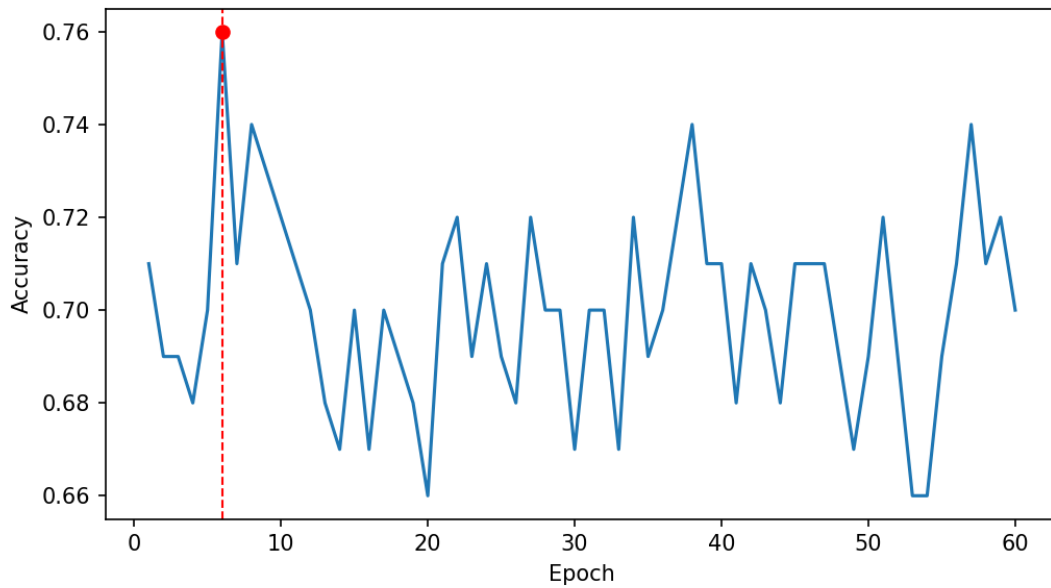


FIGURE 5 – S002 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.76) se présente à l'epoch 6.

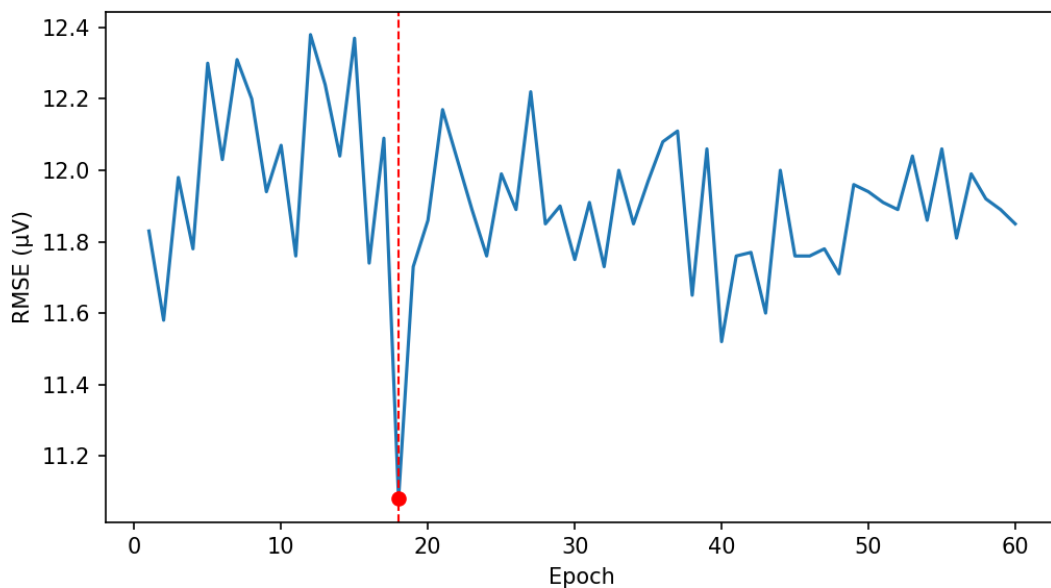


FIGURE 6 – S002 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (11.1  $\mu V$ ) se présente à l'epoch 18.

### 2.3 Sujet S003

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.56 \pm 0.16$ |
| ICUNet      | $0.43 \pm 0.15$ |
| ICUNet++    | $0.44 \pm 0.11$ |
| ICUNet_attn | $0.50 \pm 0.06$ |
| DuoCL       | $0.63 \pm 0.12$ |
| GCTNet      | $0.46 \pm 0.17$ |
| ICLabel     | $0.66 \pm 0.13$ |

TABLE 5 – S003 — Accuracy de classification.

| Comparaison | RMSE ( $\mu V$ ) | RMS_Méthode | RMS_ICLABEL |
|-------------|------------------|-------------|-------------|
| ICUNet      | 21.51            | 26.67       |             |
| ICUNet++    | 18.88            | 19.68       |             |
| ICUNet_attn | 23.69            | 19.69       | 25.1        |
| DuoCL       | 24.52            | 17.49       |             |
| GCTNet      | 23.60            | 14.51       |             |

TABLE 6 – S003 — Erreur (RMSE,  $\mu V$ ) par rapport au signal nettoyé par ICLabel.

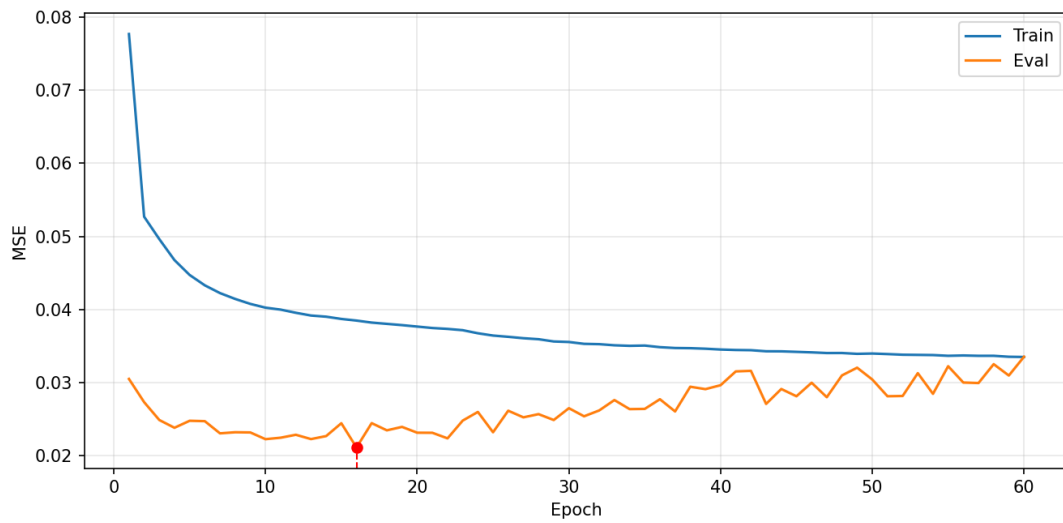


FIGURE 7 – S003 — Évolution du MSE.

Le MSE d'évaluation minimal (0.021) se présente à l'époch 16.

## Sujet S003 — ART par epoch

| Epoch | Accuracy                          | RMSE ( $\mu\text{V}$ , vs ICLabel) |
|-------|-----------------------------------|------------------------------------|
| 1     | $0.51 \pm 0.17$                   | 50.67                              |
| 2     | $0.44 \pm 0.13$                   | 47.82                              |
| 3     | $0.54 \pm 0.17$                   | 48.68                              |
| 4     | $0.51 \pm 0.13$                   | 48.15                              |
| 5     | $0.52 \pm 0.16$                   | 43.85                              |
| 6     | $0.54 \pm 0.17$                   | 47.50                              |
| 7     | $0.56 \pm 0.18$                   | 48.33                              |
| 8     | $0.53 \pm 0.11$                   | 48.73                              |
| 9     | $0.43 \pm 0.15$                   | 49.51                              |
| 10    | $0.51 \pm 0.15$                   | 50.43                              |
| 11    | $0.53 \pm 0.12$                   | 48.05                              |
| 12    | $0.52 \pm 0.16$                   | 46.54                              |
| 13    | $0.52 \pm 0.17$                   | 49.86                              |
| 14    | $0.54 \pm 0.13$                   | 48.71                              |
| 15    | $0.49 \pm 0.17$                   | 47.48                              |
| 16    | $0.43 \pm 0.12$                   | 49.94                              |
| 17    | $0.42 \pm 0.15$                   | 48.90                              |
| 18    | $0.48 \pm 0.13$                   | 48.36                              |
| 19    | $0.54 \pm 0.19$                   | 49.03                              |
| 20    | $0.50 \pm 0.21$                   | 46.65                              |
| 21    | $0.56 \pm 0.19$                   | 47.88                              |
| 22    | $0.48 \pm 0.17$                   | 47.11                              |
| 23    | $0.52 \pm 0.16$                   | 49.45                              |
| 24    | $0.46 \pm 0.16$                   | 48.71                              |
| 25    | $0.51 \pm 0.17$                   | 47.85                              |
| 26    | $0.52 \pm 0.22$                   | 46.97                              |
| 27    | $0.50 \pm 0.12$                   | 47.73                              |
| 28    | $0.53 \pm 0.10$                   | 45.04                              |
| 29    | $0.47 \pm 0.16$                   | 48.31                              |
| 30    | $0.43 \pm 0.15$                   | 44.25                              |
| 31    | $0.48 \pm 0.11$                   | 46.30                              |
| 32    | $0.47 \pm 0.16$                   | 45.41                              |
| 33    | $0.50 \pm 0.13$                   | 44.78                              |
| 34    | $0.43 \pm 0.15$                   | 47.42                              |
| 35    | $0.50 \pm 0.15$                   | 45.32                              |
| 36    | $0.56 \pm 0.16$                   | 46.59                              |
| 37    | $0.54 \pm 0.14$                   | 43.58                              |
| 38    | $0.51 \pm 0.11$                   | 45.17                              |
| 39    | <b><math>0.59 \pm 0.16</math></b> | 45.30                              |
| 40    | $0.49 \pm 0.12$                   | 44.15                              |
| 41    | $0.49 \pm 0.17$                   | 46.66                              |
| 42    | $0.58 \pm 0.14$                   | 44.28                              |
| 43    | $0.51 \pm 0.09$                   | 45.13                              |
| 44    | $0.47 \pm 0.13$                   | 45.22                              |
| 45    | $0.52 \pm 0.12$                   | 44.18                              |
| 46    | $0.47 \pm 0.21$                   | 49.24                              |
| 47    | $0.51 \pm 0.14$                   | 46.87                              |
| 48    | $0.42 \pm 0.15$                   | 45.71                              |
| 49    | $0.53 \pm 0.11$                   | 44.75                              |
| 50    | $0.47 \pm 0.20$                   | 43.86                              |
| 51    | $0.48 \pm 0.17$                   | 44.49                              |
| 52    | $0.50 \pm 0.12$                   | 44.14                              |
| 53    | $0.58 \pm 0.13$                   | 43.77                              |
| 54    | $0.51 \pm 0.19$                   | 46.30                              |
| 55    | $0.49 \pm 0.17$                   | 44.14                              |
| 56    | $0.50 \pm 0.13$                   | <b>42.76</b>                       |
| 57    | $0.54 \pm 0.18$                   | 46.63                              |
| 58    | $0.53 \pm 0.16$                   | 43.52                              |
| 59    | $0.52 \pm 0.19$                   | 45.43                              |
| 60    | $0.48 \pm 0.10$                   | 44.67                              |

Le RMS de référence (signal nettoyé par ICLabel) pour S003 est de  $25.1 \mu\text{V}$ .

## Sujet S003 - ART : Evolution des critères d'évaluation

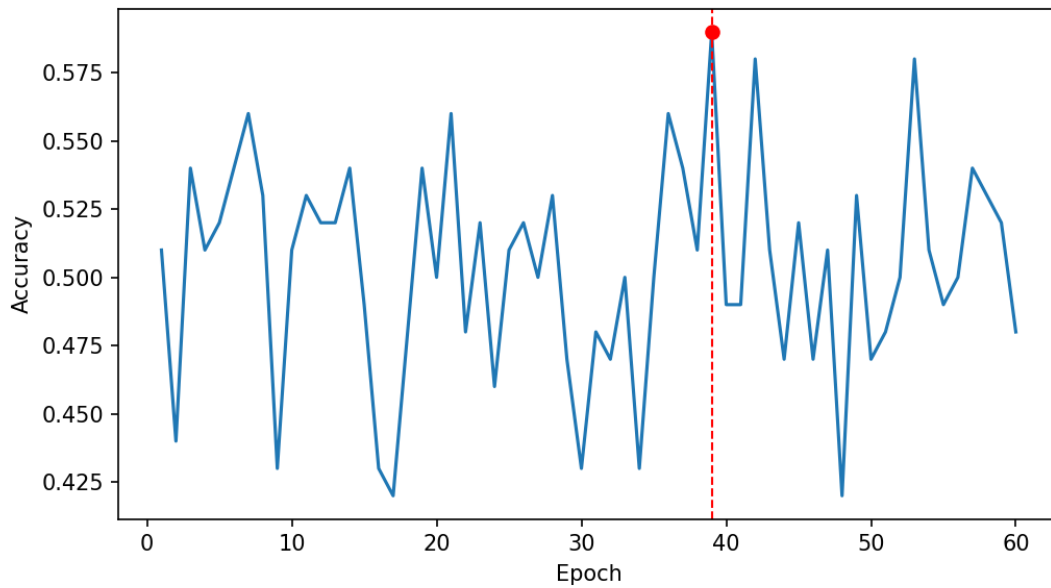


FIGURE 8 – S003 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.59) se présente à l'epoch 39.

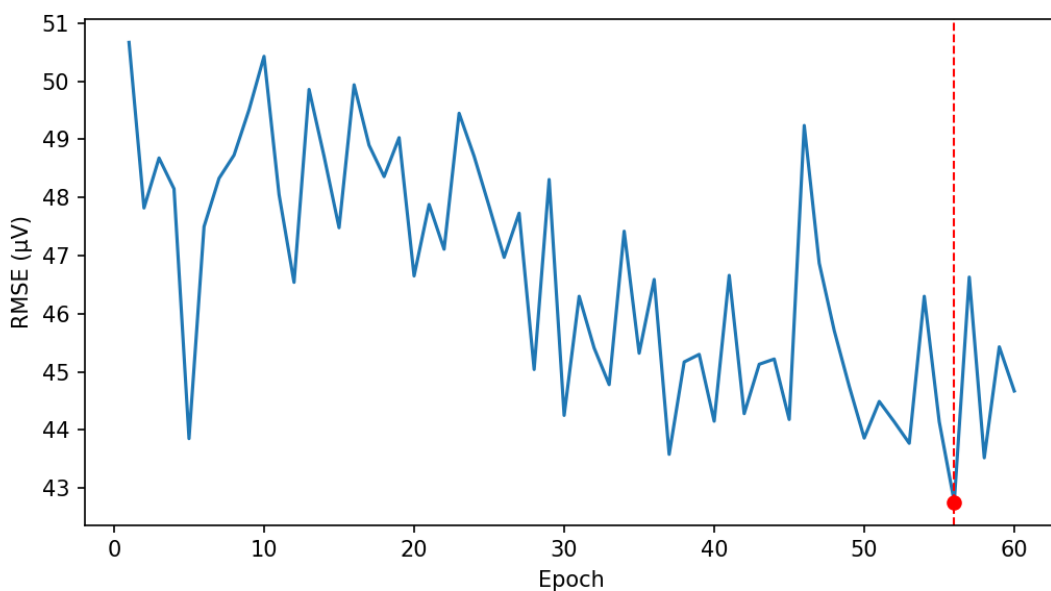


FIGURE 9 – S003 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (42.8  $\mu V$ ) se présente à l'epoch 56.

## 2.4 Sujet S004

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.43 \pm 0.16$ |
| ICUNet      | $0.58 \pm 0.17$ |
| ICUNet++    | $0.66 \pm 0.13$ |
| ICUNet_attn | $0.36 \pm 0.15$ |
| DuoCL       | $0.60 \pm 0.19$ |
| GCTNet      | $0.54 \pm 0.12$ |
| ICLabel     | $0.44 \pm 0.11$ |

TABLE 7 – S004 — Accuracy de classification.

| Comparaison | RMSE ( $\mu V$ ) | RMS_Méthode | RMS_ICLABEL |
|-------------|------------------|-------------|-------------|
| ICUNet      | 23.58            | 12.21       |             |
| ICUNet++    | 25.08            | 7.53        |             |
| ICUNet_attn | 26.24            | 10.50       | 26.1        |
| DuoCL       | 23.81            | 8.72        |             |
| GCTNet      | 24.33            | 7.30        |             |

TABLE 8 – S004 — Erreur (RMSE,  $\mu V$ ) par rapport au signal nettoyé par ICLabel.

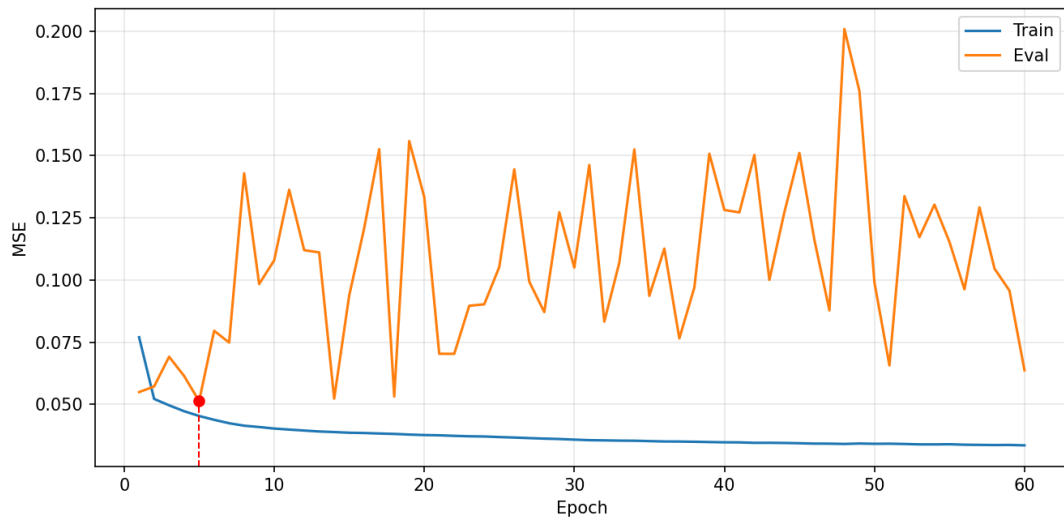


FIGURE 10 – S004 — Évolution du MSE.

Le MSE d'évaluation minimal (0.051) se présente à l'époch 5.

## Sujet S004 — ART par epoch

| Epoch | Accuracy                          | RMSE ( $\mu\text{V}$ , vs ICLabel) |
|-------|-----------------------------------|------------------------------------|
| 1     | $0.41 \pm 0.16$                   | 21.82                              |
| 2     | $0.46 \pm 0.14$                   | 21.18                              |
| 3     | $0.51 \pm 0.15$                   | 21.32                              |
| 4     | $0.38 \pm 0.12$                   | 19.69                              |
| 5     | $0.48 \pm 0.18$                   | 19.28                              |
| 6     | $0.48 \pm 0.16$                   | 19.40                              |
| 7     | $0.42 \pm 0.16$                   | 20.08                              |
| 8     | $0.43 \pm 0.16$                   | 21.25                              |
| 9     | $0.44 \pm 0.15$                   | 20.45                              |
| 10    | $0.51 \pm 0.17$                   | 21.94                              |
| 11    | $0.51 \pm 0.17$                   | 21.17                              |
| 12    | $0.48 \pm 0.20$                   | 20.93                              |
| 13    | $0.48 \pm 0.13$                   | 20.80                              |
| 14    | $0.51 \pm 0.22$                   | <b>19.11</b>                       |
| 15    | $0.49 \pm 0.19$                   | 21.40                              |
| 16    | $0.48 \pm 0.17$                   | 20.97                              |
| 17    | $0.43 \pm 0.19$                   | 22.66                              |
| 18    | $0.49 \pm 0.18$                   | 20.54                              |
| 19    | $0.53 \pm 0.20$                   | 22.67                              |
| 20    | $0.52 \pm 0.13$                   | 23.05                              |
| 21    | $0.44 \pm 0.21$                   | 22.00                              |
| 22    | $0.44 \pm 0.12$                   | 21.59                              |
| 23    | $0.40 \pm 0.14$                   | 22.07                              |
| 24    | $0.48 \pm 0.10$                   | 20.68                              |
| 25    | $0.54 \pm 0.12$                   | 22.31                              |
| 26    | $0.42 \pm 0.15$                   | 22.71                              |
| 27    | $0.42 \pm 0.13$                   | 21.56                              |
| 28    | $0.56 \pm 0.18$                   | 21.73                              |
| 29    | $0.42 \pm 0.18$                   | 22.59                              |
| 30    | $0.42 \pm 0.23$                   | 22.48                              |
| 31    | $0.41 \pm 0.14$                   | 21.70                              |
| 32    | $0.48 \pm 0.14$                   | 20.40                              |
| 33    | $0.43 \pm 0.14$                   | 22.05                              |
| 34    | $0.53 \pm 0.13$                   | 21.60                              |
| 35    | $0.48 \pm 0.11$                   | 21.94                              |
| 36    | $0.38 \pm 0.13$                   | 21.30                              |
| 37    | $0.38 \pm 0.10$                   | 20.56                              |
| 38    | $0.52 \pm 0.12$                   | 21.79                              |
| 39    | $0.47 \pm 0.16$                   | 23.36                              |
| 40    | <b><math>0.70 \pm 0.10</math></b> | 21.90                              |
| 41    | $0.50 \pm 0.10$                   | 21.85                              |
| 42    | $0.47 \pm 0.12$                   | 22.47                              |
| 43    | $0.52 \pm 0.20$                   | 21.42                              |
| 44    | $0.52 \pm 0.16$                   | 21.85                              |
| 45    | $0.56 \pm 0.13$                   | 22.46                              |
| 46    | $0.48 \pm 0.09$                   | 20.87                              |
| 47    | $0.51 \pm 0.09$                   | 20.71                              |
| 48    | $0.46 \pm 0.13$                   | 22.73                              |
| 49    | $0.41 \pm 0.14$                   | 22.72                              |
| 50    | $0.48 \pm 0.19$                   | 20.52                              |
| 51    | $0.37 \pm 0.17$                   | 21.17                              |
| 52    | $0.52 \pm 0.14$                   | 21.30                              |
| 53    | $0.50 \pm 0.11$                   | 21.74                              |
| 54    | $0.46 \pm 0.15$                   | 21.23                              |
| 55    | $0.42 \pm 0.15$                   | 21.59                              |
| 56    | $0.51 \pm 0.17$                   | 20.87                              |
| 57    | $0.58 \pm 0.22$                   | 20.56                              |
| 58    | $0.54 \pm 0.14$                   | 20.70                              |
| 59    | $0.44 \pm 0.12$                   | 20.86                              |
| 60    | $0.60 \pm 0.10$                   | 20.63                              |

Le RMS de référence (signal nettoyé par ICLabel) pour S004 est de  $26.1 \mu\text{V}$ .

## Sujet S004 - ART : Evolution des critères d'évaluation

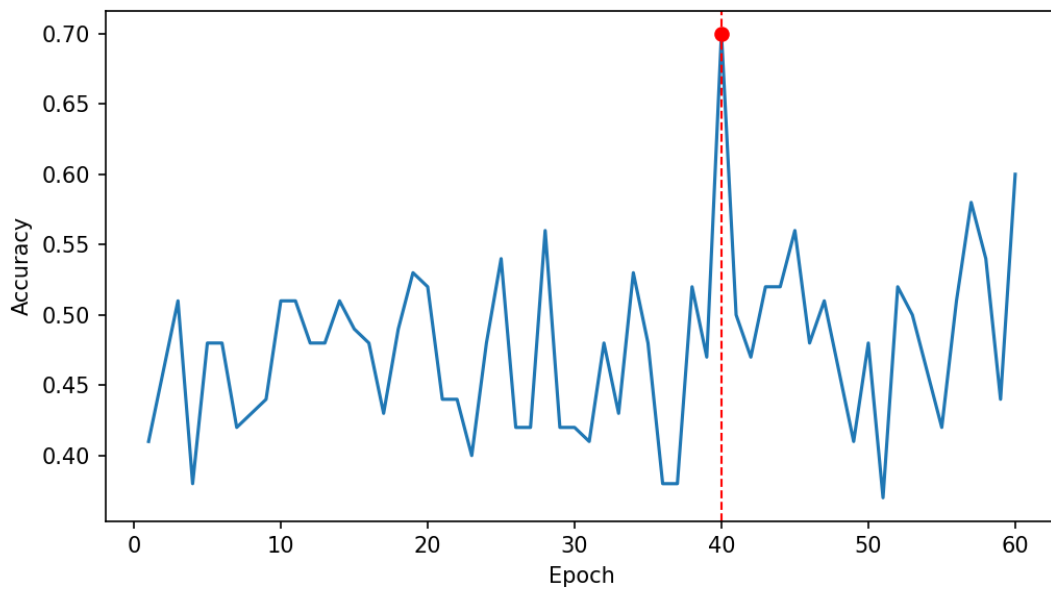


FIGURE 11 – S004 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.70) se présente à l'epoch 40.

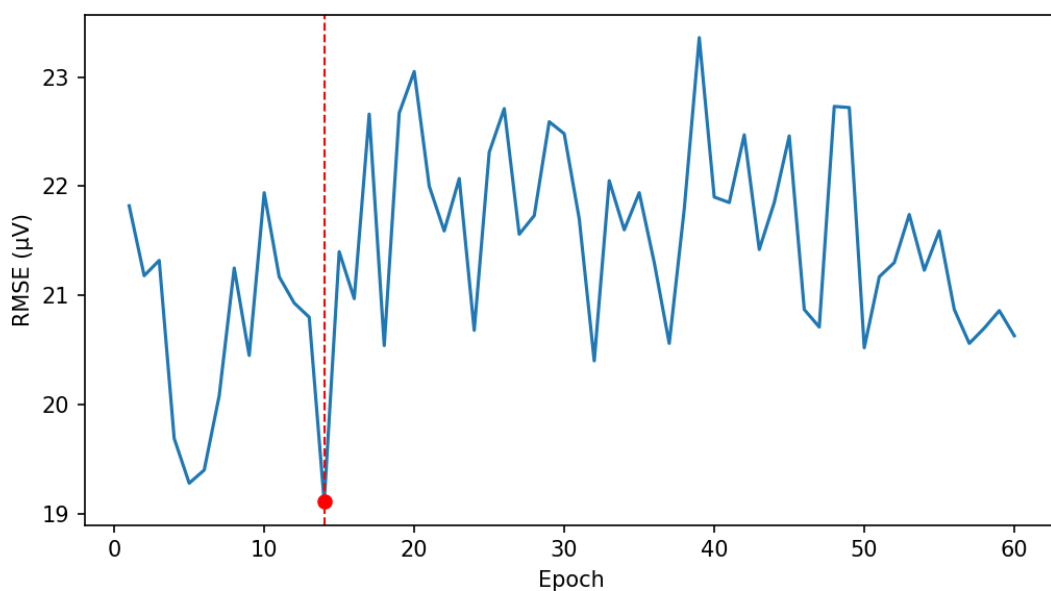


FIGURE 12 – S004 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (19.1 μV) se présente à l'epoch 14.

## 2.5 Sujet S005

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.49 \pm 0.14$ |
| ICUNet      | $0.48 \pm 0.14$ |
| ICUNet++    | $0.44 \pm 0.16$ |
| ICUNet_attn | $0.33 \pm 0.16$ |
| DuoCL       | $0.37 \pm 0.17$ |
| GCTNet      | $0.60 \pm 0.10$ |
| ICLabel     | $0.49 \pm 0.12$ |

TABLE 9 – S005 — Accuracy de classification.

| Comparaison | RMSE ( $\mu\text{V}$ ) | RMS_Méthode | RMS_ICLABEL |
|-------------|------------------------|-------------|-------------|
| ICUNet      | 12.13                  | 9.05        |             |
| ICUNet++    | 12.64                  | 8.55        |             |
| ICUNet_attn | 14.66                  | 7.83        | 16.7        |
| DuoCL       | 14.80                  | 8.48        |             |
| GCTNet      | 14.59                  | 7.01        |             |

TABLE 10 – S005 — Erreur (RMSE,  $\mu\text{V}$ ) par rapport au signal nettoyé par ICLabel.

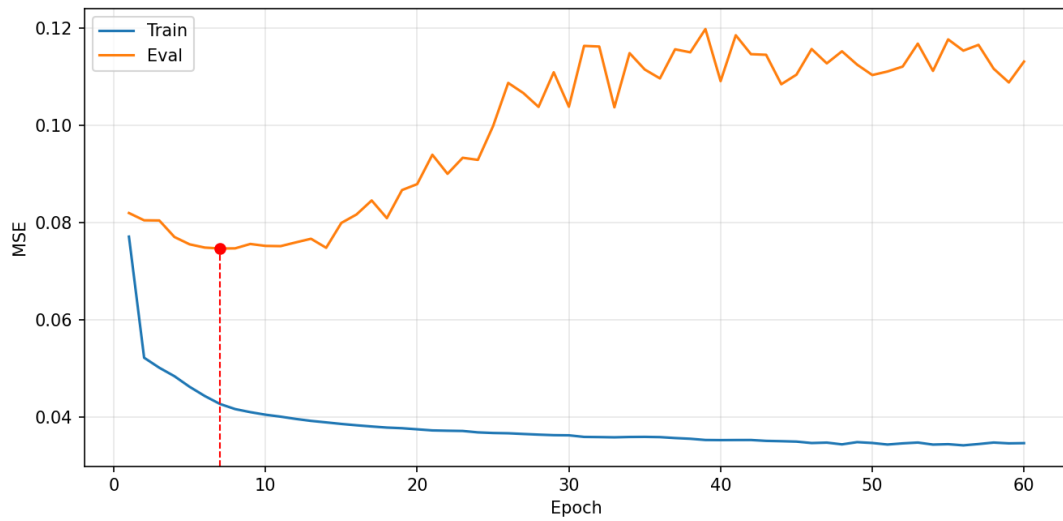


FIGURE 13 – S005 — Évolution du MSE.

Le MSE d'évaluation minimal (0.075) se présente à l'époch 7.

## Sujet S005 — ART par epoch

| Epoch | Accuracy                          | RMSE ( $\mu\text{V}$ , vs ICLabel) |
|-------|-----------------------------------|------------------------------------|
| 1     | $0.52 \pm 0.13$                   | 16.98                              |
| 2     | $0.51 \pm 0.15$                   | 16.09                              |
| 3     | $0.53 \pm 0.20$                   | 16.17                              |
| 4     | $0.54 \pm 0.14$                   | 15.61                              |
| 5     | $0.52 \pm 0.11$                   | 16.20                              |
| 6     | $0.51 \pm 0.16$                   | 16.30                              |
| 7     | $0.47 \pm 0.10$                   | 16.14                              |
| 8     | $0.52 \pm 0.17$                   | 16.08                              |
| 9     | $0.50 \pm 0.13$                   | 15.20                              |
| 10    | $0.49 \pm 0.14$                   | 15.18                              |
| 11    | <b><math>0.58 \pm 0.17</math></b> | 15.81                              |
| 12    | $0.52 \pm 0.19$                   | 15.85                              |
| 13    | $0.50 \pm 0.16$                   | 16.34                              |
| 14    | $0.51 \pm 0.17$                   | 15.74                              |
| 15    | $0.54 \pm 0.17$                   | 14.76                              |
| 16    | $0.52 \pm 0.19$                   | 14.51                              |
| 17    | $0.54 \pm 0.08$                   | 14.69                              |
| 18    | $0.54 \pm 0.10$                   | 14.94                              |
| 19    | $0.54 \pm 0.10$                   | 14.15                              |
| 20    | $0.49 \pm 0.14$                   | 14.32                              |
| 21    | $0.51 \pm 0.12$                   | 14.05                              |
| 22    | $0.50 \pm 0.14$                   | 13.86                              |
| 23    | $0.50 \pm 0.14$                   | 14.55                              |
| 24    | $0.53 \pm 0.13$                   | 14.17                              |
| 25    | $0.53 \pm 0.11$                   | 13.91                              |
| 26    | $0.50 \pm 0.06$                   | 13.66                              |
| 27    | $0.48 \pm 0.14$                   | 13.40                              |
| 28    | $0.53 \pm 0.11$                   | 13.75                              |
| 29    | $0.48 \pm 0.12$                   | 13.69                              |
| 30    | $0.52 \pm 0.10$                   | 13.91                              |
| 31    | $0.47 \pm 0.10$                   | 13.25                              |
| 32    | $0.50 \pm 0.12$                   | 13.30                              |
| 33    | $0.52 \pm 0.10$                   | 13.88                              |
| 34    | $0.49 \pm 0.10$                   | 13.71                              |
| 35    | $0.49 \pm 0.12$                   | 13.69                              |
| 36    | $0.48 \pm 0.12$                   | 13.49                              |
| 37    | $0.47 \pm 0.12$                   | 13.35                              |
| 38    | $0.53 \pm 0.11$                   | 13.35                              |
| 39    | $0.52 \pm 0.11$                   | <b>13.09</b>                       |
| 40    | $0.54 \pm 0.13$                   | 13.54                              |
| 41    | $0.51 \pm 0.14$                   | 13.28                              |
| 42    | $0.52 \pm 0.09$                   | 13.86                              |
| 43    | $0.52 \pm 0.07$                   | 13.57                              |
| 44    | $0.50 \pm 0.12$                   | 13.84                              |
| 45    | $0.48 \pm 0.11$                   | 13.63                              |
| 46    | <b><math>0.58 \pm 0.08</math></b> | 13.82                              |
| 47    | $0.52 \pm 0.14$                   | 13.48                              |
| 48    | $0.52 \pm 0.07$                   | 14.04                              |
| 49    | $0.46 \pm 0.12$                   | 13.39                              |
| 50    | $0.53 \pm 0.14$                   | 13.53                              |
| 51    | $0.49 \pm 0.14$                   | 13.47                              |
| 52    | $0.50 \pm 0.10$                   | 13.70                              |
| 53    | $0.52 \pm 0.10$                   | 13.19                              |
| 54    | $0.53 \pm 0.15$                   | 13.93                              |
| 55    | $0.51 \pm 0.12$                   | 13.72                              |
| 56    | $0.52 \pm 0.09$                   | 13.47                              |
| 57    | $0.52 \pm 0.13$                   | 13.55                              |
| 58    | $0.54 \pm 0.09$                   | 13.58                              |
| 59    | $0.52 \pm 0.10$                   | 13.63                              |
| 60    | $0.50 \pm 0.11$                   | 13.47                              |

Le RMS de référence (signal nettoyé par ICLabel) pour S005 est de  $16.7 \mu\text{V}$ .

## Sujet S005 - ART : Evolution des critères d'évaluation

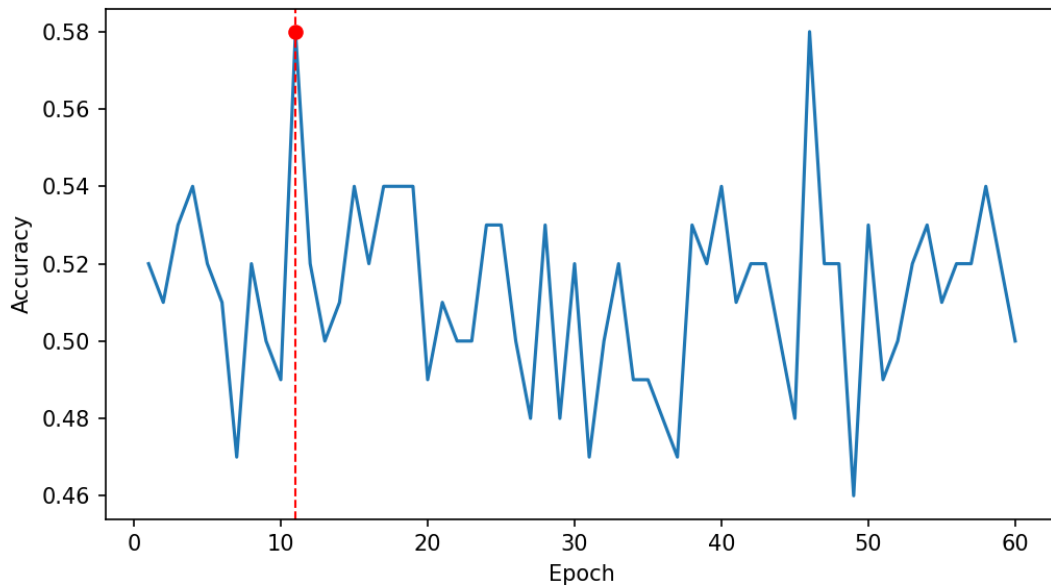


FIGURE 14 – S005 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.58) se présente à l'epoch 11.

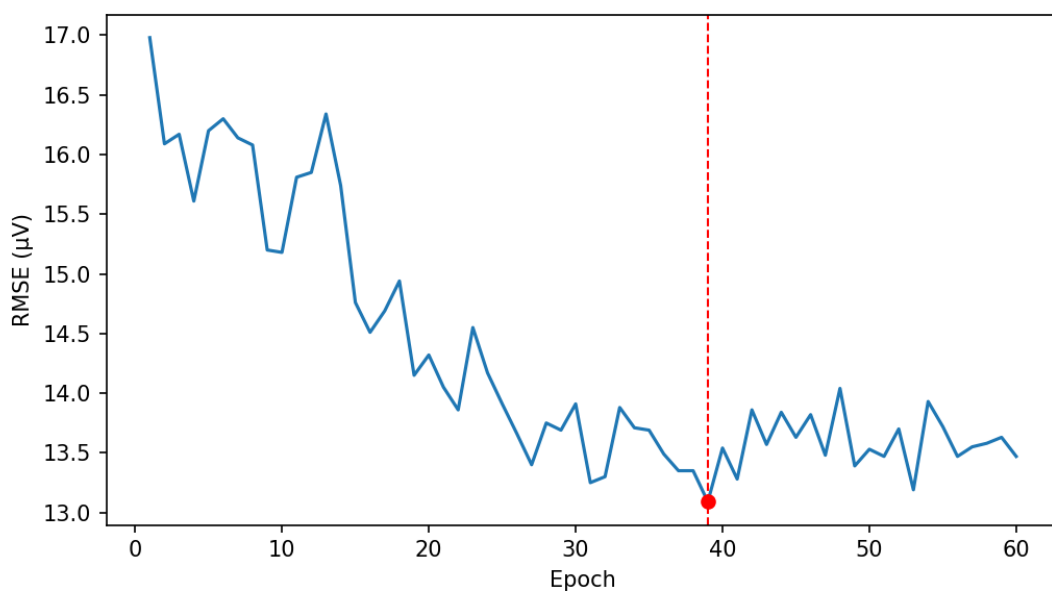


FIGURE 15 – S005 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (13.1  $\mu V$ ) se présente à l'epoch 39.

## 2.6 Sujet S006

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.50 \pm 0.11$ |
| ICUNet      | $0.53 \pm 0.12$ |
| ICUNet++    | $0.49 \pm 0.18$ |
| ICUNet_attn | $0.57 \pm 0.18$ |
| DuoCL       | $0.52 \pm 0.09$ |
| GCTNet      | $0.47 \pm 0.17$ |
| ICLabel     | $0.51 \pm 0.15$ |

TABLE 11 – S006 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 8.05            | 7.56        |             |
| ICUNet++    | 8.33            | 6.93        |             |
| ICUNet_attn | 9.89            | 8.13        | 11.8        |
| DuoCL       | 11.05           | 6.80        |             |
| GCTNet      | 10.91           | 6.22        |             |

TABLE 12 – S006 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

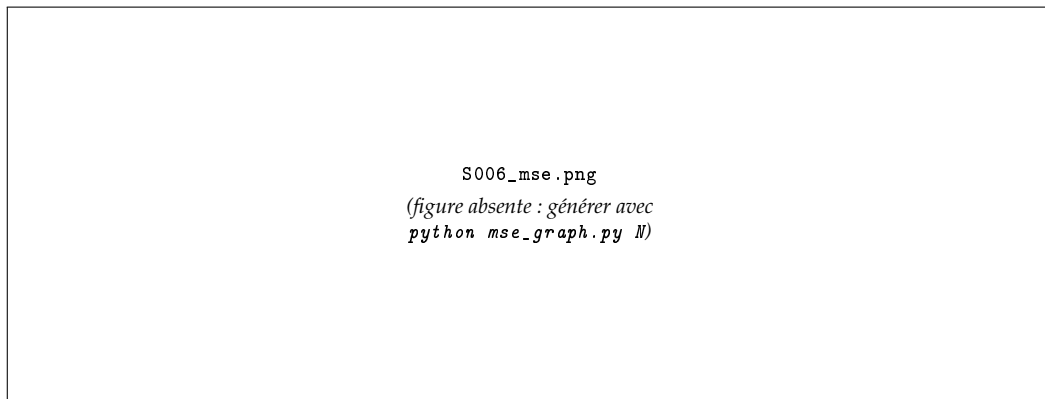


FIGURE 16 – S006 — Évolution du MSE.

## 2.7 Sujet S007

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.96 \pm 0.07$ |
| ICUNet      | $0.82 \pm 0.10$ |
| ICUNet++    | $0.77 \pm 0.09$ |
| ICUNet_attn | $0.61 \pm 0.17$ |
| DuoCL       | $0.81 \pm 0.10$ |
| GCTNet      | $0.91 \pm 0.08$ |
| ICLabel     | $0.98 \pm 0.04$ |

TABLE 13 – S007 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 8.20            | 19.50       |             |
| ICUNet++    | 7.88            | 19.25       |             |
| ICUNet_attn | 13.22           | 19.46       | 21.3        |
| DuoCL       | 15.56           | 15.70       |             |
| GCTNet      | 15.00           | 13.48       |             |

TABLE 14 – S007 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

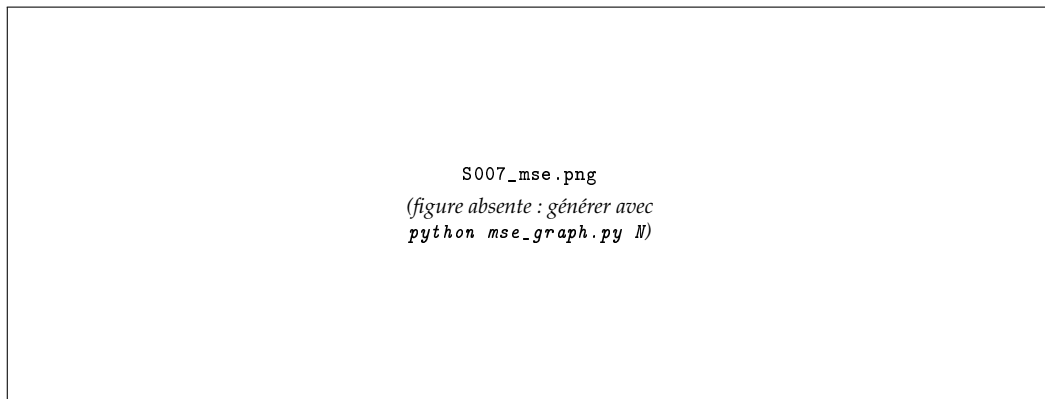


FIGURE 17 – S007 — Évolution du MSE.

## 2.8 Sujet S008

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.48 \pm 0.15$ |
| ICUNet      | $0.59 \pm 0.11$ |
| ICUNet++    | $0.52 \pm 0.15$ |
| ICUNet_attn | $0.46 \pm 0.13$ |
| DuoCL       | $0.42 \pm 0.18$ |
| GCTNet      | $0.38 \pm 0.15$ |
| ICLabel     | $0.52 \pm 0.15$ |

TABLE 15 – S008 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 9.57            | 13.02       |             |
| ICUNet++    | 10.14           | 11.23       |             |
| ICUNet_attn | 12.96           | 10.53       | 16.1        |
| DuoCL       | 14.04           | 9.52        |             |
| GCTNet      | 13.75           | 7.75        |             |

TABLE 16 – S008 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

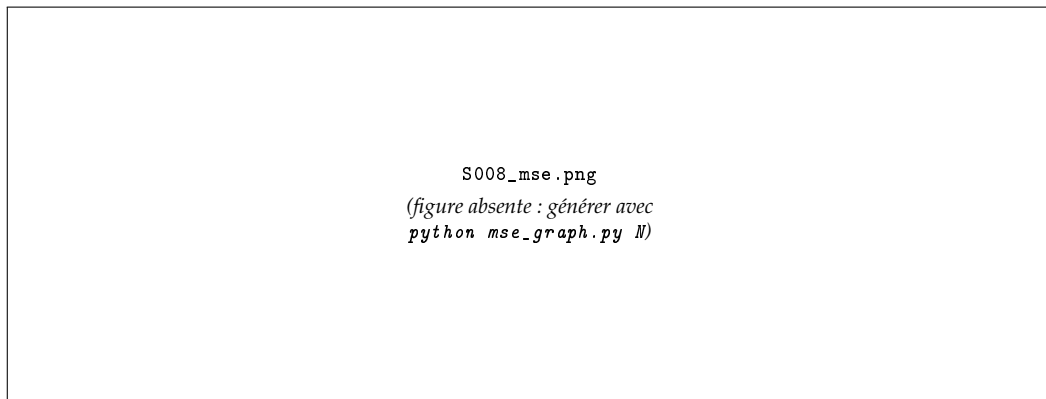


FIGURE 18 – S008 — Évolution du MSE.

## 2.9 Sujet S009

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.38 \pm 0.19$ |
| ICUNet      | $0.42 \pm 0.17$ |
| ICUNet++    | $0.42 \pm 0.16$ |
| ICUNet_attn | $0.39 \pm 0.16$ |
| DuoCL       | $0.29 \pm 0.11$ |
| GCTNet      | $0.37 \pm 0.17$ |
| ICLabel     | $0.56 \pm 0.17$ |

TABLE 17 – S009 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 19.74           | 17.67       |             |
| ICUNet++    | 19.79           | 15.67       |             |
| ICUNet_attn | 25.21           | 19.55       | 27.0        |
| DuoCL       | 25.73           | 17.79       |             |
| GCTNet      | 25.04           | 16.05       |             |

TABLE 18 – S009 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

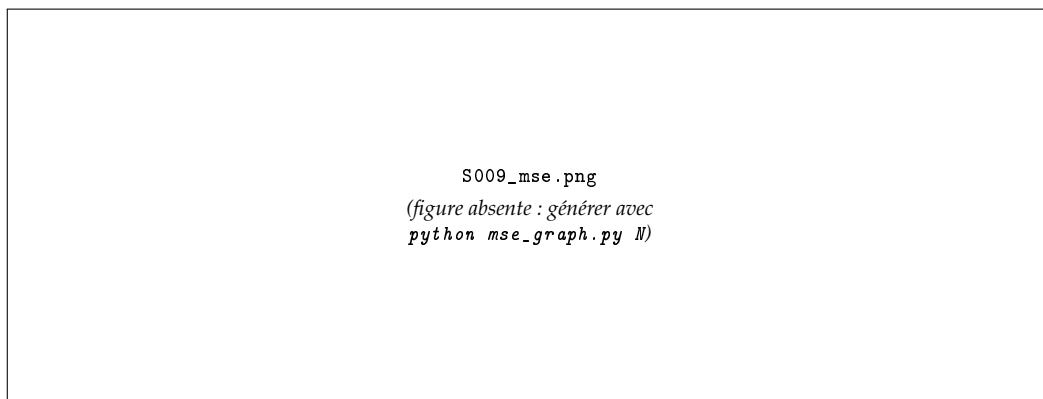


FIGURE 19 – S009 — Évolution du MSE.

## 2.10 Sujet S010

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.58 \pm 0.16$ |
| ICUNet      | $0.42 \pm 0.11$ |
| ICUNet++    | $0.42 \pm 0.13$ |
| ICUNet_attn | $0.41 \pm 0.07$ |
| DuoCL       | $0.59 \pm 0.16$ |
| GCTNet      | $0.58 \pm 0.11$ |
| ICLabel     | $0.60 \pm 0.12$ |

TABLE 19 – S010 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 18.67           | 25.23       |             |
| ICUNet++    | 18.73           | 24.23       |             |
| ICUNet_attn | 22.48           | 22.89       | 27.6        |
| DuoCL       | 25.61           | 16.38       |             |
| GCTNet      | 25.34           | 12.70       |             |

TABLE 20 – S010 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

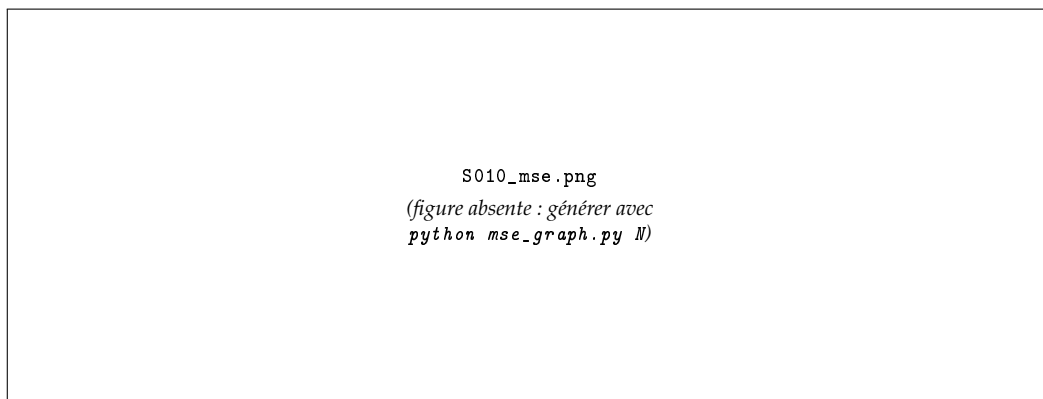


FIGURE 20 – S010 — Évolution du MSE.

## 2.11 Sujet S011

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.42 \pm 0.14$ |
| ICUNet      | $0.52 \pm 0.15$ |
| ICUNet++    | $0.42 \pm 0.18$ |
| ICUNet_attn | $0.44 \pm 0.19$ |
| DuoCL       | $0.44 \pm 0.12$ |
| GCTNet      | $0.41 \pm 0.16$ |
| ICLabel     | $0.50 \pm 0.20$ |

TABLE 21 – S011 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 5.73            | 10.48       |             |
| ICUNet++    | 6.73            | 9.09        |             |
| ICUNet_attn | 9.51            | 8.06        | 13.3        |
| DuoCL       | 11.53           | 6.32        |             |
| GCTNet      | 11.49           | 5.27        |             |

TABLE 22 – S011 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

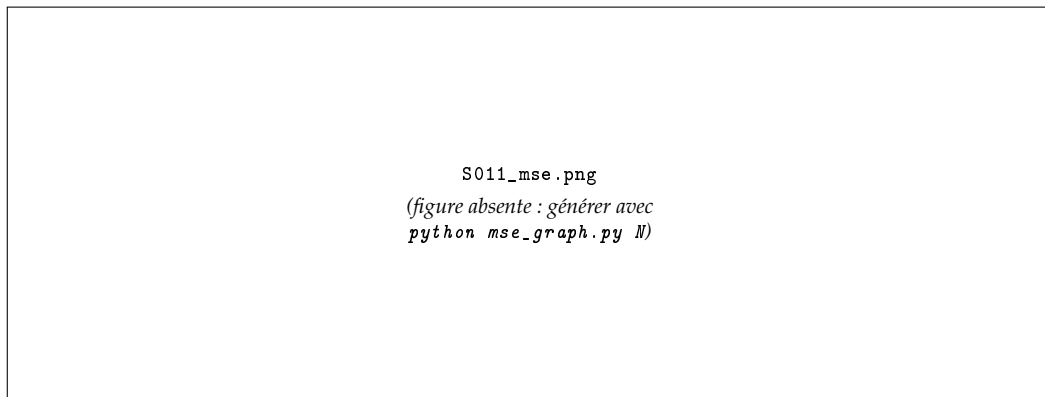


FIGURE 21 – S011 — Évolution du MSE.

## 2.12 Sujet S012

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.53 \pm 0.16$ |
| ICUNet      | $0.50 \pm 0.12$ |
| ICUNet++    | $0.37 \pm 0.09$ |
| ICUNet_attn | $0.48 \pm 0.17$ |
| DuoCL       | $0.59 \pm 0.17$ |
| GCTNet      | $0.50 \pm 0.16$ |
| ICLabel     | $0.50 \pm 0.17$ |

TABLE 23 – S012 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 12.19           | 15.40       |             |
| ICUNet++    | 10.97           | 12.50       |             |
| ICUNet_attn | 14.73           | 13.43       | 15.7        |
| DuoCL       | 14.73           | 8.41        |             |
| GCTNet      | 14.56           | 7.05        |             |

TABLE 24 – S012 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

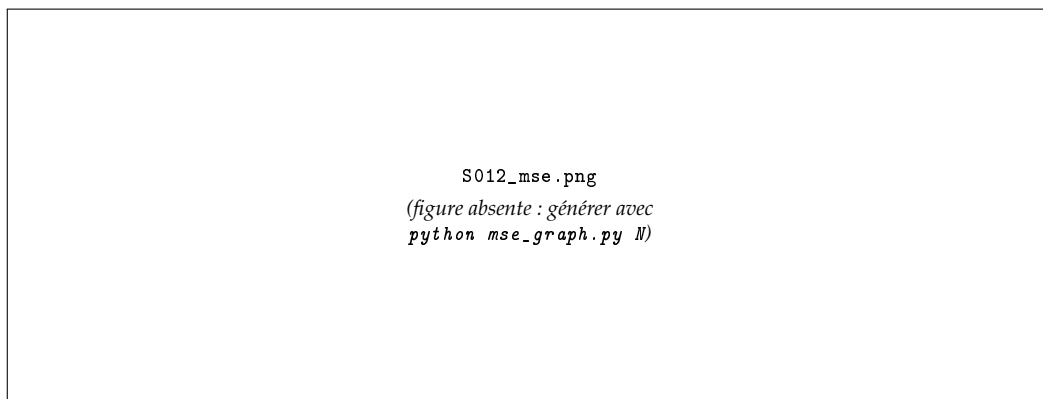


FIGURE 22 – S012 — Évolution du MSE.

### 2.13 Sujet S013

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.50 \pm 0.17$ |
| ICUNet      | $0.41 \pm 0.17$ |
| ICUNet++    | $0.32 \pm 0.14$ |
| ICUNet_attn | $0.58 \pm 0.17$ |
| DuoCL       | $0.42 \pm 0.14$ |
| GCTNet      | $0.49 \pm 0.16$ |
| ICLabel     | $0.51 \pm 0.18$ |

TABLE 25 – S013 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 19.97           | 21.23       |             |
| ICUNet++    | 20.72           | 21.08       |             |
| ICUNet_attn | 25.48           | 23.50       | 29.6        |
| DuoCL       | 27.68           | 21.10       |             |
| GCTNet      | 27.29           | 18.26       |             |

TABLE 26 – S013 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

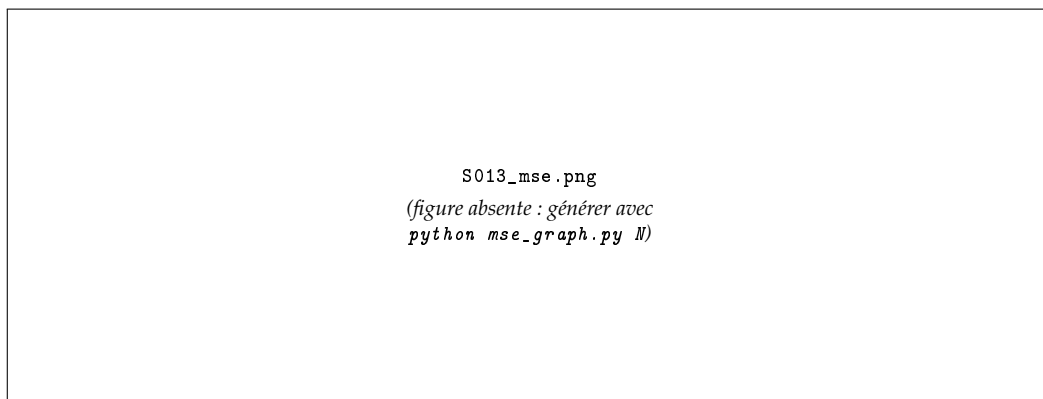


FIGURE 23 – S013 — Évolution du MSE.

## 2.14 Sujet S014

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.47 \pm 0.12$ |
| ICUNet      | $0.54 \pm 0.12$ |
| ICUNet++    | $0.61 \pm 0.11$ |
| ICUNet_attn | $0.48 \pm 0.14$ |
| DuoCL       | $0.38 \pm 0.11$ |
| GCTNet      | $0.53 \pm 0.13$ |
| ICLabel     | $0.54 \pm 0.13$ |

TABLE 27 – S014 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 9.85            | 16.66       |             |
| ICUNet++    | 9.80            | 16.98       |             |
| ICUNet_attn | 13.28           | 16.83       | 20.7        |
| DuoCL       | 16.61           | 13.84       |             |
| GCTNet      | 16.34           | 11.03       |             |

TABLE 28 – S014 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

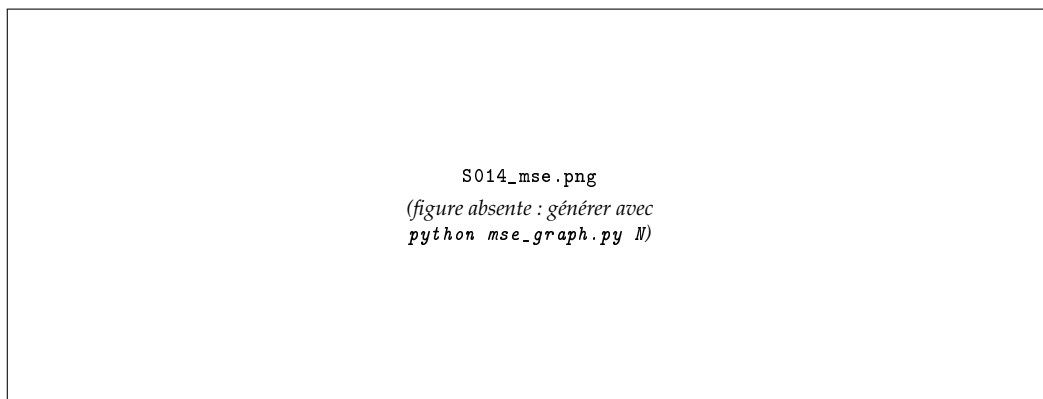


FIGURE 24 – S014 — Évolution du MSE.

## 2.15 Sujet S015

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.66 \pm 0.12$ |
| ICUNet      | $0.63 \pm 0.16$ |
| ICUNet++    | $0.47 \pm 0.14$ |
| ICUNet_attn | $0.48 \pm 0.13$ |
| DuoCL       | $0.63 \pm 0.14$ |
| GCTNet      | $0.44 \pm 0.19$ |
| ICLabel     | $0.69 \pm 0.16$ |

TABLE 29 – S015 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 14.63           | 27.60       |             |
| ICUNet++    | 16.03           | 25.33       |             |
| ICUNet_attn | 22.80           | 24.58       | 34.7        |
| DuoCL       | 28.46           | 20.62       |             |
| GCTNet      | 27.64           | 17.06       |             |

TABLE 30 – S015 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

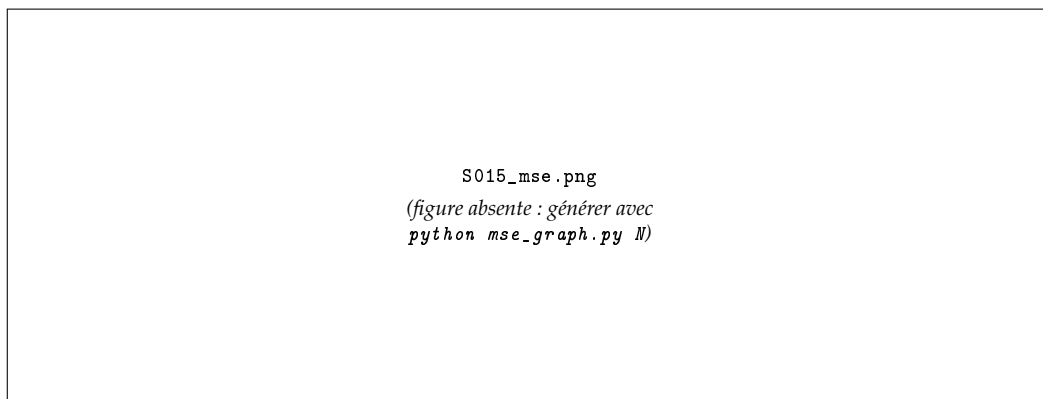


FIGURE 25 – S015 — Évolution du MSE.

## 2.16 Sujet S016

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.47 \pm 0.15$ |
| ICUNet      | $0.51 \pm 0.09$ |
| ICUNet++    | $0.54 \pm 0.14$ |
| ICUNet_attn | $0.49 \pm 0.16$ |
| DuoCL       | $0.60 \pm 0.11$ |
| GCTNet      | $0.58 \pm 0.16$ |
| ICLabel     | $0.66 \pm 0.15$ |

TABLE 31 – S016 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 10.84           | 14.00       |             |
| ICUNet++    | 9.64            | 12.24       |             |
| ICUNet_attn | 11.34           | 11.16       | 12.6        |
| DuoCL       | 12.25           | 10.08       |             |
| GCTNet      | 11.45           | 8.10        |             |

TABLE 32 – S016 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

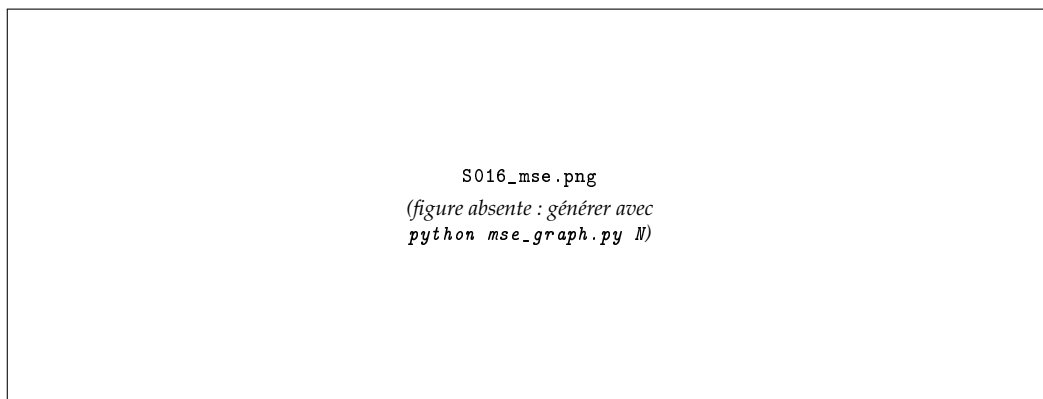


FIGURE 26 – S016 — Évolution du MSE.

## 2.17 Sujet S017

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.40 \pm 0.18$ |
| ICUNet      | $0.37 \pm 0.20$ |
| ICUNet++    | $0.41 \pm 0.07$ |
| ICUNet_attn | $0.54 \pm 0.17$ |
| DuoCL       | $0.52 \pm 0.15$ |
| GCTNet      | $0.36 \pm 0.16$ |
| ICLabel     | $0.34 \pm 0.16$ |

TABLE 33 – S017 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 27.77           | 22.06       |             |
| ICUNet++    | 28.16           | 19.09       |             |
| ICUNet_attn | 33.72           | 21.18       | 36.9        |
| DuoCL       | 34.35           | 19.61       |             |
| GCTNet      | 33.17           | 17.73       |             |

TABLE 34 – S017 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

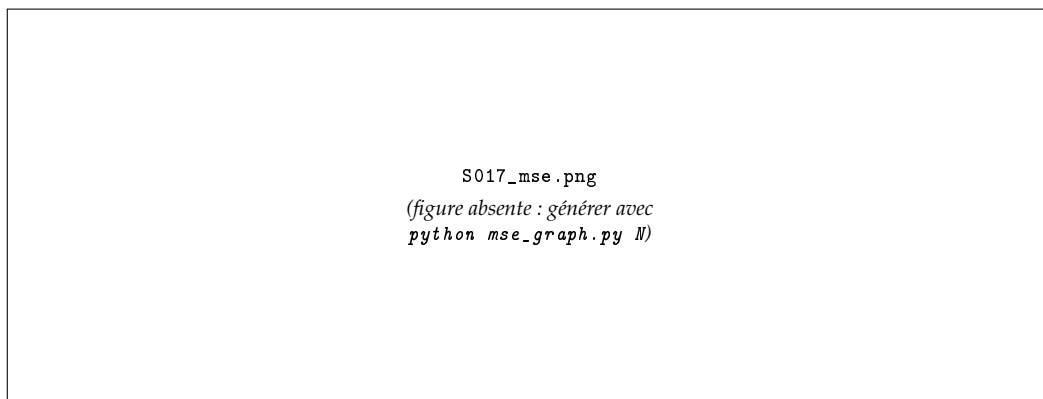


FIGURE 27 – S017 — Évolution du MSE.

## 2.18 Sujet S018

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.39 \pm 0.13$ |
| ICUNet      | $0.32 \pm 0.10$ |
| ICUNet++    | $0.52 \pm 0.17$ |
| ICUNet_attn | $0.52 \pm 0.15$ |
| DuoCL       | $0.49 \pm 0.11$ |
| GCTNet      | $0.51 \pm 0.17$ |
| ICLabel     | $0.36 \pm 0.16$ |

TABLE 35 – S018 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 5.01            | 6.30        |             |
| ICUNet++    | 4.27            | 4.99        |             |
| ICUNet_attn | 5.68            | 5.32        | 5.8         |
| DuoCL       | 5.73            | 3.83        |             |
| GCTNet      | 5.50            | 3.15        |             |

TABLE 36 – S018 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

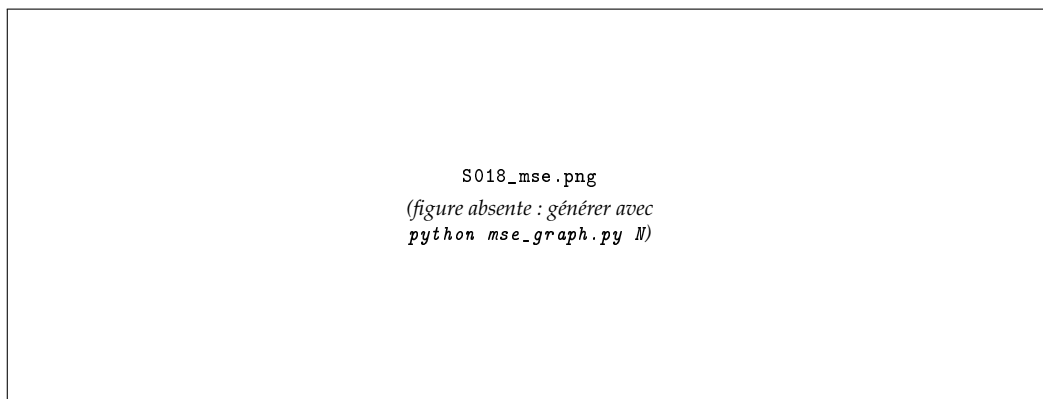


FIGURE 28 – S018 — Évolution du MSE.

## 2.19 Sujet S019

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.60 \pm 0.07$ |
| ICUNet      | $0.69 \pm 0.13$ |
| ICUNet++    | $0.29 \pm 0.16$ |
| ICUNet_attn | $0.59 \pm 0.12$ |
| DuoCL       | $0.62 \pm 0.07$ |
| GCTNet      | $0.68 \pm 0.12$ |
| ICLabel     | $0.68 \pm 0.08$ |

TABLE 37 – S019 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 15.80           | 15.01       |             |
| ICUNet++    | 16.70           | 14.04       |             |
| ICUNet_attn | 20.16           | 14.38       | 24.9        |
| DuoCL       | 21.84           | 14.07       |             |
| GCTNet      | 21.61           | 12.49       |             |

TABLE 38 – S019 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

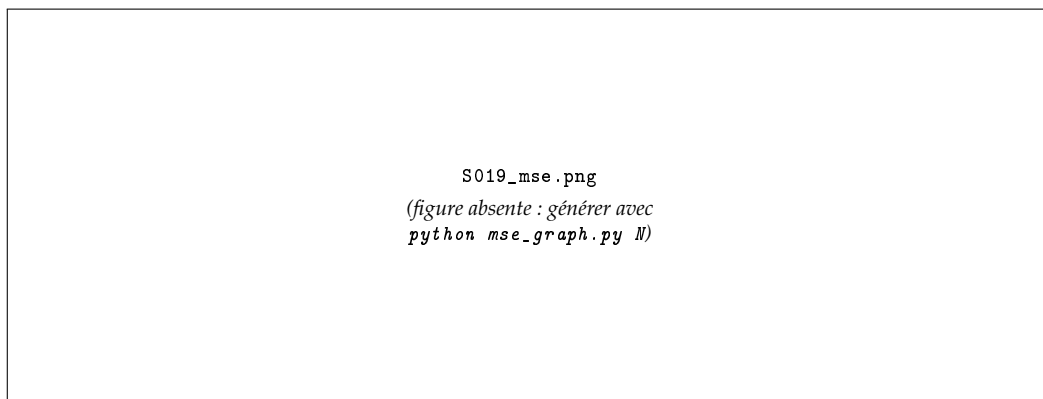


FIGURE 29 – S019 — Évolution du MSE.

## 2.20 Sujet S020

| Signal      | Accuracy        |
|-------------|-----------------|
| Brut        | $0.62 \pm 0.16$ |
| ICUNet      | $0.49 \pm 0.17$ |
| ICUNet++    | $0.40 \pm 0.23$ |
| ICUNet_attn | $0.48 \pm 0.10$ |
| DuoCL       | $0.39 \pm 0.15$ |
| GCTNet      | $0.61 \pm 0.11$ |
| ICLabel     | $0.66 \pm 0.18$ |

TABLE 39 – S020 — Accuracy de classification.

| Comparaison | RMSE ( $\mu$ V) | RMS_Méthode | RMS_ICLABEL |
|-------------|-----------------|-------------|-------------|
| ICUNet      | 5.00            | 5.78        |             |
| ICUNet++    | 4.84            | 5.28        |             |
| ICUNet_attn | 5.48            | 5.05        | 5.4         |
| DuoCL       | 5.32            | 3.59        |             |
| GCTNet      | 5.15            | 2.88        |             |

TABLE 40 – S020 — Erreur (RMSE,  $\mu$ V) par rapport au signal nettoyé par ICLabel.

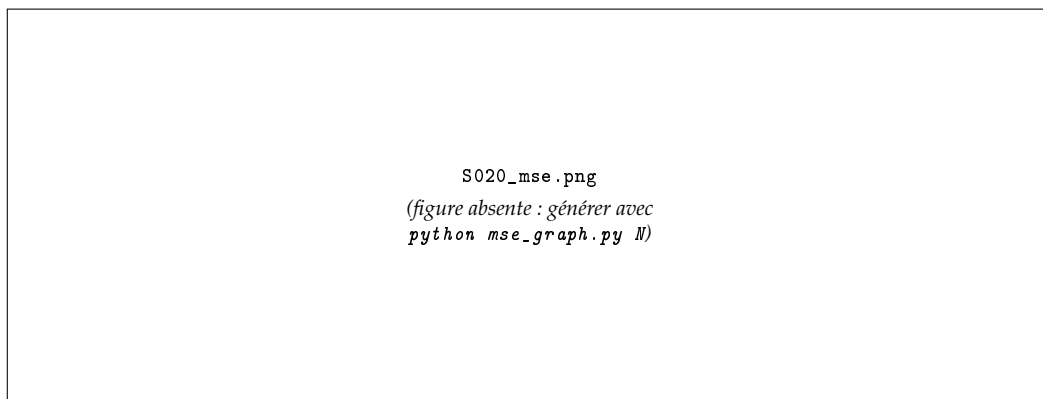


FIGURE 30 – S020 — Évolution du MSE.

### 3 Récapitulatif

TABLE 41 – Récapitulatif — accuracy CSP+LDA, les 109 sujets, par méthode (« - » = non calculé).

| Sujet | Brut        | ICLabel     | DuoCL       | GCTNet      | ICUNet      | ICUNet++    | ICUNet_attn |
|-------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| S001  | 0.63 ± 0.21 | 0.67 ± 0.13 | 0.60 ± 0.11 | 0.62 ± 0.07 | 0.51 ± 0.14 | 0.42 ± 0.16 | 0.49 ± 0.17 |
| S002  | 0.66 ± 0.18 | 0.70 ± 0.19 | 0.67 ± 0.12 | 0.68 ± 0.14 | 0.66 ± 0.14 | 0.54 ± 0.20 | 0.64 ± 0.13 |
| S003  | 0.56 ± 0.16 | 0.66 ± 0.13 | 0.63 ± 0.12 | 0.46 ± 0.17 | 0.43 ± 0.15 | 0.44 ± 0.11 | 0.50 ± 0.06 |
| S004  | 0.43 ± 0.16 | 0.44 ± 0.11 | 0.60 ± 0.19 | 0.54 ± 0.12 | 0.58 ± 0.17 | 0.66 ± 0.13 | 0.36 ± 0.15 |
| S005  | 0.49 ± 0.14 | 0.49 ± 0.12 | 0.37 ± 0.17 | 0.60 ± 0.10 | 0.48 ± 0.14 | 0.44 ± 0.16 | 0.33 ± 0.16 |
| S006  | 0.50 ± 0.11 | 0.51 ± 0.15 | 0.52 ± 0.09 | 0.47 ± 0.17 | 0.53 ± 0.12 | 0.49 ± 0.18 | 0.57 ± 0.18 |
| S007  | 0.96 ± 0.07 | 0.98 ± 0.04 | 0.81 ± 0.10 | 0.91 ± 0.08 | 0.82 ± 0.10 | 0.77 ± 0.09 | 0.61 ± 0.17 |
| S008  | 0.48 ± 0.15 | 0.52 ± 0.15 | 0.42 ± 0.18 | 0.38 ± 0.15 | 0.59 ± 0.11 | 0.52 ± 0.15 | 0.46 ± 0.13 |
| S009  | 0.38 ± 0.19 | 0.56 ± 0.17 | 0.29 ± 0.11 | 0.37 ± 0.17 | 0.42 ± 0.17 | 0.42 ± 0.16 | 0.39 ± 0.16 |
| S010  | 0.58 ± 0.16 | 0.60 ± 0.12 | 0.59 ± 0.16 | 0.58 ± 0.11 | 0.42 ± 0.11 | 0.42 ± 0.13 | 0.41 ± 0.07 |
| S011  | 0.42 ± 0.14 | 0.50 ± 0.20 | 0.44 ± 0.12 | 0.41 ± 0.16 | 0.52 ± 0.15 | 0.42 ± 0.18 | 0.44 ± 0.19 |
| S012  | 0.53 ± 0.16 | 0.50 ± 0.17 | 0.59 ± 0.17 | 0.50 ± 0.16 | 0.50 ± 0.12 | 0.37 ± 0.09 | 0.48 ± 0.17 |
| S013  | 0.50 ± 0.17 | 0.51 ± 0.18 | 0.42 ± 0.14 | 0.49 ± 0.16 | 0.41 ± 0.17 | 0.32 ± 0.14 | 0.58 ± 0.17 |
| S014  | 0.47 ± 0.12 | 0.54 ± 0.13 | 0.38 ± 0.11 | 0.53 ± 0.13 | 0.54 ± 0.12 | 0.61 ± 0.11 | 0.48 ± 0.14 |
| S015  | 0.66 ± 0.12 | 0.69 ± 0.16 | 0.63 ± 0.14 | 0.44 ± 0.19 | 0.63 ± 0.16 | 0.47 ± 0.14 | 0.48 ± 0.13 |
| S016  | 0.47 ± 0.15 | 0.66 ± 0.15 | 0.60 ± 0.11 | 0.58 ± 0.16 | 0.51 ± 0.09 | 0.54 ± 0.14 | 0.49 ± 0.16 |
| S017  | 0.40 ± 0.18 | 0.34 ± 0.16 | 0.52 ± 0.15 | 0.36 ± 0.16 | 0.37 ± 0.20 | 0.41 ± 0.07 | 0.54 ± 0.17 |
| S018  | 0.39 ± 0.13 | 0.36 ± 0.16 | 0.49 ± 0.11 | 0.51 ± 0.17 | 0.32 ± 0.10 | 0.52 ± 0.17 | 0.52 ± 0.15 |
| S019  | 0.60 ± 0.07 | 0.68 ± 0.08 | 0.62 ± 0.07 | 0.68 ± 0.12 | 0.69 ± 0.13 | 0.29 ± 0.16 | 0.59 ± 0.12 |
| S020  | 0.62 ± 0.16 | 0.66 ± 0.18 | 0.39 ± 0.15 | 0.61 ± 0.11 | 0.49 ± 0.17 | 0.40 ± 0.23 | 0.48 ± 0.10 |
| S021  | 0.64 ± 0.13 | 0.60 ± 0.11 | 0.60 ± 0.19 | 0.52 ± 0.19 | 0.59 ± 0.11 | 0.58 ± 0.18 | 0.46 ± 0.10 |
| S022  | 0.50 ± 0.14 | 0.60 ± 0.14 | 0.62 ± 0.11 | 0.58 ± 0.11 | 0.57 ± 0.18 | 0.62 ± 0.10 | 0.44 ± 0.13 |
| S023  | 0.54 ± 0.14 | 0.63 ± 0.15 | 0.50 ± 0.16 | 0.58 ± 0.16 | 0.56 ± 0.15 | 0.47 ± 0.19 | 0.44 ± 0.13 |
| S024  | 0.56 ± 0.14 | 0.51 ± 0.12 | 0.47 ± 0.15 | 0.61 ± 0.17 | 0.51 ± 0.12 | 0.58 ± 0.18 | 0.37 ± 0.07 |
| S025  | 0.48 ± 0.22 | 0.48 ± 0.16 | 0.39 ± 0.22 | 0.58 ± 0.20 | 0.39 ± 0.21 | 0.42 ± 0.10 | 0.44 ± 0.09 |
| S026  | 0.68 ± 0.10 | 0.76 ± 0.13 | 0.48 ± 0.21 | 0.50 ± 0.14 | 0.69 ± 0.16 | 0.43 ± 0.15 | 0.43 ± 0.18 |
| S027  | 0.54 ± 0.20 | 0.56 ± 0.14 | 0.56 ± 0.15 | 0.58 ± 0.20 | 0.58 ± 0.18 | 0.39 ± 0.13 | 0.46 ± 0.13 |
| S028  | 0.47 ± 0.07 | 0.39 ± 0.07 | 0.44 ± 0.16 | 0.48 ± 0.15 | 0.44 ± 0.17 | 0.44 ± 0.12 | 0.42 ± 0.08 |
| S029  | 0.96 ± 0.05 | 0.97 ± 0.05 | 0.83 ± 0.07 | 0.96 ± 0.07 | 0.90 ± 0.10 | 0.79 ± 0.13 | 0.51 ± 0.15 |
| S030  | 0.60 ± 0.17 | 0.59 ± 0.19 | 0.51 ± 0.18 | 0.43 ± 0.18 | 0.48 ± 0.16 | 0.53 ± 0.13 | 0.50 ± 0.14 |
| S031  | 0.74 ± 0.11 | 0.71 ± 0.14 | 0.56 ± 0.09 | 0.72 ± 0.15 | 0.59 ± 0.17 | 0.58 ± 0.18 | 0.57 ± 0.18 |
| S032  | 0.52 ± 0.12 | 0.67 ± 0.15 | 0.57 ± 0.19 | 0.48 ± 0.12 | 0.62 ± 0.09 | 0.61 ± 0.12 | 0.42 ± 0.13 |
| S033  | 0.53 ± 0.13 | 0.58 ± 0.18 | 0.54 ± 0.18 | 0.68 ± 0.14 | 0.59 ± 0.19 | 0.50 ± 0.20 | 0.40 ± 0.14 |
| S034  | 0.61 ± 0.12 | 0.58 ± 0.13 | 0.53 ± 0.08 | 0.49 ± 0.21 | 0.56 ± 0.10 | 0.68 ± 0.22 | 0.57 ± 0.14 |
| S035  | 0.54 ± 0.14 | 0.60 ± 0.16 | 0.56 ± 0.13 | 0.68 ± 0.16 | 0.59 ± 0.16 | 0.56 ± 0.15 | 0.58 ± 0.16 |
| S036  | 0.40 ± 0.12 | 0.47 ± 0.13 | 0.58 ± 0.18 | 0.51 ± 0.11 | 0.49 ± 0.22 | 0.53 ± 0.12 | 0.57 ± 0.16 |
| S037  | 0.42 ± 0.12 | 0.47 ± 0.15 | 0.48 ± 0.11 | 0.53 ± 0.18 | 0.21 ± 0.14 | 0.54 ± 0.14 | 0.49 ± 0.20 |
| S038  | 0.40 ± 0.12 | 0.40 ± 0.11 | 0.36 ± 0.11 | 0.41 ± 0.11 | 0.42 ± 0.14 | 0.51 ± 0.16 | 0.41 ± 0.09 |
| S039  | 0.59 ± 0.13 | 0.49 ± 0.17 | 0.50 ± 0.14 | 0.41 ± 0.10 | 0.60 ± 0.16 | 0.43 ± 0.14 | 0.49 ± 0.16 |
| S040  | 0.40 ± 0.11 | 0.47 ± 0.11 | 0.47 ± 0.15 | 0.51 ± 0.11 | 0.40 ± 0.11 | 0.43 ± 0.14 | 0.54 ± 0.14 |
| S041  | 0.51 ± 0.14 | 0.48 ± 0.11 | 0.63 ± 0.14 | 0.72 ± 0.12 | 0.46 ± 0.15 | 0.58 ± 0.18 | 0.56 ± 0.14 |
| S042  | 0.78 ± 0.12 | 0.80 ± 0.11 | 0.79 ± 0.13 | 0.73 ± 0.10 | 0.74 ± 0.13 | 0.70 ± 0.14 | 0.72 ± 0.17 |
| S043  | 0.54 ± 0.14 | 0.57 ± 0.21 | 0.57 ± 0.09 | 0.58 ± 0.15 | 0.66 ± 0.20 | 0.44 ± 0.20 | 0.60 ± 0.13 |
| S044  | 0.54 ± 0.14 | 0.60 ± 0.12 | 0.58 ± 0.11 | 0.66 ± 0.14 | 0.62 ± 0.16 | 0.60 ± 0.13 | 0.53 ± 0.12 |
| S045  | 0.44 ± 0.10 | 0.49 ± 0.14 | 0.50 ± 0.14 | 0.49 ± 0.18 | 0.43 ± 0.13 | 0.43 ± 0.16 | 0.58 ± 0.14 |
| S046  | 0.64 ± 0.21 | 0.60 ± 0.24 | 0.44 ± 0.15 | 0.41 ± 0.09 | 0.50 ± 0.16 | 0.60 ± 0.17 | 0.50 ± 0.14 |
| S047  | 0.61 ± 0.09 | 0.58 ± 0.14 | 0.50 ± 0.12 | 0.41 ± 0.17 | 0.51 ± 0.14 | 0.64 ± 0.14 | 0.51 ± 0.10 |
| S048  | 0.69 ± 0.11 | 0.72 ± 0.14 | 0.63 ± 0.13 | 0.73 ± 0.18 | 0.57 ± 0.17 | 0.62 ± 0.15 | 0.67 ± 0.15 |
| S049  | 0.86 ± 0.11 | 0.78 ± 0.10 | 0.59 ± 0.19 | 0.64 ± 0.10 | 0.67 ± 0.11 | 0.61 ± 0.11 | 0.58 ± 0.11 |
| S050  | 0.51 ± 0.15 | 0.51 ± 0.07 | 0.43 ± 0.14 | 0.58 ± 0.11 | 0.59 ± 0.17 | 0.54 ± 0.13 | 0.62 ± 0.21 |
| S051  | 0.51 ± 0.12 | 0.42 ± 0.13 | 0.54 ± 0.10 | 0.57 ± 0.10 | 0.36 ± 0.16 | 0.40 ± 0.07 | 0.36 ± 0.12 |
| S052  | 0.57 ± 0.16 | 0.61 ± 0.14 | 0.63 ± 0.17 | 0.58 ± 0.15 | 0.60 ± 0.14 | 0.68 ± 0.17 | 0.51 ± 0.14 |
| S053  | 0.72 ± 0.10 | 0.72 ± 0.11 | 0.53 ± 0.16 | 0.46 ± 0.20 | 0.53 ± 0.11 | 0.69 ± 0.12 | 0.53 ± 0.14 |
| S054  | 0.79 ± 0.14 | 0.78 ± 0.15 | 0.68 ± 0.10 | 0.62 ± 0.10 | 0.56 ± 0.15 | 0.60 ± 0.10 | 0.63 ± 0.15 |
| S055  | 0.64 ± 0.16 | 0.72 ± 0.13 | 0.63 ± 0.10 | 0.61 ± 0.14 | 0.77 ± 0.16 | 0.53 ± 0.15 | 0.47 ± 0.11 |
| S056  | 0.71 ± 0.11 | 0.78 ± 0.11 | 0.74 ± 0.17 | 0.73 ± 0.15 | 0.60 ± 0.15 | 0.59 ± 0.14 | 0.62 ± 0.13 |
| S057  | 0.36 ± 0.13 | 0.37 ± 0.20 | 0.56 ± 0.16 | 0.51 ± 0.11 | 0.53 ± 0.13 | 0.42 ± 0.15 | 0.53 ± 0.19 |
| S058  | 0.44 ± 0.16 | 0.52 ± 0.12 | 0.39 ± 0.14 | 0.46 ± 0.16 | 0.62 ± 0.12 | 0.47 ± 0.18 | 0.48 ± 0.12 |
| S059  | 0.56 ± 0.10 | 0.64 ± 0.12 | 0.58 ± 0.08 | 0.60 ± 0.10 | 0.50 ± 0.14 | 0.54 ± 0.12 | 0.43 ± 0.10 |
| S060  | 0.76 ± 0.15 | 0.71 ± 0.15 | 0.83 ± 0.14 | 0.77 ± 0.13 | 0.67 ± 0.15 | 0.62 ± 0.18 | 0.64 ± 0.14 |
| S061  | 0.68 ± 0.09 | 0.70 ± 0.19 | 0.66 ± 0.18 | 0.56 ± 0.14 | 0.63 ± 0.18 | 0.52 ± 0.19 | 0.49 ± 0.15 |
| S062  | 0.91 ± 0.07 | 0.91 ± 0.07 | 0.79 ± 0.08 | 0.80 ± 0.14 | 0.76 ± 0.12 | 0.71 ± 0.07 | 0.43 ± 0.18 |
| S063  | 0.41 ± 0.11 | 0.58 ± 0.13 | 0.42 ± 0.11 | 0.63 ± 0.14 | 0.42 ± 0.14 | 0.39 ± 0.12 | 0.54 ± 0.18 |
| S064  | 0.47 ± 0.18 | 0.50 ± 0.17 | 0.50 ± 0.11 | 0.48 ± 0.17 | 0.61 ± 0.12 | 0.56 ± 0.12 | 0.42 ± 0.18 |
| S065  | 0.44 ± 0.19 | 0.51 ± 0.15 | 0.40 ± 0.17 | 0.47 ± 0.17 | 0.49 ± 0.17 | 0.39 ± 0.18 | 0.48 ± 0.14 |
| S066  | 0.53 ± 0.13 | 0.56 ± 0.16 | 0.52 ± 0.22 | 0.44 ± 0.13 | 0.60 ± 0.11 | 0.40 ± 0.12 | 0.52 ± 0.15 |
| S067  | 0.49 ± 0.11 | 0.44 ± 0.15 | 0.58 ± 0.18 | 0.66 ± 0.15 | 0.44 ± 0.12 | 0.47 ± 0.16 | 0.50 ± 0.21 |
| S068  | 0.52 ± 0.13 | 0.60 ± 0.15 | 0.51 ± 0.20 | 0.52 ± 0.12 | 0.53 ± 0.11 | 0.48 ± 0.11 | 0.44 ± 0.14 |
| S069  | 0.47 ± 0.13 | 0.43 ± 0.18 | 0.52 ± 0.18 | 0.48 ± 0.11 | 0.41 ± 0.15 | 0.57 ± 0.12 | 0.46 ± 0.20 |
| S070  | 0.67 ± 0.18 | 0.72 ± 0.19 | 0.47 ± 0.08 | 0.44 ± 0.16 | 0.57 ± 0.13 | 0.53 ± 0.14 | 0.43 ± 0.12 |
| S071  | 0.59 ± 0.11 | 0.60 ± 0.13 | 0.53 ± 0.14 | 0.63 ± 0.14 | 0.40 ± 0.19 | 0.44 ± 0.10 | 0.53 ± 0.12 |
| S072  | 0.74 ± 0.09 | 0.72 ± 0.11 | 0.61 ± 0.17 | 0.72 ± 0.13 | 0.64 ± 0.14 | 0.62 ± 0.13 | 0.57 ± 0.14 |
| S073  | 0.31 ± 0.20 | 0.48 ± 0.17 | 0.48 ± 0.14 | 0.41 ± 0.17 | 0.68 ± 0.16 | 0.34 ± 0.13 | 0.52 ± 0.12 |
| S074  | 0.63 ± 0.11 | 0.59 ± 0.17 | 0.60 ± 0.18 | 0.48 ± 0.23 | 0.63 ± 0.11 | 0.48 ± 0.16 | 0.59 ± 0.14 |
| S075  | 0.53 ± 0.11 | 0.56 ± 0.16 | 0.49 ± 0.22 | 0.58 ± 0.16 | 0.53 ± 0.19 | 0.44 ± 0.18 | 0.52 ± 0.21 |
| S076  | 0.52 ± 0.14 | 0.51 ± 0.11 | 0.37 ± 0.11 | 0.46 ± 0.20 | 0.54 ± 0.12 | 0.49 ± 0.14 | 0.42 ± 0.18 |
| S077  | 0.43 ± 0.09 | 0.41 ± 0.12 | 0.49 ± 0.16 | 0.46 ± 0.13 | 0.44 ± 0.15 | 0.49 ± 0.16 | 0.46 ± 0.14 |
| S078  | 0.31 ± 0.15 | 0.48 ± 0.13 | 0.48 ± 0.16 | 0.42 ± 0.12 | 0.42 ± 0.10 | 0.42 ± 0.10 | 0.57 ± 0.10 |
| S079  | 0.52 ± 0.19 | 0.57 ± 0.19 | 0.53 ± 0.16 | 0.47 ± 0.17 | 0.62 ± 0.15 | 0.61 ± 0.14 | 0.50 ± 0.17 |
| S080  | 0.54 ± 0.18 | 0.60 ± 0.14 | 0.47 ± 0.14 | 0.53 ± 0.14 | 0.59 ± 0.16 | 0.56 ± 0.15 | 0.54 ± 0.24 |
| S081  | 0.46 ± 0.12 | 0.51 ± 0.17 | 0.49 ± 0.16 | 0.50 ± 0.21 | 0.39 ± 0.14 | 0.39 ± 0.13 | 0.43 ± 0.14 |
| S082  | 0.66 ± 0.13 | 0.70 ± 0.12 | 0.61 ± 0.16 | 0.57 ± 0.18 | 0.70 ± 0.16 | 0.63 ± 0.11 | 0.56 ± 0.11 |
| S083  | 0.58 ± 0.15 | 0.51 ± 0.15 | 0.51 ± 0.13 | 0.58 ± 0.19 | 0.41 ± 0.19 | 0.40 ± 0.16 | 0.51 ± 0.11 |
| S084  | 0.43 ± 0.13 | 0.57 ± 0.12 | 0.63 ± 0.13 | 0.59 ± 0.17 | 0.47 ± 0.08 | 0.41 ± 0.12 | 0.51 ± 0.11 |

| Sujet          | Brut        | ICLabel            | DuoCL       | GCTNet      | ICUNet      | ICUNet++    | ICUNet_attn |
|----------------|-------------|--------------------|-------------|-------------|-------------|-------------|-------------|
| S085           | 0.66 ± 0.09 | 0.82 ± 0.10        | 0.68 ± 0.16 | 0.73 ± 0.12 | 0.71 ± 0.17 | 0.70 ± 0.16 | 0.56 ± 0.10 |
| S086           | 0.64 ± 0.16 | 0.59 ± 0.14        | 0.46 ± 0.12 | 0.41 ± 0.17 | 0.57 ± 0.13 | 0.37 ± 0.16 | 0.42 ± 0.14 |
| S087           | 0.54 ± 0.10 | 0.49 ± 0.09        | 0.43 ± 0.15 | 0.40 ± 0.11 | 0.51 ± 0.15 | 0.53 ± 0.16 | 0.59 ± 0.12 |
| S088           | 0.40 ± 0.18 | 0.40 ± 0.12        | 0.48 ± 0.14 | 0.44 ± 0.17 | 0.50 ± 0.13 | 0.43 ± 0.13 | 0.38 ± 0.10 |
| S089           | 0.31 ± 0.13 | 0.41 ± 0.14        | 0.41 ± 0.15 | 0.36 ± 0.13 | 0.57 ± 0.10 | 0.54 ± 0.13 | 0.54 ± 0.15 |
| S090           | 0.48 ± 0.14 | 0.49 ± 0.18        | 0.34 ± 0.18 | 0.48 ± 0.09 | 0.50 ± 0.20 | 0.50 ± 0.17 | 0.32 ± 0.13 |
| S091           | 0.61 ± 0.09 | 0.54 ± 0.17        | 0.57 ± 0.18 | 0.46 ± 0.14 | 0.49 ± 0.12 | 0.41 ± 0.13 | 0.43 ± 0.18 |
| S092           | 0.50 ± 0.09 | 0.56 ± 0.16        | 0.55 ± 0.10 | 0.62 ± 0.14 | 0.49 ± 0.16 | 0.50 ± 0.14 | 0.53 ± 0.18 |
| S093           | 0.68 ± 0.15 | 0.67 ± 0.19        | 0.71 ± 0.15 | 0.80 ± 0.14 | 0.42 ± 0.19 | 0.61 ± 0.14 | 0.60 ± 0.20 |
| S094           | 0.71 ± 0.13 | 0.73 ± 0.17        | 0.74 ± 0.13 | 0.77 ± 0.10 | 0.64 ± 0.16 | 0.58 ± 0.18 | 0.50 ± 0.18 |
| S095           | 0.49 ± 0.14 | 0.53 ± 0.10        | 0.50 ± 0.18 | 0.46 ± 0.16 | 0.44 ± 0.19 | 0.43 ± 0.10 | 0.51 ± 0.13 |
| S096           | 0.52 ± 0.13 | 0.46 ± 0.18        | 0.61 ± 0.13 | 0.57 ± 0.20 | 0.47 ± 0.12 | 0.58 ± 0.16 | 0.51 ± 0.14 |
| S097           | 0.50 ± 0.14 | 0.43 ± 0.15        | 0.63 ± 0.09 | 0.47 ± 0.13 | 0.59 ± 0.17 | 0.60 ± 0.15 | 0.53 ± 0.13 |
| S098           | 0.58 ± 0.20 | 0.62 ± 0.16        | 0.49 ± 0.19 | 0.42 ± 0.15 | 0.58 ± 0.15 | 0.61 ± 0.11 | 0.56 ± 0.16 |
| S099           | 0.50 ± 0.14 | 0.41 ± 0.17        | 0.51 ± 0.14 | 0.50 ± 0.17 | 0.48 ± 0.13 | 0.48 ± 0.19 | 0.37 ± 0.19 |
| S100           | 0.50 ± 0.15 | 0.62 ± 0.17        | 0.41 ± 0.18 | 0.46 ± 0.15 | 0.54 ± 0.14 | 0.45 ± 0.15 | 0.56 ± 0.13 |
| S101           | 0.47 ± 0.14 | 0.36 ± 0.10        | 0.41 ± 0.17 | 0.50 ± 0.20 | 0.30 ± 0.10 | 0.40 ± 0.14 | 0.41 ± 0.21 |
| S102           | 0.62 ± 0.16 | 0.66 ± 0.18        | 0.38 ± 0.13 | 0.51 ± 0.13 | 0.53 ± 0.18 | 0.49 ± 0.16 | 0.59 ± 0.10 |
| S103           | 0.51 ± 0.13 | 0.50 ± 0.12        | 0.50 ± 0.13 | 0.53 ± 0.15 | 0.58 ± 0.16 | 0.50 ± 0.12 | 0.51 ± 0.10 |
| S104           | 0.48 ± 0.13 | 0.46 ± 0.16        | 0.57 ± 0.13 | 0.53 ± 0.12 | 0.50 ± 0.18 | 0.44 ± 0.14 | 0.57 ± 0.10 |
| S105           | 0.51 ± 0.11 | 0.49 ± 0.16        | 0.47 ± 0.17 | 0.52 ± 0.17 | 0.48 ± 0.15 | 0.54 ± 0.10 | 0.51 ± 0.19 |
| S106           | 0.52 ± 0.19 | 0.46 ± 0.15        | 0.63 ± 0.13 | 0.59 ± 0.19 | 0.37 ± 0.17 | 0.56 ± 0.16 | 0.58 ± 0.15 |
| S107           | 0.43 ± 0.16 | 0.53 ± 0.13        | 0.47 ± 0.11 | 0.48 ± 0.14 | 0.48 ± 0.11 | 0.38 ± 0.14 | 0.33 ± 0.20 |
| S108           | 0.57 ± 0.18 | 0.54 ± 0.22        | 0.60 ± 0.13 | 0.63 ± 0.16 | 0.44 ± 0.19 | 0.33 ± 0.19 | 0.51 ± 0.10 |
| S109           | 0.51 ± 0.05 | 0.52 ± 0.18        | 0.53 ± 0.04 | 0.51 ± 0.11 | 0.44 ± 0.07 | 0.39 ± 0.17 | 0.41 ± 0.19 |
| <b>Moyenne</b> | 0.55 ± 0.13 | <b>0.57 ± 0.13</b> | 0.54 ± 0.11 | 0.55 ± 0.12 | 0.53 ± 0.11 | 0.51 ± 0.10 | 0.50 ± 0.08 |

TABLE 42 – Récapitulatif — RMSE, RMS\_Brut, RMS\_Méthode et RMS de référence ICLabel (en  $\mu V$ ), les 109 sujets (4 lignes par sujet), par méthode — comparable à l’ordre de grandeur des signaux (« - » = non calculé).

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S001 RMSE        | 24.8  | 24.3   | 17.2   | 16.0     | 19.4        |
| S001 RMS_Brut    | 36.2  | 36.2   | 36.2   | 36.2     | 36.2        |
| S001 RMS_Méthode | 16.0  | 12.5   | 27.7   | 25.9     | 23.0        |
| S001 RMS_ICLABEL | 26.5  | 26.5   | 26.5   | 26.5     | 26.5        |
| S002 RMSE        | 12.1  | 11.8   | 7.1    | 6.7      | 9.6         |
| S002 RMS_Brut    | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S002 RMS_Méthode | 9.2   | 7.8    | 11.8   | 12.0     | 12.1        |
| S002 RMS_ICLABEL | 14.7  | 14.7   | 14.7   | 14.7     | 14.7        |
| S003 RMSE        | 24.5  | 23.6   | 21.5   | 18.9     | 23.7        |
| S003 RMS_Brut    | 54.1  | 54.1   | 54.1   | 54.1     | 54.1        |
| S003 RMS_Méthode | 17.5  | 14.5   | 26.7   | 19.7     | 19.7        |
| S003 RMS_ICLABEL | 25.1  | 25.1   | 25.1   | 25.1     | 25.1        |
| S004 RMSE        | 23.8  | 24.3   | 23.6   | 25.1     | 26.2        |
| S004 RMS_Brut    | 32.7  | 32.7   | 32.7   | 32.7     | 32.7        |
| S004 RMS_Méthode | 8.7   | 7.3    | 12.2   | 7.5      | 10.5        |
| S004 RMS_ICLABEL | 26.1  | 26.1   | 26.1   | 26.1     | 26.1        |
| S005 RMSE        | 14.8  | 14.6   | 12.1   | 12.6     | 14.7        |
| S005 RMS_Brut    | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S005 RMS_Méthode | 8.5   | 7.0    | 9.0    | 8.5      | 7.8         |
| S005 RMS_ICLABEL | 16.7  | 16.7   | 16.7   | 16.7     | 16.7        |
| S006 RMSE        | 11.0  | 10.9   | 8.0    | 8.3      | 9.9         |
| S006 RMS_Brut    | 20.6  | 20.6   | 20.6   | 20.6     | 20.6        |
| S006 RMS_Méthode | 6.8   | 6.2    | 7.6    | 6.9      | 8.1         |
| S006 RMS_ICLABEL | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S007 RMSE        | 15.6  | 15.0   | 8.2    | 7.9      | 13.2        |
| S007 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S007 RMS_Méthode | 15.7  | 13.5   | 19.5   | 19.3     | 19.5        |
| S007 RMS_ICLABEL | 21.3  | 21.3   | 21.3   | 21.3     | 21.3        |
| S008 RMSE        | 14.0  | 13.8   | 9.6    | 10.1     | 13.0        |
| S008 RMS_Brut    | 25.3  | 25.3   | 25.3   | 25.3     | 25.3        |
| S008 RMS_Méthode | 9.5   | 7.8    | 13.0   | 11.2     | 10.5        |
| S008 RMS_ICLABEL | 16.1  | 16.1   | 16.1   | 16.1     | 16.1        |
| S009 RMSE        | 25.7  | 25.0   | 19.7   | 19.8     | 25.2        |
| S009 RMS_Brut    | 56.7  | 56.7   | 56.7   | 56.7     | 56.7        |
| S009 RMS_Méthode | 17.8  | 16.0   | 17.7   | 15.7     | 19.6        |
| S009 RMS_ICLABEL | 27.0  | 27.0   | 27.0   | 27.0     | 27.0        |
| S010 RMSE        | 25.6  | 25.3   | 18.7   | 18.7     | 22.5        |
| S010 RMS_Brut    | 39.7  | 39.7   | 39.7   | 39.7     | 39.7        |
| S010 RMS_Méthode | 16.4  | 12.7   | 25.2   | 24.2     | 22.9        |
| S010 RMS_ICLABEL | 27.6  | 27.6   | 27.6   | 27.6     | 27.6        |
| S011 RMSE        | 11.5  | 11.5   | 5.7    | 6.7      | 9.5         |
| S011 RMS_Brut    | 15.8  | 15.8   | 15.8   | 15.8     | 15.8        |
| S011 RMS_Méthode | 6.3   | 5.3    | 10.5   | 9.1      | 8.1         |
| S011 RMS_ICLABEL | 13.3  | 13.3   | 13.3   | 13.3     | 13.3        |
| S012 RMSE        | 14.7  | 14.6   | 12.2   | 11.0     | 14.7        |
| S012 RMS_Brut    | 36.5  | 36.5   | 36.5   | 36.5     | 36.5        |
| S012 RMS_Méthode | 8.4   | 7.0    | 15.4   | 12.5     | 13.4        |
| S012 RMS_ICLABEL | 15.7  | 15.7   | 15.7   | 15.7     | 15.7        |
| S013 RMSE        | 27.7  | 27.3   | 20.0   | 20.7     | 25.5        |
| S013 RMS_Brut    | 56.3  | 56.3   | 56.3   | 56.3     | 56.3        |
| S013 RMS_Méthode | 21.1  | 18.3   | 21.2   | 21.1     | 23.5        |
| S013 RMS_ICLABEL | 29.6  | 29.6   | 29.6   | 29.6     | 29.6        |
| S014 RMSE        | 16.6  | 16.3   | 9.9    | 9.8      | 13.3        |
| S014 RMS_Brut    | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S014 RMS_Méthode | 13.8  | 11.0   | 16.7   | 17.0     | 16.8        |
| S014 RMS_ICLABEL | 20.7  | 20.7   | 20.7   | 20.7     | 20.7        |
| S015 RMSE        | 28.5  | 27.6   | 14.6   | 16.0     | 22.8        |
| S015 RMS_Brut    | 42.3  | 42.3   | 42.3   | 42.3     | 42.3        |
| S015 RMS_Méthode | 20.6  | 17.1   | 27.6   | 25.3     | 24.6        |
| S015 RMS_ICLABEL | 34.7  | 34.7   | 34.7   | 34.7     | 34.7        |
| S016 RMSE        | 12.2  | 11.5   | 10.8   | 9.6      | 11.3        |
| S016 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S016 RMS_Méthode | 10.1  | 8.1    | 14.0   | 12.2     | 11.2        |
| S016 RMS_ICLABEL | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S017 RMSE        | 34.4  | 33.2   | 27.8   | 28.2     | 33.7        |
| S017 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S017 RMS_Méthode | 19.6  | 17.7   | 22.1   | 19.1     | 21.2        |
| S017 RMS_ICLABEL | 36.9  | 36.9   | 36.9   | 36.9     | 36.9        |
| S018 RMSE        | 5.7   | 5.5    | 5.0    | 4.3      | 5.7         |
| S018 RMS_Brut    | 13.8  | 13.8   | 13.8   | 13.8     | 13.8        |
| S018 RMS_Méthode | 3.8   | 3.1    | 6.3    | 5.0      | 5.3         |
| S018 RMS_ICLABEL | 5.8   | 5.8    | 5.8    | 5.8      | 5.8         |
| S019 RMSE        | 21.8  | 21.6   | 15.8   | 16.7     | 20.2        |
| S019 RMS_Brut    | 37.4  | 37.4   | 37.4   | 37.4     | 37.4        |
| S019 RMS_Méthode | 14.1  | 12.5   | 15.0   | 14.0     | 14.4        |
| S019 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S020 RMSE        | 5.3   | 5.1    | 5.0    | 4.8      | 5.5         |
| S020 RMS_Brut    | 11.1  | 11.1   | 11.1   | 11.1     | 11.1        |
| S020 RMS_Méthode | 3.6   | 2.9    | 5.8    | 5.3      | 5.1         |
| S020 RMS_ICLABEL | 5.4   | 5.4    | 5.4    | 5.4      | 5.4         |
| S021 RMSE        | 11.9  | 11.1   | 9.1    | 8.5      | 10.9        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S021 RMS_Brut    | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S021 RMS_Methode | 8.9   | 6.9    | 12.2   | 9.8      | 9.0         |
| S021 RMS_ICLABEL | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S022 RMSE        | 25.3  | 24.8   | 20.4   | 18.2     | 27.2        |
| S022 RMS_Brut    | 68.5  | 68.5   | 68.5   | 68.5     | 68.5        |
| S022 RMS_Methode | 16.5  | 14.0   | 27.5   | 21.7     | 25.8        |
| S022 RMS_ICLABEL | 27.7  | 27.7   | 27.7   | 27.7     | 27.7        |
| S023 RMSE        | 31.7  | 31.1   | 20.0   | 20.6     | 28.3        |
| S023 RMS_Brut    | 56.2  | 56.2   | 56.2   | 56.2     | 56.2        |
| S023 RMS_Methode | 21.1  | 17.6   | 25.3   | 24.8     | 26.1        |
| S023 RMS_ICLABEL | 37.1  | 37.1   | 37.1   | 37.1     | 37.1        |
| S024 RMSE        | 15.5  | 15.5   | 11.6   | 13.0     | 14.3        |
| S024 RMS_Brut    | 34.6  | 34.6   | 34.6   | 34.6     | 34.6        |
| S024 RMS_Methode | 12.4  | 11.9   | 8.6    | 4.5      | 11.0        |
| S024 RMS_ICLABEL | 14.4  | 14.4   | 14.4   | 14.4     | 14.4        |
| S025 RMSE        | 7.5   | 7.1    | 4.9    | 4.8      | 6.3         |
| S025 RMS_Brut    | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S025 RMS_Methode | 6.3   | 5.4    | 7.4    | 7.6      | 7.3         |
| S025 RMS_ICLABEL | 9.1   | 9.1    | 9.1    | 9.1      | 9.1         |
| S026 RMSE        | 5.4   | 5.0    | 5.6    | 4.9      | 5.2         |
| S026 RMS_Brut    | 11.9  | 11.9   | 11.9   | 11.9     | 11.9        |
| S026 RMS_Methode | 4.4   | 3.7    | 6.2    | 5.2      | 4.6         |
| S026 RMS_ICLABEL | 4.9   | 4.9    | 4.9    | 4.9      | 4.9         |
| S027 RMSE        | 24.5  | 24.0   | 15.0   | 15.9     | 21.2        |
| S027 RMS_Brut    | 41.7  | 41.7   | 41.7   | 41.7     | 41.7        |
| S027 RMS_Methode | 17.4  | 14.3   | 22.1   | 20.7     | 22.3        |
| S027 RMS_ICLABEL | 29.3  | 29.3   | 29.3   | 29.3     | 29.3        |
| S028 RMSE        | 34.2  | 34.0   | 19.3   | 19.8     | 29.1        |
| S028 RMS_Brut    | 55.1  | 55.1   | 55.1   | 55.1     | 55.1        |
| S028 RMS_Methode | 22.7  | 17.5   | 31.6   | 30.3     | 28.5        |
| S028 RMS_ICLABEL | 40.2  | 40.2   | 40.2   | 40.2     | 40.2        |
| S029 RMSE        | 10.0  | 9.8    | 4.8    | 5.0      | 7.8         |
| S029 RMS_Brut    | 14.5  | 14.5   | 14.5   | 14.5     | 14.5        |
| S029 RMS_Methode | 8.4   | 7.1    | 11.3   | 11.0     | 10.1        |
| S029 RMS_ICLABEL | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S030 RMSE        | 8.1   | 8.1    | 4.7    | 5.1      | 7.1         |
| S030 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S030 RMS_Methode | 4.7   | 3.6    | 6.8    | 6.1      | 5.4         |
| S030 RMS_ICLABEL | 9.3   | 9.3    | 9.3    | 9.3      | 9.3         |
| S031 RMSE        | 21.9  | 21.9   | 12.0   | 11.9     | 17.2        |
| S031 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S031 RMS_Methode | 18.4  | 13.6   | 27.3   | 27.9     | 27.8        |
| S031 RMS_ICLABEL | 27.9  | 27.9   | 27.9   | 27.9     | 27.9        |
| S032 RMSE        | 30.3  | 30.4   | 20.3   | 23.1     | 28.6        |
| S032 RMS_Brut    | 53.9  | 53.9   | 53.9   | 53.9     | 53.9        |
| S032 RMS_Methode | 15.1  | 13.2   | 22.7   | 18.2     | 19.4        |
| S032 RMS_ICLABEL | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S033 RMSE        | 12.8  | 12.6   | 7.9    | 7.7      | 11.8        |
| S033 RMS_Brut    | 25.8  | 25.8   | 25.8   | 25.8     | 25.8        |
| S033 RMS_Methode | 8.2   | 7.0    | 12.4   | 11.3     | 12.7        |
| S033 RMS_ICLABEL | 14.8  | 14.8   | 14.8   | 14.8     | 14.8        |
| S034 RMSE        | 9.2   | 9.2    | 4.6    | 5.0      | 7.0         |
| S034 RMS_Brut    | 12.4  | 12.4   | 12.4   | 12.4     | 12.4        |
| S034 RMS_Methode | 6.3   | 5.0    | 9.0    | 9.2      | 9.5         |
| S034 RMS_ICLABEL | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S035 RMSE        | 10.5  | 10.4   | 5.5    | 5.3      | 8.8         |
| S035 RMS_Brut    | 18.8  | 18.8   | 18.8   | 18.8     | 18.8        |
| S035 RMS_Methode | 8.2   | 6.6    | 12.1   | 11.9     | 12.1        |
| S035 RMS_ICLABEL | 13.1  | 13.1   | 13.1   | 13.1     | 13.1        |
| S036 RMSE        | 27.7  | 26.3   | 21.4   | 20.5     | 25.6        |
| S036 RMS_Brut    | 53.4  | 53.4   | 53.4   | 53.4     | 53.4        |
| S036 RMS_Methode | 22.5  | 18.2   | 32.1   | 30.0     | 28.5        |
| S036 RMS_ICLABEL | 31.4  | 31.4   | 31.4   | 31.4     | 31.4        |
| S037 RMSE        | 5.8   | 5.7    | 3.9    | 4.1      | 5.0         |
| S037 RMS_Brut    | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S037 RMS_Methode | 3.5   | 3.0    | 4.0    | 3.7      | 3.8         |
| S037 RMS_ICLABEL | 6.6   | 6.6    | 6.6    | 6.6      | 6.6         |
| S038 RMSE        | 19.1  | 19.1   | 10.5   | 10.5     | 13.9        |
| S038 RMS_Brut    | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S038 RMS_Methode | 12.7  | 10.4   | 16.4   | 17.6     | 17.2        |
| S038 RMS_ICLABEL | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S039 RMSE        | 35.0  | 35.1   | 19.4   | 19.5     | 27.5        |
| S039 RMS_Brut    | 51.4  | 51.4   | 51.4   | 51.4     | 51.4        |
| S039 RMS_Methode | 20.5  | 16.0   | 31.2   | 32.5     | 32.0        |
| S039 RMS_ICLABEL | 40.6  | 40.6   | 40.6   | 40.6     | 40.6        |
| S040 RMSE        | 18.5  | 18.5   | 14.1   | 15.1     | 17.3        |
| S040 RMS_Brut    | 26.4  | 26.4   | 26.4   | 26.4     | 26.4        |
| S040 RMS_Methode | 10.0  | 8.0    | 15.1   | 12.6     | 11.4        |
| S040 RMS_ICLABEL | 20.2  | 20.2   | 20.2   | 20.2     | 20.2        |
| S041 RMSE        | 6.8   | 6.7    | 3.8    | 4.0      | 5.8         |
| S041 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S041 RMS_Methode | 4.0   | 3.2    | 6.3    | 6.0      | 6.0         |
| S041 RMS_ICLABEL | 7.9   | 7.9    | 7.9    | 7.9      | 7.9         |
| S042 RMSE        | 8.2   | 8.0    | 3.4    | 3.2      | 6.0         |
| S042 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S042 RMS_Methode | 7.4   | 6.4    | 10.2   | 10.3     | 10.4        |
| S042 RMS_ICLABEL | 11.0  | 11.0   | 11.0   | 11.0     | 11.0        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S043 RMSE        | 24.8  | 24.4   | 20.3   | 20.8     | 24.5        |
| S043 RMS_Brut    | 48.0  | 48.0   | 48.0   | 48.0     | 48.0        |
| S043 RMS_Methode | 15.4  | 13.0   | 21.8   | 19.5     | 19.8        |
| S043 RMS_ICLABEL | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S044 RMSE        | 33.0  | 31.7   | 23.2   | 21.1     | 31.4        |
| S044 RMS_Brut    | 67.9  | 67.9   | 67.9   | 67.9     | 67.9        |
| S044 RMS_Methode | 30.3  | 25.3   | 40.4   | 38.9     | 39.9        |
| S044 RMS_ICLABEL | 43.0  | 43.0   | 43.0   | 43.0     | 43.0        |
| S045 RMSE        | 27.5  | 26.4   | 20.3   | 19.3     | 24.3        |
| S045 RMS_Brut    | 46.4  | 46.4   | 46.4   | 46.4     | 46.4        |
| S045 RMS_Methode | 18.0  | 14.6   | 24.8   | 23.3     | 23.0        |
| S045 RMS_ICLABEL | 29.0  | 29.0   | 29.0   | 29.0     | 29.0        |
| S046 RMSE        | 22.8  | 22.3   | 15.4   | 14.5     | 19.0        |
| S046 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S046 RMS_Methode | 15.1  | 11.9   | 23.9   | 21.6     | 18.8        |
| S046 RMS_ICLABEL | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S047 RMSE        | 7.0   | 6.7    | 4.6    | 4.4      | 6.2         |
| S047 RMS_Brut    | 12.0  | 12.0   | 12.0   | 12.0     | 12.0        |
| S047 RMS_Methode | 5.6   | 4.4    | 7.6    | 7.2      | 6.2         |
| S047 RMS_ICLABEL | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S048 RMSE        | 42.1  | 40.8   | 21.1   | 20.8     | 32.4        |
| S048 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S048 RMS_Methode | 39.8  | 32.5   | 49.5   | 50.8     | 49.7        |
| S048 RMS_ICLABEL | 58.2  | 58.2   | 58.2   | 58.2     | 58.2        |
| S049 RMSE        | 31.5  | 31.3   | 17.2   | 17.9     | 25.2        |
| S049 RMS_Brut    | 47.9  | 47.9   | 47.9   | 47.9     | 47.9        |
| S049 RMS_Methode | 16.8  | 14.2   | 27.9   | 26.2     | 26.0        |
| S049 RMS_ICLABEL | 36.0  | 36.0   | 36.0   | 36.0     | 36.0        |
| S050 RMSE        | 10.1  | 9.5    | 8.1    | 7.7      | 9.1         |
| S050 RMS_Brut    | 17.3  | 17.3   | 17.3   | 17.3     | 17.3        |
| S050 RMS_Methode | 7.6   | 6.3    | 9.5    | 8.2      | 7.5         |
| S050 RMS_ICLABEL | 10.2  | 10.2   | 10.2   | 10.2     | 10.2        |
| S051 RMSE        | 3.9   | 3.8    | 3.5    | 3.2      | 4.3         |
| S051 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S051 RMS_Methode | 2.2   | 1.9    | 4.4    | 3.6      | 3.9         |
| S051 RMS_ICLABEL | 4.1   | 4.1    | 4.1    | 4.1      | 4.1         |
| S052 RMSE        | 30.7  | 30.5   | 17.5   | 17.7     | 24.4        |
| S052 RMS_Brut    | 45.7  | 45.7   | 45.7   | 45.7     | 45.7        |
| S052 RMS_Methode | 23.6  | 19.4   | 31.1   | 32.1     | 32.4        |
| S052 RMS_ICLABEL | 38.6  | 38.6   | 38.6   | 38.6     | 38.6        |
| S053 RMSE        | 29.3  | 29.0   | 20.2   | 21.1     | 28.3        |
| S053 RMS_Brut    | 62.9  | 62.9   | 62.9   | 62.9     | 62.9        |
| S053 RMS_Methode | 18.2  | 15.5   | 24.0   | 20.0     | 24.4        |
| S053 RMS_ICLABEL | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S054 RMSE        | 28.0  | 27.2   | 18.0   | 16.9     | 26.7        |
| S054 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S054 RMS_Methode | 24.7  | 20.4   | 35.4   | 35.2     | 36.7        |
| S054 RMS_ICLABEL | 35.9  | 35.9   | 35.9   | 35.9     | 35.9        |
| S055 RMSE        | 7.4   | 7.3    | 4.7    | 4.9      | 7.0         |
| S055 RMS_Brut    | 14.6  | 14.6   | 14.6   | 14.6     | 14.6        |
| S055 RMS_Methode | 4.1   | 3.5    | 6.9    | 5.7      | 5.2         |
| S055 RMS_ICLABEL | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S056 RMSE        | 33.7  | 32.3   | 23.2   | 23.1     | 29.8        |
| S056 RMS_Brut    | 61.0  | 61.0   | 61.0   | 61.0     | 61.0        |
| S056 RMS_Methode | 30.3  | 25.2   | 31.9   | 31.3     | 31.6        |
| S056 RMS_ICLABEL | 39.8  | 39.8   | 39.8   | 39.8     | 39.8        |
| S057 RMSE        | 19.4  | 18.6   | 14.3   | 13.7     | 16.2        |
| S057 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S057 RMS_Methode | 14.7  | 12.3   | 16.3   | 15.8     | 14.2        |
| S057 RMS_ICLABEL | 19.9  | 19.9   | 19.9   | 19.9     | 19.9        |
| S058 RMSE        | 13.9  | 13.3   | 11.4   | 11.0     | 12.3        |
| S058 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S058 RMS_Methode | 11.2  | 10.2   | 9.8    | 8.4      | 8.9         |
| S058 RMS_ICLABEL | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S059 RMSE        | 22.4  | 21.1   | 21.7   | 21.1     | 21.9        |
| S059 RMS_Brut    | 39.1  | 39.1   | 39.1   | 39.1     | 39.1        |
| S059 RMS_Methode | 17.8  | 14.2   | 27.2   | 26.3     | 22.9        |
| S059 RMS_ICLABEL | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S060 RMSE        | 28.3  | 27.9   | 16.9   | 16.3     | 24.5        |
| S060 RMS_Brut    | 49.3  | 49.3   | 49.3   | 49.3     | 49.3        |
| S060 RMS_Methode | 18.4  | 15.2   | 24.8   | 25.2     | 25.2        |
| S060 RMS_ICLABEL | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S061 RMSE        | 37.2  | 36.4   | 22.4   | 23.7     | 34.3        |
| S061 RMS_Brut    | 65.7  | 65.7   | 65.7   | 65.7     | 65.7        |
| S061 RMS_Methode | 24.2  | 20.5   | 34.6   | 31.2     | 28.3        |
| S061 RMS_ICLABEL | 43.4  | 43.4   | 43.4   | 43.4     | 43.4        |
| S062 RMSE        | 13.1  | 13.0   | 8.0    | 7.6      | 12.3        |
| S062 RMS_Brut    | 28.0  | 28.0   | 28.0   | 28.0     | 28.0        |
| S062 RMS_Methode | 8.8   | 7.3    | 14.7   | 13.7     | 13.3        |
| S062 RMS_ICLABEL | 15.2  | 15.2   | 15.2   | 15.2     | 15.2        |
| S063 RMSE        | 12.4  | 11.7   | 10.1   | 9.7      | 11.2        |
| S063 RMS_Brut    | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S063 RMS_Methode | 8.8   | 7.8    | 8.8    | 7.7      | 8.1         |
| S063 RMS_ICLABEL | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S064 RMSE        | 7.5   | 7.4    | 4.9    | 4.7      | 6.5         |
| S064 RMS_Brut    | 13.4  | 13.4   | 13.4   | 13.4     | 13.4        |
| S064 RMS_Methode | 4.7   | 3.7    | 8.1    | 7.5      | 6.5         |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S064 RMS_ICLABEL | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S065 RMSE        | 20.9  | 20.2   | 16.1   | 16.3     | 19.1        |
| S065 RMS_Brut    | 32.6  | 32.6   | 32.6   | 32.6     | 32.6        |
| S065 RMS_Methode | 16.2  | 12.7   | 20.8   | 20.2     | 18.5        |
| S065 RMS_ICLABEL | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S066 RMSE        | 17.7  | 17.4   | 8.6    | 8.4      | 13.9        |
| S066 RMS_Brut    | 28.9  | 28.9   | 28.9   | 28.9     | 28.9        |
| S066 RMS_Methode | 13.4  | 11.0   | 18.7   | 18.3     | 18.7        |
| S066 RMS_ICLABEL | 21.9  | 21.9   | 21.9   | 21.9     | 21.9        |
| S067 RMSE        | 6.8   | 6.9    | 5.4    | 5.7      | 6.5         |
| S067 RMS_Brut    | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S067 RMS_Methode | 3.7   | 3.5    | 3.9    | 3.3      | 3.7         |
| S067 RMS_ICLABEL | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S068 RMSE        | 22.6  | 21.7   | 18.3   | 17.9     | 21.2        |
| S068 RMS_Brut    | 39.6  | 39.6   | 39.6   | 39.6     | 39.6        |
| S068 RMS_Methode | 15.6  | 12.4   | 21.4   | 19.9     | 19.6        |
| S068 RMS_ICLABEL | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S069 RMSE        | 40.7  | 41.8   | 23.8   | 36.6     | 41.3        |
| S069 RMS_Brut    | 52.8  | 52.8   | 52.8   | 52.8     | 52.8        |
| S069 RMS_Methode | 12.2  | 10.4   | 26.4   | 12.9     | 12.8        |
| S069 RMS_ICLABEL | 44.2  | 44.2   | 44.2   | 44.2     | 44.2        |
| S070 RMSE        | 27.0  | 26.3   | 18.3   | 18.0     | 23.0        |
| S070 RMS_Brut    | 43.6  | 43.6   | 43.6   | 43.6     | 43.6        |
| S070 RMS_Methode | 20.5  | 15.9   | 29.6   | 29.6     | 27.9        |
| S070 RMS_ICLABEL | 31.0  | 31.0   | 31.0   | 31.0     | 31.0        |
| S071 RMSE        | 18.5  | 17.5   | 14.0   | 13.5     | 15.9        |
| S071 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S071 RMS_Methode | 14.9  | 11.7   | 17.3   | 16.5     | 15.2        |
| S071 RMS_ICLABEL | 18.9  | 18.9   | 18.9   | 18.9     | 18.9        |
| S072 RMSE        | 5.5   | 5.5    | 3.9    | 4.0      | 5.1         |
| S072 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S072 RMS_Methode | 3.2   | 2.7    | 5.7    | 5.3      | 5.2         |
| S072 RMS_ICLABEL | 6.1   | 6.1    | 6.1    | 6.1      | 6.1         |
| S073 RMSE        | 7.1   | 6.7    | 5.4    | 5.2      | 6.3         |
| S073 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S073 RMS_Methode | 5.2   | 4.2    | 7.0    | 6.4      | 6.2         |
| S073 RMS_ICLABEL | 7.4   | 7.4    | 7.4    | 7.4      | 7.4         |
| S074 RMSE        | 8.8   | 8.6    | 4.2    | 4.0      | 6.1         |
| S074 RMS_Brut    | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S074 RMS_Methode | 5.9   | 4.8    | 9.0    | 9.1      | 8.7         |
| S074 RMS_ICLABEL | 10.6  | 10.6   | 10.6   | 10.6     | 10.6        |
| S075 RMSE        | 44.5  | 44.3   | 23.2   | 24.9     | 36.6        |
| S075 RMS_Brut    | 68.1  | 68.1   | 68.1   | 68.1     | 68.1        |
| S075 RMS_Methode | 30.0  | 22.5   | 42.9   | 40.4     | 39.0        |
| S075 RMS_ICLABEL | 53.5  | 53.5   | 53.5   | 53.5     | 53.5        |
| S076 RMSE        | 6.6   | 6.4    | 4.5    | 4.7      | 5.8         |
| S076 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S076 RMS_Methode | 4.9   | 4.2    | 6.0    | 5.2      | 4.8         |
| S076 RMS_ICLABEL | 7.5   | 7.5    | 7.5    | 7.5      | 7.5         |
| S077 RMSE        | 30.7  | 29.9   | 21.7   | 21.1     | 27.3        |
| S077 RMS_Brut    | 55.8  | 55.8   | 55.8   | 55.8     | 55.8        |
| S077 RMS_Methode | 21.2  | 16.1   | 31.9   | 30.0     | 28.4        |
| S077 RMS_ICLABEL | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S078 RMSE        | 15.5  | 15.3   | 8.3    | 8.5      | 12.7        |
| S078 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S078 RMS_Methode | 10.8  | 8.6    | 14.9   | 14.0     | 13.5        |
| S078 RMS_ICLABEL | 18.5  | 18.5   | 18.5   | 18.5     | 18.5        |
| S079 RMSE        | 35.8  | 35.7   | 20.9   | 25.7     | 30.2        |
| S079 RMS_Brut    | 44.1  | 44.1   | 44.1   | 44.1     | 44.1        |
| S079 RMS_Methode | 21.7  | 17.4   | 30.0   | 26.6     | 25.2        |
| S079 RMS_ICLABEL | 42.1  | 42.1   | 42.1   | 42.1     | 42.1        |
| S080 RMSE        | 19.4  | 19.2   | 11.8   | 12.8     | 17.3        |
| S080 RMS_Brut    | 32.0  | 32.0   | 32.0   | 32.0     | 32.0        |
| S080 RMS_Methode | 11.4  | 9.4    | 16.5   | 14.4     | 14.1        |
| S080 RMS_ICLABEL | 22.2  | 22.2   | 22.2   | 22.2     | 22.2        |
| S081 RMSE        | 16.3  | 16.3   | 10.1   | 10.1     | 13.6        |
| S081 RMS_Brut    | 22.8  | 22.8   | 22.8   | 22.8     | 22.8        |
| S081 RMS_Methode | 7.1   | 5.9    | 13.1   | 12.3     | 10.7        |
| S081 RMS_ICLABEL | 17.9  | 17.9   | 17.9   | 17.9     | 17.9        |
| S082 RMSE        | 16.0  | 15.4   | 11.7   | 11.5     | 14.0        |
| S082 RMS_Brut    | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S082 RMS_Methode | 13.4  | 10.8   | 17.7   | 17.6     | 16.5        |
| S082 RMS_ICLABEL | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S083 RMSE        | 28.9  | 28.5   | 17.5   | 18.4     | 25.0        |
| S083 RMS_Brut    | 45.6  | 45.6   | 45.6   | 45.6     | 45.6        |
| S083 RMS_Methode | 19.4  | 16.4   | 23.2   | 23.3     | 22.2        |
| S083 RMS_ICLABEL | 34.2  | 34.2   | 34.2   | 34.2     | 34.2        |
| S084 RMSE        | 24.0  | 23.9   | 17.2   | 17.9     | 21.4        |
| S084 RMS_Brut    | 33.0  | 33.0   | 33.0   | 33.0     | 33.0        |
| S084 RMS_Methode | 16.6  | 13.3   | 19.2   | 20.1     | 20.8        |
| S084 RMS_ICLABEL | 28.8  | 28.8   | 28.8   | 28.8     | 28.8        |
| S085 RMSE        | 15.5  | 15.5   | 10.3   | 11.1     | 14.2        |
| S085 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S085 RMS_Methode | 10.4  | 8.4    | 12.9   | 12.8     | 12.8        |
| S085 RMS_ICLABEL | 18.6  | 18.6   | 18.6   | 18.6     | 18.6        |
| S086 RMSE        | 39.1  | 38.7   | 27.3   | 29.2     | 34.0        |
| S086 RMS_Brut    | 59.4  | 59.4   | 59.4   | 59.4     | 59.4        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S086 RMS_Methode | 14.2  | 12.2   | 31.4   | 27.3     | 29.6        |
| S086 RMS_ICLABEL | 42.0  | 42.0   | 42.0   | 42.0     | 42.0        |
| S087 RMSE        | 22.9  | 21.5   | 17.4   | 16.4     | 19.6        |
| S087 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S087 RMS_Methode | 19.2  | 15.5   | 23.8   | 23.6     | 22.7        |
| S087 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S088 RMSE        | 46.9  | 48.2   | 25.8   | 40.2     | 47.0        |
| S088 RMS_Brut    | 64.4  | 64.4   | 64.4   | 64.4     | 64.4        |
| S088 RMS_Methode | 15.2  | 12.5   | 31.8   | 18.5     | 20.0        |
| S088 RMS_ICLABEL | 51.1  | 51.1   | 51.1   | 51.1     | 51.1        |
| S089 RMSE        | 4.9   | 4.8    | 2.6    | 2.5      | 4.0         |
| S089 RMS_Brut    | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S089 RMS_Methode | 2.9   | 2.3    | 4.8    | 4.5      | 4.4         |
| S089 RMS_ICLABEL | 5.5   | 5.5    | 5.5    | 5.5      | 5.5         |
| S090 RMSE        | 20.7  | 19.9   | 19.4   | 18.0     | 22.0        |
| S090 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S090 RMS_Methode | 17.6  | 15.1   | 22.8   | 20.7     | 21.3        |
| S090 RMS_ICLABEL | 21.7  | 21.7   | 21.7   | 21.7     | 21.7        |
| S091 RMSE        | 27.9  | 25.7   | 38.8   | 35.6     | 35.5        |
| S091 RMS_Brut    | 84.1  | 84.1   | 84.1   | 84.1     | 84.1        |
| S091 RMS_Methode | 21.9  | 18.8   | 41.9   | 37.8     | 33.8        |
| S091 RMS_ICLABEL | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S092 RMSE        | 25.8  | 25.0   | 17.6   | 17.5     | 22.1        |
| S092 RMS_Brut    | 40.3  | 40.3   | 40.3   | 40.3     | 40.3        |
| S092 RMS_Methode | 23.0  | 18.1   | 29.1   | 30.3     | 29.8        |
| S092 RMS_ICLABEL | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S093 RMSE        | 13.1  | 13.1   | 9.6    | 10.2     | 11.5        |
| S093 RMS_Brut    | 21.8  | 21.8   | 21.8   | 21.8     | 21.8        |
| S093 RMS_Methode | 8.2   | 6.6    | 14.8   | 15.0     | 14.0        |
| S093 RMS_ICLABEL | 14.9  | 14.9   | 14.9   | 14.9     | 14.9        |
| S094 RMSE        | 30.3  | 30.0   | 18.5   | 18.5     | 26.7        |
| S094 RMS_Brut    | 52.4  | 52.4   | 52.4   | 52.4     | 52.4        |
| S094 RMS_Methode | 22.3  | 18.3   | 30.2   | 29.7     | 30.2        |
| S094 RMS_ICLABEL | 36.8  | 36.8   | 36.8   | 36.8     | 36.8        |
| S095 RMSE        | 47.6  | 48.5   | 27.1   | 40.2     | 48.1        |
| S095 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S095 RMS_Methode | 24.4  | 19.5   | 34.2   | 21.0     | 16.9        |
| S095 RMS_ICLABEL | 54.5  | 54.5   | 54.5   | 54.5     | 54.5        |
| S096 RMSE        | 34.7  | 33.8   | 27.0   | 27.8     | 33.7        |
| S096 RMS_Brut    | 69.0  | 69.0   | 69.0   | 69.0     | 69.0        |
| S096 RMS_Methode | 22.0  | 18.5   | 35.4   | 32.8     | 30.2        |
| S096 RMS_ICLABEL | 38.4  | 38.4   | 38.4   | 38.4     | 38.4        |
| S097 RMSE        | 22.0  | 21.3   | 19.7   | 17.5     | 23.3        |
| S097 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S097 RMS_Methode | 15.4  | 12.7   | 23.7   | 19.4     | 21.4        |
| S097 RMS_ICLABEL | 22.9  | 22.9   | 22.9   | 22.9     | 22.9        |
| S098 RMSE        | 8.5   | 8.3    | 4.1    | 3.9      | 6.1         |
| S098 RMS_Brut    | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S098 RMS_Methode | 8.6   | 7.4    | 9.9    | 10.4     | 10.7        |
| S098 RMS_ICLABEL | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S099 RMSE        | 30.6  | 28.1   | 24.8   | 25.1     | 28.3        |
| S099 RMS_Brut    | 54.6  | 54.6   | 54.6   | 54.6     | 54.6        |
| S099 RMS_Methode | 26.1  | 21.8   | 22.8   | 20.1     | 21.2        |
| S099 RMS_ICLABEL | 29.9  | 29.9   | 29.9   | 29.9     | 29.9        |
| S100 RMSE        | 23.8  | 23.2   | 17.6   | 17.6     | 22.3        |
| S100 RMS_Brut    | 45.9  | 45.9   | 45.9   | 45.9     | 45.9        |
| S100 RMS_Methode | 16.4  | 14.1   | 20.1   | 21.9     | 21.8        |
| S100 RMS_ICLABEL | 26.2  | 26.2   | 26.2   | 26.2     | 26.2        |
| S101 RMSE        | 23.1  | 22.7   | 17.0   | 17.0     | 20.0        |
| S101 RMS_Brut    | 32.3  | 32.3   | 32.3   | 32.3     | 32.3        |
| S101 RMS_Methode | 14.1  | 11.3   | 20.3   | 18.5     | 16.7        |
| S101 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S102 RMSE        | 6.7   | 6.5    | 4.5    | 4.3      | 5.6         |
| S102 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S102 RMS_Methode | 4.7   | 3.7    | 7.0    | 6.6      | 6.2         |
| S102 RMS_ICLABEL | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S103 RMSE        | 31.5  | 30.2   | 25.6   | 24.5     | 27.7        |
| S103 RMS_Brut    | 48.8  | 48.8   | 48.8   | 48.8     | 48.8        |
| S103 RMS_Methode | 26.3  | 20.6   | 37.5   | 36.1     | 32.8        |
| S103 RMS_ICLABEL | 35.7  | 35.7   | 35.7   | 35.7     | 35.7        |
| S104 RMSE        | 16.9  | 16.7   | 9.0    | 9.3      | 14.9        |
| S104 RMS_Brut    | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S104 RMS_Methode | 11.7  | 9.5    | 16.8   | 15.3     | 14.7        |
| S104 RMS_ICLABEL | 20.0  | 20.0   | 20.0   | 20.0     | 20.0        |
| S105 RMSE        | 15.7  | 15.5   | 10.5   | 10.0     | 13.8        |
| S105 RMS_Brut    | 30.1  | 30.1   | 30.1   | 30.1     | 30.1        |
| S105 RMS_Methode | 11.7  | 8.9    | 18.8   | 18.2     | 17.7        |
| S105 RMS_ICLABEL | 18.0  | 18.0   | 18.0   | 18.0     | 18.0        |
| S106 RMSE        | 25.4  | 25.1   | 13.7   | 14.1     | 20.8        |
| S106 RMS_Brut    | 41.8  | 41.8   | 41.8   | 41.8     | 41.8        |
| S106 RMS_Methode | 18.9  | 15.2   | 25.4   | 25.3     | 25.0        |
| S106 RMS_ICLABEL | 31.2  | 31.2   | 31.2   | 31.2     | 31.2        |
| S107 RMSE        | 26.7  | 26.0   | 17.1   | 17.4     | 20.7        |
| S107 RMS_Brut    | 38.7  | 38.7   | 38.7   | 38.7     | 38.7        |
| S107 RMS_Methode | 20.4  | 16.0   | 25.8   | 24.3     | 22.4        |
| S107 RMS_ICLABEL | 29.2  | 29.2   | 29.2   | 29.2     | 29.2        |
| S108 RMSE        | 21.4  | 21.2   | 18.7   | 20.1     | 22.4        |

| Sujet               | DuoCL | GCTNet | ICUNet      | ICUNet++ | ICUNet_attn |
|---------------------|-------|--------|-------------|----------|-------------|
| S108 RMS_Brut       | 51.9  | 51.9   | 51.9        | 51.9     | 51.9        |
| S108 RMS_Methode    | 16.5  | 14.6   | 23.5        | 21.7     | 21.2        |
| S108 RMS_ICLABEL    | 23.9  | 23.9   | 23.9        | 23.9     | 23.9        |
| S109 RMSE           | 35.9  | 36.2   | 35.5        | 37.4     | 40.0        |
| S109 RMS_Brut       | 46.7  | 46.7   | 46.7        | 46.7     | 46.7        |
| S109 RMS_Methode    | 24.0  | 19.1   | 20.4        | 19.5     | 20.1        |
| S109 RMS_ICLABEL    | 44.4  | 44.4   | 44.4        | 44.4     | 44.4        |
| <b>Moyenne RMSE</b> | 20.8  | 20.4   | <b>14.4</b> | 14.8     | 18.8        |
| Moyenne RMS_Brut    | 35.8  | 35.8   | 35.8        | 35.8     | 35.8        |
| Moyenne RMS_Methode | 14.2  | 11.7   | 19.5        | 18.0     | 17.8        |
| Moyenne RMS_ICLABEL | 23.6  | 23.6   | 23.6        | 23.6     | 23.6        |

L'ordre de grandeur des signaux nettoyés par ICLabel est de 23.6  $\mu\text{V}$ , contre 35.8  $\mu\text{V}$  pour le signal brut (RMS moyen sur les 109 sujets).

TABLE 43 – Récapitulatif — RMSE, RMS\_Brut, RMS\_Méthode et RMS de référence ICLabel (en  $\mu V$ ), les 109 sujets (4 lignes par sujet), par méthode — signaux de Nettoyé\_Normalisé (chaque essai renormalisé avec la moyenne/écart-type du signal brut Prétraité\_Total du même sujet, « - » = non calculé).

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S001 RMSE        | 39.5  | 39.4   | 22.8   | 22.0     | 27.7        |
| S001 RMS_Brut    | 36.2  | 36.2   | 36.2   | 36.2     | 36.2        |
| S001 RMS_Méthode | 36.2  | 36.2   | 36.2   | 36.2     | 36.2        |
| S001 RMS_ICLABEL | 36.2  | 36.2   | 36.2   | 36.2     | 36.2        |
| S002 RMSE        | 17.5  | 16.7   | 9.3    | 8.8      | 13.0        |
| S002 RMS_Brut    | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S002 RMS_Méthode | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S002 RMS_ICLABEL | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S003 RMSE        | 60.5  | 60.1   | 44.8   | 43.4     | 56.2        |
| S003 RMS_Brut    | 54.1  | 54.1   | 54.1   | 54.1     | 54.1        |
| S003 RMS_Méthode | 54.1  | 54.1   | 54.1   | 54.1     | 54.1        |
| S003 RMS_ICLABEL | 54.1  | 54.1   | 54.1   | 54.1     | 54.1        |
| S004 RMSE        | 35.3  | 36.6   | 34.8   | 39.1     | 41.7        |
| S004 RMS_Brut    | 32.7  | 32.7   | 32.7   | 32.7     | 32.7        |
| S004 RMS_Méthode | 32.7  | 32.7   | 32.7   | 32.7     | 32.7        |
| S004 RMS_ICLABEL | 32.7  | 32.7   | 32.7   | 32.7     | 32.7        |
| S005 RMSE        | 23.3  | 22.7   | 16.7   | 17.9     | 22.9        |
| S005 RMS_Brut    | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S005 RMS_Méthode | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S005 RMS_ICLABEL | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S006 RMSE        | 22.6  | 22.6   | 14.9   | 15.5     | 19.5        |
| S006 RMS_Brut    | 20.6  | 20.6   | 20.6   | 20.6     | 20.6        |
| S006 RMS_Méthode | 20.6  | 20.6   | 20.6   | 20.6     | 20.6        |
| S006 RMS_ICLABEL | 20.6  | 20.6   | 20.6   | 20.6     | 20.6        |
| S007 RMSE        | 23.1  | 22.1   | 11.6   | 11.2     | 18.7        |
| S007 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S007 RMS_Méthode | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S007 RMS_ICLABEL | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S008 RMSE        | 25.6  | 25.0   | 16.1   | 16.8     | 23.4        |
| S008 RMS_Brut    | 25.3  | 25.3   | 25.3   | 25.3     | 25.3        |
| S008 RMS_Méthode | 25.3  | 25.3   | 25.3   | 25.3     | 25.3        |
| S008 RMS_ICLABEL | 25.3  | 25.3   | 25.3   | 25.3     | 25.3        |
| S009 RMSE        | 62.3  | 61.4   | 45.9   | 45.6     | 60.6        |
| S009 RMS_Brut    | 56.7  | 56.7   | 56.7   | 56.7     | 56.7        |
| S009 RMS_Méthode | 56.7  | 56.7   | 56.7   | 56.7     | 56.7        |
| S009 RMS_ICLABEL | 56.7  | 56.7   | 56.7   | 56.7     | 56.7        |
| S010 RMSE        | 43.1  | 43.6   | 27.8   | 28.4     | 35.3        |
| S010 RMS_Brut    | 39.7  | 39.7   | 39.7   | 39.7     | 39.7        |
| S010 RMS_Méthode | 39.7  | 39.7   | 39.7   | 39.7     | 39.7        |
| S010 RMS_ICLABEL | 39.7  | 39.7   | 39.7   | 39.7     | 39.7        |
| S011 RMSE        | 16.0  | 15.9   | 6.9    | 7.7      | 12.5        |
| S011 RMS_Brut    | 15.8  | 15.8   | 15.8   | 15.8     | 15.8        |
| S011 RMS_Méthode | 15.8  | 15.8   | 15.8   | 15.8     | 15.8        |
| S011 RMS_ICLABEL | 15.8  | 15.8   | 15.8   | 15.8     | 15.8        |
| S012 RMSE        | 40.9  | 40.9   | 29.2   | 27.9     | 37.4        |
| S012 RMS_Brut    | 36.5  | 36.5   | 36.5   | 36.5     | 36.5        |
| S012 RMS_Méthode | 36.5  | 36.5   | 36.5   | 36.5     | 36.5        |
| S012 RMS_ICLABEL | 36.5  | 36.5   | 36.5   | 36.5     | 36.5        |
| S013 RMSE        | 60.0  | 60.5   | 41.3   | 43.6     | 53.8        |
| S013 RMS_Brut    | 56.3  | 56.3   | 56.3   | 56.3     | 56.3        |
| S013 RMS_Méthode | 56.3  | 56.3   | 56.3   | 56.3     | 56.3        |
| S013 RMS_ICLABEL | 56.3  | 56.3   | 56.3   | 56.3     | 56.3        |
| S014 RMSE        | 22.9  | 22.4   | 12.3   | 12.4     | 17.3        |
| S014 RMS_Brut    | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S014 RMS_Méthode | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S014 RMS_ICLABEL | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S015 RMSE        | 39.4  | 37.4   | 17.3   | 18.3     | 29.9        |
| S015 RMS_Brut    | 42.3  | 42.3   | 42.3   | 42.3     | 42.3        |
| S015 RMS_Méthode | 42.3  | 42.3   | 42.3   | 42.3     | 42.3        |
| S015 RMS_ICLABEL | 42.3  | 42.3   | 42.3   | 42.3     | 42.3        |
| S016 RMSE        | 26.4  | 25.9   | 20.1   | 19.4     | 23.9        |
| S016 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S016 RMS_Méthode | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S016 RMS_ICLABEL | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S017 RMSE        | 70.1  | 67.2   | 52.3   | 52.2     | 68.0        |
| S017 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S017 RMS_Méthode | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S017 RMS_ICLABEL | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S018 RMSE        | 15.8  | 15.5   | 11.5   | 10.8     | 14.2        |
| S018 RMS_Brut    | 13.8  | 13.8   | 13.8   | 13.8     | 13.8        |
| S018 RMS_Méthode | 13.8  | 13.8   | 13.8   | 13.8     | 13.8        |
| S018 RMS_ICLABEL | 13.8  | 13.8   | 13.8   | 13.8     | 13.8        |
| S019 RMSE        | 38.1  | 37.7   | 24.3   | 26.0     | 34.5        |
| S019 RMS_Brut    | 37.4  | 37.4   | 37.4   | 37.4     | 37.4        |
| S019 RMS_Méthode | 37.4  | 37.4   | 37.4   | 37.4     | 37.4        |
| S019 RMS_ICLABEL | 37.4  | 37.4   | 37.4   | 37.4     | 37.4        |
| S020 RMSE        | 12.7  | 12.7   | 9.9    | 10.1     | 11.8        |
| S020 RMS_Brut    | 11.1  | 11.1   | 11.1   | 11.1     | 11.1        |
| S020 RMS_Méthode | 11.1  | 11.1   | 11.1   | 11.1     | 11.1        |
| S020 RMS_ICLABEL | 11.1  | 11.1   | 11.1   | 11.1     | 11.1        |
| S021 RMSE        | 26.7  | 26.4   | 18.0   | 18.2     | 24.3        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S021 RMS_Brut    | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S021 RMS_Methode | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S021 RMS_ICLABEL | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S022 RMSE        | 73.6  | 72.2   | 52.7   | 50.3     | 71.4        |
| S022 RMS_Brut    | 68.5  | 68.5   | 68.5   | 68.5     | 68.5        |
| S022 RMS_Methode | 68.5  | 68.5   | 68.5   | 68.5     | 68.5        |
| S022 RMS_ICLABEL | 68.5  | 68.5   | 68.5   | 68.5     | 68.5        |
| S023 RMSE        | 55.9  | 54.2   | 30.5   | 31.2     | 48.9        |
| S023 RMS_Brut    | 56.2  | 56.2   | 56.2   | 56.2     | 56.2        |
| S023 RMS_Methode | 56.2  | 56.2   | 56.2   | 56.2     | 56.2        |
| S023 RMS_ICLABEL | 56.2  | 56.2   | 56.2   | 56.2     | 56.2        |
| S024 RMSE        | 39.8  | 40.5   | 31.3   | 36.5     | 38.2        |
| S024 RMS_Brut    | 34.6  | 34.6   | 34.6   | 34.6     | 34.6        |
| S024 RMS_Methode | 34.6  | 34.6   | 34.6   | 34.6     | 34.6        |
| S024 RMS_ICLABEL | 34.6  | 34.6   | 34.6   | 34.6     | 34.6        |
| S025 RMSE        | 11.1  | 10.4   | 6.7    | 6.6      | 8.9         |
| S025 RMS_Brut    | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S025 RMS_Methode | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S025 RMS_ICLABEL | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S026 RMSE        | 13.8  | 13.4   | 11.7   | 11.6     | 13.0        |
| S026 RMS_Brut    | 11.9  | 11.9   | 11.9   | 11.9     | 11.9        |
| S026 RMS_Methode | 11.9  | 11.9   | 11.9   | 11.9     | 11.9        |
| S026 RMS_ICLABEL | 11.9  | 11.9   | 11.9   | 11.9     | 11.9        |
| S027 RMSE        | 40.3  | 39.0   | 22.2   | 23.6     | 34.5        |
| S027 RMS_Brut    | 41.7  | 41.7   | 41.7   | 41.7     | 41.7        |
| S027 RMS_Methode | 41.7  | 41.7   | 41.7   | 41.7     | 41.7        |
| S027 RMS_ICLABEL | 41.7  | 41.7   | 41.7   | 41.7     | 41.7        |
| S028 RMSE        | 54.5  | 53.3   | 27.3   | 27.9     | 45.1        |
| S028 RMS_Brut    | 55.1  | 55.1   | 55.1   | 55.1     | 55.1        |
| S028 RMS_Methode | 55.1  | 55.1   | 55.1   | 55.1     | 55.1        |
| S028 RMS_ICLABEL | 55.1  | 55.1   | 55.1   | 55.1     | 55.1        |
| S029 RMSE        | 12.7  | 12.4   | 5.5    | 5.6      | 9.3         |
| S029 RMS_Brut    | 14.5  | 14.5   | 14.5   | 14.5     | 14.5        |
| S029 RMS_Methode | 14.5  | 14.5   | 14.5   | 14.5     | 14.5        |
| S029 RMS_ICLABEL | 14.5  | 14.5   | 14.5   | 14.5     | 14.5        |
| S030 RMSE        | 11.0  | 10.9   | 5.3    | 5.5      | 8.8         |
| S030 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S030 RMS_Methode | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S030 RMS_ICLABEL | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S031 RMSE        | 28.8  | 28.2   | 14.4   | 14.2     | 19.9        |
| S031 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S031 RMS_Methode | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S031 RMS_ICLABEL | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S032 RMSE        | 58.6  | 59.1   | 34.3   | 38.2     | 53.5        |
| S032 RMS_Brut    | 53.9  | 53.9   | 53.9   | 53.9     | 53.9        |
| S032 RMS_Methode | 53.9  | 53.9   | 53.9   | 53.9     | 53.9        |
| S032 RMS_ICLABEL | 53.9  | 53.9   | 53.9   | 53.9     | 53.9        |
| S033 RMSE        | 25.8  | 25.3   | 14.1   | 13.7     | 22.2        |
| S033 RMS_Brut    | 25.8  | 25.8   | 25.8   | 25.8     | 25.8        |
| S033 RMS_Methode | 25.8  | 25.8   | 25.8   | 25.8     | 25.8        |
| S033 RMS_ICLABEL | 25.8  | 25.8   | 25.8   | 25.8     | 25.8        |
| S034 RMSE        | 11.3  | 11.2   | 4.8    | 5.3      | 7.8         |
| S034 RMS_Brut    | 12.4  | 12.4   | 12.4   | 12.4     | 12.4        |
| S034 RMS_Methode | 12.4  | 12.4   | 12.4   | 12.4     | 12.4        |
| S034 RMS_ICLABEL | 12.4  | 12.4   | 12.4   | 12.4     | 12.4        |
| S035 RMSE        | 17.1  | 16.7   | 8.3    | 8.0      | 13.3        |
| S035 RMS_Brut    | 18.8  | 18.8   | 18.8   | 18.8     | 18.8        |
| S035 RMS_Methode | 18.8  | 18.8   | 18.8   | 18.8     | 18.8        |
| S035 RMS_ICLABEL | 18.8  | 18.8   | 18.8   | 18.8     | 18.8        |
| S036 RMSE        | 53.0  | 50.9   | 36.4   | 35.8     | 45.9        |
| S036 RMS_Brut    | 53.4  | 53.4   | 53.4   | 53.4     | 53.4        |
| S036 RMS_Methode | 53.4  | 53.4   | 53.4   | 53.4     | 53.4        |
| S036 RMS_ICLABEL | 53.4  | 53.4   | 53.4   | 53.4     | 53.4        |
| S037 RMSE        | 8.4   | 8.2    | 4.6    | 5.0      | 6.9         |
| S037 RMS_Brut    | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S037 RMS_Methode | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S037 RMS_ICLABEL | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S038 RMSE        | 26.1  | 26.0   | 12.1   | 12.8     | 17.9        |
| S038 RMS_Brut    | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S038 RMS_Methode | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S038 RMS_ICLABEL | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S039 RMSE        | 51.3  | 50.7   | 24.5   | 25.2     | 37.2        |
| S039 RMS_Brut    | 51.4  | 51.4   | 51.4   | 51.4     | 51.4        |
| S039 RMS_Methode | 51.4  | 51.4   | 51.4   | 51.4     | 51.4        |
| S039 RMS_ICLABEL | 51.4  | 51.4   | 51.4   | 51.4     | 51.4        |
| S040 RMSE        | 28.7  | 28.8   | 19.7   | 21.5     | 26.0        |
| S040 RMS_Brut    | 26.4  | 26.4   | 26.4   | 26.4     | 26.4        |
| S040 RMS_Methode | 26.4  | 26.4   | 26.4   | 26.4     | 26.4        |
| S040 RMS_ICLABEL | 26.4  | 26.4   | 26.4   | 26.4     | 26.4        |
| S041 RMSE        | 10.8  | 10.5   | 5.4    | 5.6      | 8.8         |
| S041 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S041 RMS_Methode | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S041 RMS_ICLABEL | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S042 RMSE        | 10.5  | 10.1   | 3.9    | 3.8      | 6.9         |
| S042 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S042 RMS_Methode | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S042 RMS_ICLABEL | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S043 RMSE        | 52.7  | 52.4   | 39.1   | 41.6     | 49.4        |
| S043 RMS_Brut    | 48.0  | 48.0   | 48.0   | 48.0     | 48.0        |
| S043 RMS_Methode | 48.0  | 48.0   | 48.0   | 48.0     | 48.0        |
| S043 RMS_ICLABEL | 48.0  | 48.0   | 48.0   | 48.0     | 48.0        |
| S044 RMSE        | 59.1  | 56.3   | 39.4   | 36.5     | 53.3        |
| S044 RMS_Brut    | 67.9  | 67.9   | 67.9   | 67.9     | 67.9        |
| S044 RMS_Methode | 67.9  | 67.9   | 67.9   | 67.9     | 67.9        |
| S044 RMS_ICLABEL | 67.9  | 67.9   | 67.9   | 67.9     | 67.9        |
| S045 RMSE        | 51.4  | 50.2   | 34.3   | 33.0     | 42.3        |
| S045 RMS_Brut    | 46.4  | 46.4   | 46.4   | 46.4     | 46.4        |
| S045 RMS_Methode | 46.4  | 46.4   | 46.4   | 46.4     | 46.4        |
| S045 RMS_ICLABEL | 46.4  | 46.4   | 46.4   | 46.4     | 46.4        |
| S046 RMSE        | 38.5  | 38.1   | 22.9   | 22.1     | 30.4        |
| S046 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S046 RMS_Methode | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S046 RMS_ICLABEL | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S047 RMSE        | 11.5  | 11.1   | 6.9    | 6.7      | 9.9         |
| S047 RMS_Brut    | 12.0  | 12.0   | 12.0   | 12.0     | 12.0        |
| S047 RMS_Methode | 12.0  | 12.0   | 12.0   | 12.0     | 12.0        |
| S047 RMS_ICLABEL | 12.0  | 12.0   | 12.0   | 12.0     | 12.0        |
| S048 RMSE        | 49.8  | 46.1   | 22.4   | 22.4     | 36.1        |
| S048 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S048 RMS_Methode | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S048 RMS_ICLABEL | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S049 RMSE        | 48.7  | 47.5   | 23.1   | 23.6     | 36.9        |
| S049 RMS_Brut    | 47.9  | 47.9   | 47.9   | 47.9     | 47.9        |
| S049 RMS_Methode | 47.9  | 47.9   | 47.9   | 47.9     | 47.9        |
| S049 RMS_ICLABEL | 47.9  | 47.9   | 47.9   | 47.9     | 47.9        |
| S050 RMSE        | 19.2  | 18.7   | 14.2   | 14.2     | 17.3        |
| S050 RMS_Brut    | 17.3  | 17.3   | 17.3   | 17.3     | 17.3        |
| S050 RMS_Methode | 17.3  | 17.3   | 17.3   | 17.3     | 17.3        |
| S050 RMS_ICLABEL | 17.3  | 17.3   | 17.3   | 17.3     | 17.3        |
| S051 RMSE        | 12.3  | 12.2   | 9.0    | 9.2      | 12.1        |
| S051 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S051 RMS_Methode | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S051 RMS_ICLABEL | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S052 RMSE        | 41.0  | 40.1   | 21.1   | 21.6     | 30.7        |
| S052 RMS_Brut    | 45.7  | 45.7   | 45.7   | 45.7     | 45.7        |
| S052 RMS_Methode | 45.7  | 45.7   | 45.7   | 45.7     | 45.7        |
| S052 RMS_ICLABEL | 45.7  | 45.7   | 45.7   | 45.7     | 45.7        |
| S053 RMSE        | 65.4  | 64.9   | 40.7   | 41.2     | 60.3        |
| S053 RMS_Brut    | 62.9  | 62.9   | 62.9   | 62.9     | 62.9        |
| S053 RMS_Methode | 62.9  | 62.9   | 62.9   | 62.9     | 62.9        |
| S053 RMS_ICLABEL | 62.9  | 62.9   | 62.9   | 62.9     | 62.9        |
| S054 RMSE        | 50.6  | 48.8   | 30.7   | 29.0     | 44.5        |
| S054 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S054 RMS_Methode | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S054 RMS_ICLABEL | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S055 RMSE        | 15.3  | 15.1   | 8.6    | 8.8      | 14.2        |
| S055 RMS_Brut    | 14.6  | 14.6   | 14.6   | 14.6     | 14.6        |
| S055 RMS_Methode | 14.6  | 14.6   | 14.6   | 14.6     | 14.6        |
| S055 RMS_ICLABEL | 14.6  | 14.6   | 14.6   | 14.6     | 14.6        |
| S056 RMSE        | 57.0  | 55.6   | 37.4   | 37.1     | 49.6        |
| S056 RMS_Brut    | 61.0  | 61.0   | 61.0   | 61.0     | 61.0        |
| S056 RMS_Methode | 61.0  | 61.0   | 61.0   | 61.0     | 61.0        |
| S056 RMS_ICLABEL | 61.0  | 61.0   | 61.0   | 61.0     | 61.0        |
| S057 RMSE        | 36.2  | 36.0   | 24.8   | 24.3     | 30.0        |
| S057 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S057 RMS_Methode | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S057 RMS_ICLABEL | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S058 RMSE        | 30.7  | 30.5   | 26.3   | 26.6     | 29.7        |
| S058 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S058 RMS_Methode | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S058 RMS_ICLABEL | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S059 RMSE        | 41.8  | 41.0   | 33.0   | 33.0     | 37.1        |
| S059 RMS_Brut    | 39.1  | 39.1   | 39.1   | 39.1     | 39.1        |
| S059 RMS_Methode | 39.1  | 39.1   | 39.1   | 39.1     | 39.1        |
| S059 RMS_ICLABEL | 39.1  | 39.1   | 39.1   | 39.1     | 39.1        |
| S060 RMSE        | 48.9  | 47.7   | 25.8   | 24.7     | 40.5        |
| S060 RMS_Brut    | 49.3  | 49.3   | 49.3   | 49.3     | 49.3        |
| S060 RMS_Methode | 49.3  | 49.3   | 49.3   | 49.3     | 49.3        |
| S060 RMS_ICLABEL | 49.3  | 49.3   | 49.3   | 49.3     | 49.3        |
| S061 RMSE        | 64.9  | 62.5   | 35.3   | 35.9     | 57.8        |
| S061 RMS_Brut    | 65.7  | 65.7   | 65.7   | 65.7     | 65.7        |
| S061 RMS_Methode | 65.7  | 65.7   | 65.7   | 65.7     | 65.7        |
| S061 RMS_ICLABEL | 65.7  | 65.7   | 65.7   | 65.7     | 65.7        |
| S062 RMSE        | 27.7  | 27.5   | 15.1   | 14.6     | 24.7        |
| S062 RMS_Brut    | 28.0  | 28.0   | 28.0   | 28.0     | 28.0        |
| S062 RMS_Methode | 28.0  | 28.0   | 28.0   | 28.0     | 28.0        |
| S062 RMS_ICLABEL | 28.0  | 28.0   | 28.0   | 28.0     | 28.0        |
| S063 RMSE        | 28.1  | 27.5   | 22.4   | 22.0     | 25.6        |
| S063 RMS_Brut    | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S063 RMS_Methode | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S063 RMS_ICLABEL | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S064 RMSE        | 13.6  | 13.4   | 8.0    | 7.9      | 11.5        |
| S064 RMS_Brut    | 13.4  | 13.4   | 13.4   | 13.4     | 13.4        |
| S064 RMS_Methode | 13.4  | 13.4   | 13.4   | 13.4     | 13.4        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S064 RMS_ICLABEL | 13.4  | 13.4   | 13.4   | 13.4     | 13.4        |
| S065 RMSE        | 33.1  | 32.7   | 23.3   | 23.9     | 29.0        |
| S065 RMS_Brut    | 32.6  | 32.6   | 32.6   | 32.6     | 32.6        |
| S065 RMS_Methode | 32.6  | 32.6   | 32.6   | 32.6     | 32.6        |
| S065 RMS_ICLABEL | 32.6  | 32.6   | 32.6   | 32.6     | 32.6        |
| S066 RMSE        | 26.8  | 25.9   | 11.7   | 11.6     | 20.3        |
| S066 RMS_Brut    | 28.9  | 28.9   | 28.9   | 28.9     | 28.9        |
| S066 RMS_Methode | 28.9  | 28.9   | 28.9   | 28.9     | 28.9        |
| S066 RMS_ICLABEL | 28.9  | 28.9   | 28.9   | 28.9     | 28.9        |
| S067 RMSE        | 12.8  | 12.9   | 8.9    | 9.5      | 11.9        |
| S067 RMS_Brut    | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S067 RMS_Methode | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S067 RMS_ICLABEL | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S068 RMSE        | 44.4  | 44.2   | 32.4   | 32.8     | 39.3        |
| S068 RMS_Brut    | 39.6  | 39.6   | 39.6   | 39.6     | 39.6        |
| S068 RMS_Methode | 39.6  | 39.6   | 39.6   | 39.6     | 39.6        |
| S068 RMS_ICLABEL | 39.6  | 39.6   | 39.6   | 39.6     | 39.6        |
| S069 RMSE        | 56.8  | 60.3   | 24.4   | 40.7     | 59.4        |
| S069 RMS_Brut    | 52.8  | 52.8   | 52.8   | 52.8     | 52.8        |
| S069 RMS_Methode | 52.8  | 52.8   | 52.8   | 52.8     | 52.8        |
| S069 RMS_ICLABEL | 52.8  | 52.8   | 52.8   | 52.8     | 52.8        |
| S070 RMSE        | 43.2  | 42.6   | 26.3   | 25.8     | 33.8        |
| S070 RMS_Brut    | 43.6  | 43.6   | 43.6   | 43.6     | 43.6        |
| S070 RMS_Methode | 43.6  | 43.6   | 43.6   | 43.6     | 43.6        |
| S070 RMS_ICLABEL | 43.6  | 43.6   | 43.6   | 43.6     | 43.6        |
| S071 RMSE        | 31.0  | 30.9   | 22.1   | 22.0     | 26.4        |
| S071 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S071 RMS_Methode | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S071 RMS_ICLABEL | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S072 RMSE        | 11.2  | 11.1   | 7.0    | 7.3      | 9.8         |
| S072 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S072 RMS_Methode | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S072 RMS_ICLABEL | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S073 RMSE        | 13.7  | 13.5   | 9.3    | 9.5      | 11.7        |
| S073 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S073 RMS_Methode | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S073 RMS_ICLABEL | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S074 RMSE        | 10.6  | 10.0   | 4.5    | 4.3      | 6.7         |
| S074 RMS_Brut    | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S074 RMS_Methode | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S074 RMS_ICLABEL | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S075 RMSE        | 65.1  | 63.0   | 29.8   | 31.3     | 51.9        |
| S075 RMS_Brut    | 68.1  | 68.1   | 68.1   | 68.1     | 68.1        |
| S075 RMS_Methode | 68.1  | 68.1   | 68.1   | 68.1     | 68.1        |
| S075 RMS_ICLABEL | 68.1  | 68.1   | 68.1   | 68.1     | 68.1        |
| S076 RMSE        | 10.6  | 10.2   | 6.6    | 6.8      | 9.2         |
| S076 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S076 RMS_Methode | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S076 RMS_ICLABEL | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S077 RMSE        | 59.4  | 59.2   | 37.0   | 36.8     | 49.0        |
| S077 RMS_Brut    | 55.8  | 55.8   | 55.8   | 55.8     | 55.8        |
| S077 RMS_Methode | 55.8  | 55.8   | 55.8   | 55.8     | 55.8        |
| S077 RMS_ICLABEL | 55.8  | 55.8   | 55.8   | 55.8     | 55.8        |
| S078 RMSE        | 24.7  | 24.0   | 11.9   | 11.9     | 19.8        |
| S078 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S078 RMS_Methode | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S078 RMS_ICLABEL | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S079 RMSE        | 43.2  | 42.1   | 20.7   | 26.2     | 33.3        |
| S079 RMS_Brut    | 44.1  | 44.1   | 44.1   | 44.1     | 44.1        |
| S079 RMS_Methode | 44.1  | 44.1   | 44.1   | 44.1     | 44.1        |
| S079 RMS_ICLABEL | 44.1  | 44.1   | 44.1   | 44.1     | 44.1        |
| S080 RMSE        | 32.7  | 32.1   | 17.2   | 18.2     | 28.1        |
| S080 RMS_Brut    | 32.0  | 32.0   | 32.0   | 32.0     | 32.0        |
| S080 RMS_Methode | 32.0  | 32.0   | 32.0   | 32.0     | 32.0        |
| S080 RMS_ICLABEL | 32.0  | 32.0   | 32.0   | 32.0     | 32.0        |
| S081 RMSE        | 24.6  | 24.3   | 13.1   | 12.8     | 19.0        |
| S081 RMS_Brut    | 22.8  | 22.8   | 22.8   | 22.8     | 22.8        |
| S081 RMS_Methode | 22.8  | 22.8   | 22.8   | 22.8     | 22.8        |
| S081 RMS_ICLABEL | 22.8  | 22.8   | 22.8   | 22.8     | 22.8        |
| S082 RMSE        | 23.9  | 23.2   | 16.0   | 15.7     | 19.6        |
| S082 RMS_Brut    | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S082 RMS_Methode | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S082 RMS_ICLABEL | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S083 RMSE        | 44.0  | 43.1   | 21.8   | 23.6     | 36.6        |
| S083 RMS_Brut    | 45.6  | 45.6   | 45.6   | 45.6     | 45.6        |
| S083 RMS_Methode | 45.6  | 45.6   | 45.6   | 45.6     | 45.6        |
| S083 RMS_ICLABEL | 45.6  | 45.6   | 45.6   | 45.6     | 45.6        |
| S084 RMSE        | 31.3  | 30.7   | 19.9   | 21.4     | 26.5        |
| S084 RMS_Brut    | 33.0  | 33.0   | 33.0   | 33.0     | 33.0        |
| S084 RMS_Methode | 33.0  | 33.0   | 33.0   | 33.0     | 33.0        |
| S084 RMS_ICLABEL | 33.0  | 33.0   | 33.0   | 33.0     | 33.0        |
| S085 RMSE        | 23.6  | 23.3   | 13.8   | 15.3     | 21.2        |
| S085 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S085 RMS_Methode | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S085 RMS_ICLABEL | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S086 RMSE        | 67.3  | 65.0   | 40.3   | 43.3     | 52.7        |
| S086 RMS_Brut    | 59.4  | 59.4   | 59.4   | 59.4     | 59.4        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S086 RMS_Methode | 59.4  | 59.4   | 59.4   | 59.4     | 59.4        |
| S086 RMS_ICLABEL | 59.4  | 59.4   | 59.4   | 59.4     | 59.4        |
| S087 RMSE        | 37.1  | 35.9   | 25.7   | 24.5     | 29.8        |
| S087 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S087 RMS_Methode | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S087 RMS_ICLABEL | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S088 RMSE        | 67.4  | 71.3   | 30.7   | 40.1     | 68.8        |
| S088 RMS_Brut    | 64.4  | 64.4   | 64.4   | 64.4     | 64.4        |
| S088 RMS_Methode | 64.4  | 64.4   | 64.4   | 64.4     | 64.4        |
| S088 RMS_ICLABEL | 64.4  | 64.4   | 64.4   | 64.4     | 64.4        |
| S089 RMSE        | 8.8   | 8.6    | 4.2    | 3.9      | 7.0         |
| S089 RMS_Brut    | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S089 RMS_Methode | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S089 RMS_ICLABEL | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S090 RMSE        | 58.2  | 58.0   | 48.6   | 47.8     | 57.5        |
| S090 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S090 RMS_Methode | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S090 RMS_ICLABEL | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S091 RMSE        | 98.9  | 95.9   | 90.9   | 91.0     | 99.3        |
| S091 RMS_Brut    | 84.1  | 84.1   | 84.1   | 84.1     | 84.1        |
| S091 RMS_Methode | 84.1  | 84.1   | 84.1   | 84.1     | 84.1        |
| S091 RMS_ICLABEL | 84.1  | 84.1   | 84.1   | 84.1     | 84.1        |
| S092 RMSE        | 37.7  | 37.2   | 23.9   | 23.5     | 29.5        |
| S092 RMS_Brut    | 40.3  | 40.3   | 40.3   | 40.3     | 40.3        |
| S092 RMS_Methode | 40.3  | 40.3   | 40.3   | 40.3     | 40.3        |
| S092 RMS_ICLABEL | 40.3  | 40.3   | 40.3   | 40.3     | 40.3        |
| S093 RMSE        | 22.5  | 22.8   | 15.6   | 16.8     | 19.2        |
| S093 RMS_Brut    | 21.8  | 21.8   | 21.8   | 21.8     | 21.8        |
| S093 RMS_Methode | 21.8  | 21.8   | 21.8   | 21.8     | 21.8        |
| S093 RMS_ICLABEL | 21.8  | 21.8   | 21.8   | 21.8     | 21.8        |
| S094 RMSE        | 48.4  | 47.5   | 26.3   | 26.2     | 41.1        |
| S094 RMS_Brut    | 52.4  | 52.4   | 52.4   | 52.4     | 52.4        |
| S094 RMS_Methode | 52.4  | 52.4   | 52.4   | 52.4     | 52.4        |
| S094 RMS_ICLABEL | 52.4  | 52.4   | 52.4   | 52.4     | 52.4        |
| S095 RMSE        | 58.6  | 59.5   | 23.9   | 36.9     | 57.0        |
| S095 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S095 RMS_Methode | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S095 RMS_ICLABEL | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S096 RMSE        | 72.4  | 70.9   | 49.5   | 51.6     | 65.8        |
| S096 RMS_Brut    | 69.0  | 69.0   | 69.0   | 69.0     | 69.0        |
| S096 RMS_Methode | 69.0  | 69.0   | 69.0   | 69.0     | 69.0        |
| S096 RMS_ICLABEL | 69.0  | 69.0   | 69.0   | 69.0     | 69.0        |
| S097 RMSE        | 61.1  | 60.4   | 47.0   | 45.6     | 58.7        |
| S097 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S097 RMS_Methode | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S097 RMS_ICLABEL | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S098 RMSE        | 9.8   | 9.2    | 3.9    | 4.0      | 6.4         |
| S098 RMS_Brut    | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S098 RMS_Methode | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S098 RMS_ICLABEL | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S099 RMSE        | 59.9  | 57.8   | 50.2   | 52.5     | 59.2        |
| S099 RMS_Brut    | 54.6  | 54.6   | 54.6   | 54.6     | 54.6        |
| S099 RMS_Methode | 54.6  | 54.6   | 54.6   | 54.6     | 54.6        |
| S099 RMS_ICLABEL | 54.6  | 54.6   | 54.6   | 54.6     | 54.6        |
| S100 RMSE        | 48.0  | 47.4   | 32.7   | 32.7     | 42.3        |
| S100 RMS_Brut    | 45.9  | 45.9   | 45.9   | 45.9     | 45.9        |
| S100 RMS_Methode | 45.9  | 45.9   | 45.9   | 45.9     | 45.9        |
| S100 RMS_ICLABEL | 45.9  | 45.9   | 45.9   | 45.9     | 45.9        |
| S101 RMSE        | 35.4  | 35.1   | 23.4   | 23.5     | 28.9        |
| S101 RMS_Brut    | 32.3  | 32.3   | 32.3   | 32.3     | 32.3        |
| S101 RMS_Methode | 32.3  | 32.3   | 32.3   | 32.3     | 32.3        |
| S101 RMS_ICLABEL | 32.3  | 32.3   | 32.3   | 32.3     | 32.3        |
| S102 RMSE        | 11.5  | 11.4   | 6.9    | 6.7      | 8.9         |
| S102 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S102 RMS_Methode | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S102 RMS_ICLABEL | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S103 RMSE        | 48.1  | 47.1   | 34.1   | 33.3     | 39.2        |
| S103 RMS_Brut    | 48.8  | 48.8   | 48.8   | 48.8     | 48.8        |
| S103 RMS_Methode | 48.8  | 48.8   | 48.8   | 48.8     | 48.8        |
| S103 RMS_ICLABEL | 48.8  | 48.8   | 48.8   | 48.8     | 48.8        |
| S104 RMSE        | 30.0  | 29.2   | 14.2   | 14.1     | 25.7        |
| S104 RMS_Brut    | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S104 RMS_Methode | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S104 RMS_ICLABEL | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S105 RMSE        | 30.2  | 30.1   | 18.0   | 17.5     | 24.1        |
| S105 RMS_Brut    | 30.1  | 30.1   | 30.1   | 30.1     | 30.1        |
| S105 RMS_Methode | 30.1  | 30.1   | 30.1   | 30.1     | 30.1        |
| S105 RMS_ICLABEL | 30.1  | 30.1   | 30.1   | 30.1     | 30.1        |
| S106 RMSE        | 38.8  | 37.9   | 18.7   | 19.4     | 30.6        |
| S106 RMS_Brut    | 41.8  | 41.8   | 41.8   | 41.8     | 41.8        |
| S106 RMS_Methode | 41.8  | 41.8   | 41.8   | 41.8     | 41.8        |
| S106 RMS_ICLABEL | 41.8  | 41.8   | 41.8   | 41.8     | 41.8        |
| S107 RMSE        | 40.1  | 40.1   | 24.1   | 24.5     | 30.3        |
| S107 RMS_Brut    | 38.7  | 38.7   | 38.7   | 38.7     | 38.7        |
| S107 RMS_Methode | 38.7  | 38.7   | 38.7   | 38.7     | 38.7        |
| S107 RMS_ICLABEL | 38.7  | 38.7   | 38.7   | 38.7     | 38.7        |
| S108 RMSE        | 52.8  | 53.0   | 40.7   | 44.4     | 52.2        |

| Sujet               | DuoCL | GCTNet | ICUNet      | ICUNet++ | ICUNet_attn |
|---------------------|-------|--------|-------------|----------|-------------|
| S108 RMS_Brut       | 51.9  | 51.9   | 51.9        | 51.9     | 51.9        |
| S108 RMS_Methode    | 51.9  | 51.9   | 51.9        | 51.9     | 51.9        |
| S108 RMS_ICLABEL    | 51.9  | 51.9   | 51.9        | 51.9     | 51.9        |
| S109 RMSE           | 42.3  | 41.5   | 40.6        | 44.1     | 49.3        |
| S109 RMS_Brut       | 46.7  | 46.7   | 46.7        | 46.7     | 46.7        |
| S109 RMS_Methode    | 46.7  | 46.7   | 46.7        | 46.7     | 46.7        |
| S109 RMS_ICLABEL    | 46.7  | 46.7   | 46.7        | 46.7     | 46.7        |
| <b>Moyenne RMSE</b> | 36.5  | 36.0   | <b>23.1</b> | 23.8     | 31.9        |
| Moyenne RMS_Brut    | 35.8  | 35.8   | 35.8        | 35.8     | 35.8        |
| Moyenne RMS_Methode | 35.8  | 35.8   | 35.8        | 35.8     | 35.8        |
| Moyenne RMS_ICLABEL | 35.8  | 35.8   | 35.8        | 35.8     | 35.8        |

center

TABLE 44 – Récapitulatif — RMSE, RMS\_Brut, RMS\_Méthode et RMS de référence ICLabel (en  $\mu V$ ), les 109 sujets (4 lignes par sujet), par méthode — signaux de Nettoyé\_Normalisé\_2 (chaque essai recalé sur la moyenne/écart-type d’ICLabel du même sujet; ICLabel lui-même inchangé, « - » = non calculé).

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S001 RMSE        | 28.9  | 28.9   | 16.7   | 16.1     | 20.2        |
| S001 RMS_Brut    | 36.2  | 36.2   | 36.2   | 36.2     | 36.2        |
| S001 RMS_Methode | 26.5  | 26.5   | 26.5   | 26.5     | 26.5        |
| S001 RMS_ICLABEL | 26.5  | 26.5   | 26.5   | 26.5     | 26.5        |
| S002 RMSE        | 13.7  | 13.1   | 7.0    | 6.7      | 9.9         |
| S002 RMS_Brut    | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S002 RMS_Methode | 14.7  | 14.7   | 14.7   | 14.7     | 14.7        |
| S002 RMS_ICLABEL | 14.7  | 14.7   | 14.7   | 14.7     | 14.7        |
| S003 RMSE        | 27.9  | 27.7   | 20.5   | 20.0     | 25.7        |
| S003 RMS_Brut    | 54.1  | 54.1   | 54.1   | 54.1     | 54.1        |
| S003 RMS_Methode | 25.1  | 25.1   | 25.1   | 25.1     | 25.1        |
| S003 RMS_ICLABEL | 25.1  | 25.1   | 25.1   | 25.1     | 25.1        |
| S004 RMSE        | 28.1  | 29.2   | 27.5   | 31.1     | 33.2        |
| S004 RMS_Brut    | 32.7  | 32.7   | 32.7   | 32.7     | 32.7        |
| S004 RMS_Methode | 26.1  | 26.1   | 26.1   | 26.1     | 26.1        |
| S004 RMS_ICLABEL | 26.1  | 26.1   | 26.1   | 26.1     | 26.1        |
| S005 RMSE        | 17.3  | 16.8   | 12.6   | 13.4     | 17.0        |
| S005 RMS_Brut    | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S005 RMS_Methode | 16.7  | 16.7   | 16.7   | 16.7     | 16.7        |
| S005 RMS_ICLABEL | 16.7  | 16.7   | 16.7   | 16.7     | 16.7        |
| S006 RMSE        | 12.9  | 12.9   | 8.4    | 8.8      | 11.0        |
| S006 RMS_Brut    | 20.6  | 20.6   | 20.6   | 20.6     | 20.6        |
| S006 RMS_Methode | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S006 RMS_ICLABEL | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S007 RMSE        | 16.9  | 16.1   | 8.3    | 8.0      | 13.4        |
| S007 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S007 RMS_Methode | 21.3  | 21.3   | 21.3   | 21.3     | 21.3        |
| S007 RMS_ICLABEL | 21.3  | 21.3   | 21.3   | 21.3     | 21.3        |
| S008 RMSE        | 16.1  | 15.8   | 9.9    | 10.5     | 14.3        |
| S008 RMS_Brut    | 25.3  | 25.3   | 25.3   | 25.3     | 25.3        |
| S008 RMS_Methode | 16.1  | 16.1   | 16.1   | 16.1     | 16.1        |
| S008 RMS_ICLABEL | 16.1  | 16.1   | 16.1   | 16.1     | 16.1        |
| S009 RMSE        | 29.6  | 29.1   | 21.3   | 21.3     | 28.0        |
| S009 RMS_Brut    | 56.7  | 56.7   | 56.7   | 56.7     | 56.7        |
| S009 RMS_Methode | 27.0  | 27.0   | 27.0   | 27.0     | 27.0        |
| S009 RMS_ICLABEL | 27.0  | 27.0   | 27.0   | 27.0     | 27.0        |
| S010 RMSE        | 29.9  | 30.2   | 19.2   | 19.6     | 23.9        |
| S010 RMS_Brut    | 39.7  | 39.7   | 39.7   | 39.7     | 39.7        |
| S010 RMS_Methode | 27.6  | 27.6   | 27.6   | 27.6     | 27.6        |
| S010 RMS_ICLABEL | 27.6  | 27.6   | 27.6   | 27.6     | 27.6        |
| S011 RMSE        | 13.3  | 13.2   | 5.6    | 6.3      | 10.1        |
| S011 RMS_Brut    | 15.8  | 15.8   | 15.8   | 15.8     | 15.8        |
| S011 RMS_Methode | 13.3  | 13.3   | 13.3   | 13.3     | 13.3        |
| S011 RMS_ICLABEL | 13.3  | 13.3   | 13.3   | 13.3     | 13.3        |
| S012 RMSE        | 17.4  | 17.4   | 11.7   | 11.4     | 15.4        |
| S012 RMS_Brut    | 36.5  | 36.5   | 36.5   | 36.5     | 36.5        |
| S012 RMS_Methode | 15.7  | 15.7   | 15.7   | 15.7     | 15.7        |
| S012 RMS_ICLABEL | 15.7  | 15.7   | 15.7   | 15.7     | 15.7        |
| S013 RMSE        | 31.2  | 31.4   | 21.3   | 22.4     | 27.6        |
| S013 RMS_Brut    | 56.3  | 56.3   | 56.3   | 56.3     | 56.3        |
| S013 RMS_Methode | 29.6  | 29.6   | 29.6   | 29.6     | 29.6        |
| S013 RMS_ICLABEL | 29.6  | 29.6   | 29.6   | 29.6     | 29.6        |
| S014 RMSE        | 18.5  | 18.1   | 9.9    | 10.0     | 13.8        |
| S014 RMS_Brut    | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S014 RMS_Methode | 20.7  | 20.7   | 20.7   | 20.7     | 20.7        |
| S014 RMS_ICLABEL | 20.7  | 20.7   | 20.7   | 20.7     | 20.7        |
| S015 RMSE        | 32.1  | 30.3   | 14.0   | 14.9     | 23.8        |
| S015 RMS_Brut    | 42.3  | 42.3   | 42.3   | 42.3     | 42.3        |
| S015 RMS_Methode | 34.7  | 34.7   | 34.7   | 34.7     | 34.7        |
| S015 RMS_ICLABEL | 34.7  | 34.7   | 34.7   | 34.7     | 34.7        |
| S016 RMSE        | 13.4  | 13.1   | 10.1   | 9.7      | 11.9        |
| S016 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S016 RMS_Methode | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S016 RMS_ICLABEL | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S017 RMSE        | 40.7  | 39.0   | 30.3   | 30.2     | 39.1        |
| S017 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S017 RMS_Methode | 36.9  | 36.9   | 36.9   | 36.9     | 36.9        |
| S017 RMS_ICLABEL | 36.9  | 36.9   | 36.9   | 36.9     | 36.9        |
| S018 RMSE        | 6.6   | 6.5    | 4.7    | 4.5      | 5.9         |
| S018 RMS_Brut    | 13.8  | 13.8   | 13.8   | 13.8     | 13.8        |
| S018 RMS_Methode | 5.8   | 5.8    | 5.8    | 5.8      | 5.8         |
| S018 RMS_ICLABEL | 5.8   | 5.8    | 5.8    | 5.8      | 5.8         |
| S019 RMSE        | 25.2  | 24.9   | 15.6   | 16.7     | 22.4        |
| S019 RMS_Brut    | 37.4  | 37.4   | 37.4   | 37.4     | 37.4        |
| S019 RMS_Methode | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S019 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S020 RMSE        | 6.1   | 6.1    | 4.7    | 4.8      | 5.6         |
| S020 RMS_Brut    | 11.1  | 11.1   | 11.1   | 11.1     | 11.1        |
| S020 RMS_Methode | 5.4   | 5.4    | 5.4    | 5.4      | 5.4         |
| S020 RMS_ICLABEL | 5.4   | 5.4    | 5.4    | 5.4      | 5.4         |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S021 RMSE        | 13.2  | 13.0   | 8.8    | 8.9      | 11.9        |
| S021 RMS_Brut    | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S021 RMS_Methode | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S021 RMS_ICLABEL | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S022 RMSE        | 29.3  | 28.9   | 20.1   | 19.3     | 27.8        |
| S022 RMS_Brut    | 68.5  | 68.5   | 68.5   | 68.5     | 68.5        |
| S022 RMS_Methode | 27.7  | 27.7   | 27.7   | 27.7     | 27.7        |
| S022 RMS_ICLABEL | 27.7  | 27.7   | 27.7   | 27.7     | 27.7        |
| S023 RMSE        | 36.4  | 35.3   | 19.5   | 20.0     | 31.0        |
| S023 RMS_Brut    | 56.2  | 56.2   | 56.2   | 56.2     | 56.2        |
| S023 RMS_Methode | 37.1  | 37.1   | 37.1   | 37.1     | 37.1        |
| S023 RMS_ICLABEL | 37.1  | 37.1   | 37.1   | 37.1     | 37.1        |
| S024 RMSE        | 16.5  | 16.8   | 12.9   | 15.0     | 15.8        |
| S024 RMS_Brut    | 34.6  | 34.6   | 34.6   | 34.6     | 34.6        |
| S024 RMS_Methode | 14.4  | 14.4   | 14.4   | 14.4     | 14.4        |
| S024 RMS_ICLABEL | 14.4  | 14.4   | 14.4   | 14.4     | 14.4        |
| S025 RMSE        | 8.4   | 7.9    | 5.1    | 5.0      | 6.7         |
| S025 RMS_Brut    | 11.8  | 11.8   | 11.8   | 11.8     | 11.8        |
| S025 RMS_Methode | 9.1   | 9.1    | 9.1    | 9.1      | 9.1         |
| S025 RMS_ICLABEL | 9.1   | 9.1    | 9.1    | 9.1      | 9.1         |
| S026 RMSE        | 5.7   | 5.5    | 4.8    | 4.8      | 5.3         |
| S026 RMS_Brut    | 11.9  | 11.9   | 11.9   | 11.9     | 11.9        |
| S026 RMS_Methode | 4.9   | 4.9    | 4.9    | 4.9      | 4.9         |
| S026 RMS_ICLABEL | 4.9   | 4.9    | 4.9    | 4.9      | 4.9         |
| S027 RMSE        | 27.9  | 27.0   | 14.9   | 15.8     | 22.7        |
| S027 RMS_Brut    | 41.7  | 41.7   | 41.7   | 41.7     | 41.7        |
| S027 RMS_Methode | 29.3  | 29.3   | 29.3   | 29.3     | 29.3        |
| S027 RMS_ICLABEL | 29.3  | 29.3   | 29.3   | 29.3     | 29.3        |
| S028 RMSE        | 39.2  | 38.3   | 19.3   | 19.7     | 31.0        |
| S028 RMS_Brut    | 55.1  | 55.1   | 55.1   | 55.1     | 55.1        |
| S028 RMS_Methode | 40.2  | 40.2   | 40.2   | 40.2     | 40.2        |
| S028 RMS_ICLABEL | 40.2  | 40.2   | 40.2   | 40.2     | 40.2        |
| S029 RMSE        | 11.2  | 10.9   | 4.8    | 4.9      | 8.1         |
| S029 RMS_Brut    | 14.5  | 14.5   | 14.5   | 14.5     | 14.5        |
| S029 RMS_Methode | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S029 RMS_ICLABEL | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S030 RMSE        | 9.5   | 9.4    | 4.5    | 4.7      | 7.6         |
| S030 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S030 RMS_Methode | 9.3   | 9.3    | 9.3    | 9.3      | 9.3         |
| S030 RMS_ICLABEL | 9.3   | 9.3    | 9.3    | 9.3      | 9.3         |
| S031 RMSE        | 24.3  | 23.8   | 12.1   | 11.9     | 16.7        |
| S031 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S031 RMS_Methode | 27.9  | 27.9   | 27.9   | 27.9     | 27.9        |
| S031 RMS_ICLABEL | 27.9  | 27.9   | 27.9   | 27.9     | 27.9        |
| S032 RMSE        | 35.9  | 36.3   | 20.7   | 23.4     | 32.6        |
| S032 RMS_Brut    | 53.9  | 53.9   | 53.9   | 53.9     | 53.9        |
| S032 RMS_Methode | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S032 RMS_ICLABEL | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S033 RMSE        | 14.8  | 14.4   | 8.0    | 7.8      | 12.4        |
| S033 RMS_Brut    | 25.8  | 25.8   | 25.8   | 25.8     | 25.8        |
| S033 RMS_Methode | 14.8  | 14.8   | 14.8   | 14.8     | 14.8        |
| S033 RMS_ICLABEL | 14.8  | 14.8   | 14.8   | 14.8     | 14.8        |
| S034 RMSE        | 10.3  | 10.2   | 4.4    | 4.8      | 7.1         |
| S034 RMS_Brut    | 12.4  | 12.4   | 12.4   | 12.4     | 12.4        |
| S034 RMS_Methode | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S034 RMS_ICLABEL | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S035 RMSE        | 11.8  | 11.5   | 5.6    | 5.4      | 8.8         |
| S035 RMS_Brut    | 18.8  | 18.8   | 18.8   | 18.8     | 18.8        |
| S035 RMS_Methode | 13.1  | 13.1   | 13.1   | 13.1     | 13.1        |
| S035 RMS_ICLABEL | 13.1  | 13.1   | 13.1   | 13.1     | 13.1        |
| S036 RMSE        | 30.9  | 29.8   | 20.8   | 20.3     | 26.2        |
| S036 RMS_Brut    | 53.4  | 53.4   | 53.4   | 53.4     | 53.4        |
| S036 RMS_Methode | 31.4  | 31.4   | 31.4   | 31.4     | 31.4        |
| S036 RMS_ICLABEL | 31.4  | 31.4   | 31.4   | 31.4     | 31.4        |
| S037 RMSE        | 6.7   | 6.5    | 3.7    | 3.9      | 5.4         |
| S037 RMS_Brut    | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S037 RMS_Methode | 6.6   | 6.6    | 6.6    | 6.6      | 6.6         |
| S037 RMS_ICLABEL | 6.6   | 6.6    | 6.6    | 6.6      | 6.6         |
| S038 RMSE        | 21.9  | 21.8   | 9.9    | 10.4     | 14.4        |
| S038 RMS_Brut    | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S038 RMS_Methode | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S038 RMS_ICLABEL | 22.4  | 22.4   | 22.4   | 22.4     | 22.4        |
| S039 RMSE        | 40.3  | 39.9   | 19.2   | 19.7     | 28.9        |
| S039 RMS_Brut    | 51.4  | 51.4   | 51.4   | 51.4     | 51.4        |
| S039 RMS_Methode | 40.6  | 40.6   | 40.6   | 40.6     | 40.6        |
| S039 RMS_ICLABEL | 40.6  | 40.6   | 40.6   | 40.6     | 40.6        |
| S040 RMSE        | 22.0  | 22.1   | 15.1   | 16.5     | 19.7        |
| S040 RMS_Brut    | 26.4  | 26.4   | 26.4   | 26.4     | 26.4        |
| S040 RMS_Methode | 20.2  | 20.2   | 20.2   | 20.2     | 20.2        |
| S040 RMS_ICLABEL | 20.2  | 20.2   | 20.2   | 20.2     | 20.2        |
| S041 RMSE        | 7.8   | 7.5    | 3.9    | 4.0      | 6.2         |
| S041 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S041 RMS_Methode | 7.9   | 7.9    | 7.9    | 7.9      | 7.9         |
| S041 RMS_ICLABEL | 7.9   | 7.9    | 7.9    | 7.9      | 7.9         |
| S042 RMSE        | 9.0   | 8.7    | 3.4    | 3.2      | 5.9         |
| S042 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S042 RMS_Methode | 11.0  | 11.0   | 11.0   | 11.0     | 11.0        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S042 RMS_ICLABEL | 11.0  | 11.0   | 11.0   | 11.0     | 11.0        |
| S043 RMSE        | 29.0  | 28.9   | 21.0   | 22.3     | 26.6        |
| S043 RMS_Brut    | 48.0  | 48.0   | 48.0   | 48.0     | 48.0        |
| S043 RMS_Methode | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S043 RMS_ICLABEL | 26.6  | 26.6   | 26.6   | 26.6     | 26.6        |
| S044 RMSE        | 36.5  | 34.6   | 23.5   | 21.8     | 32.0        |
| S044 RMS_Brut    | 67.9  | 67.9   | 67.9   | 67.9     | 67.9        |
| S044 RMS_Methode | 43.0  | 43.0   | 43.0   | 43.0     | 43.0        |
| S044 RMS_ICLABEL | 43.0  | 43.0   | 43.0   | 43.0     | 43.0        |
| S045 RMSE        | 32.0  | 31.2   | 21.2   | 20.3     | 26.0        |
| S045 RMS_Brut    | 46.4  | 46.4   | 46.4   | 46.4     | 46.4        |
| S045 RMS_Methode | 29.0  | 29.0   | 29.0   | 29.0     | 29.0        |
| S045 RMS_ICLABEL | 29.0  | 29.0   | 29.0   | 29.0     | 29.0        |
| S046 RMSE        | 26.4  | 26.1   | 15.6   | 15.1     | 20.5        |
| S046 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S046 RMS_Methode | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S046 RMS_ICLABEL | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S047 RMSE        | 7.8   | 7.6    | 4.7    | 4.6      | 6.7         |
| S047 RMS_Brut    | 12.0  | 12.0   | 12.0   | 12.0     | 12.0        |
| S047 RMS_Methode | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S047 RMS_ICLABEL | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S048 RMSE        | 45.8  | 42.3   | 20.6   | 20.7     | 32.8        |
| S048 RMS_Brut    | 63.3  | 63.3   | 63.3   | 63.3     | 63.3        |
| S048 RMS_Methode | 58.2  | 58.2   | 58.2   | 58.2     | 58.2        |
| S048 RMS_ICLABEL | 58.2  | 58.2   | 58.2   | 58.2     | 58.2        |
| S049 RMSE        | 36.5  | 35.7   | 17.0   | 17.4     | 27.1        |
| S049 RMS_Brut    | 47.9  | 47.9   | 47.9   | 47.9     | 47.9        |
| S049 RMS_Methode | 36.0  | 36.0   | 36.0   | 36.0     | 36.0        |
| S049 RMS_ICLABEL | 36.0  | 36.0   | 36.0   | 36.0     | 36.0        |
| S050 RMSE        | 11.4  | 11.1   | 8.3    | 8.3      | 10.1        |
| S050 RMS_Brut    | 17.3  | 17.3   | 17.3   | 17.3     | 17.3        |
| S050 RMS_Methode | 10.2  | 10.2   | 10.2   | 10.2     | 10.2        |
| S050 RMS_ICLABEL | 10.2  | 10.2   | 10.2   | 10.2     | 10.2        |
| S051 RMSE        | 4.6   | 4.5    | 3.2    | 3.3      | 4.3         |
| S051 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S051 RMS_Methode | 4.1   | 4.1    | 4.1    | 4.1      | 4.1         |
| S051 RMS_ICLABEL | 4.1   | 4.1    | 4.1    | 4.1      | 4.1         |
| S052 RMSE        | 34.4  | 33.6   | 17.5   | 17.9     | 25.3        |
| S052 RMS_Brut    | 45.7  | 45.7   | 45.7   | 45.7     | 45.7        |
| S052 RMS_Methode | 38.6  | 38.6   | 38.6   | 38.6     | 38.6        |
| S052 RMS_ICLABEL | 38.6  | 38.6   | 38.6   | 38.6     | 38.6        |
| S053 RMSE        | 34.1  | 33.8   | 21.0   | 21.3     | 31.0        |
| S053 RMS_Brut    | 62.9  | 62.9   | 62.9   | 62.9     | 62.9        |
| S053 RMS_Methode | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S053 RMS_ICLABEL | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S054 RMSE        | 31.0  | 29.8   | 17.9   | 17.0     | 25.9        |
| S054 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S054 RMS_Methode | 35.9  | 35.9   | 35.9   | 35.9     | 35.9        |
| S054 RMS_ICLABEL | 35.9  | 35.9   | 35.9   | 35.9     | 35.9        |
| S055 RMSE        | 8.6   | 8.5    | 4.8    | 5.0      | 7.9         |
| S055 RMS_Brut    | 14.6  | 14.6   | 14.6   | 14.6     | 14.6        |
| S055 RMS_Methode | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S055 RMS_ICLABEL | 8.3   | 8.3    | 8.3    | 8.3      | 8.3         |
| S056 RMSE        | 37.0  | 36.1   | 24.2   | 24.1     | 31.9        |
| S056 RMS_Brut    | 61.0  | 61.0   | 61.0   | 61.0     | 61.0        |
| S056 RMS_Methode | 39.8  | 39.8   | 39.8   | 39.8     | 39.8        |
| S056 RMS_ICLABEL | 39.8  | 39.8   | 39.8   | 39.8     | 39.8        |
| S057 RMSE        | 21.7  | 21.6   | 14.8   | 14.5     | 17.9        |
| S057 RMS_Brut    | 33.1  | 33.1   | 33.1   | 33.1     | 33.1        |
| S057 RMS_Methode | 19.9  | 19.9   | 19.9   | 19.9     | 19.9        |
| S057 RMS_ICLABEL | 19.9  | 19.9   | 19.9   | 19.9     | 19.9        |
| S058 RMSE        | 14.4  | 14.3   | 12.3   | 12.4     | 13.9        |
| S058 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S058 RMS_Methode | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S058 RMS_ICLABEL | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S059 RMSE        | 24.8  | 24.3   | 19.6   | 19.5     | 21.7        |
| S059 RMS_Brut    | 39.1  | 39.1   | 39.1   | 39.1     | 39.1        |
| S059 RMS_Methode | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S059 RMS_ICLABEL | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S060 RMSE        | 32.4  | 31.6   | 16.8   | 16.2     | 26.3        |
| S060 RMS_Brut    | 49.3  | 49.3   | 49.3   | 49.3     | 49.3        |
| S060 RMS_Methode | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S060 RMS_ICLABEL | 32.9  | 32.9   | 32.9   | 32.9     | 32.9        |
| S061 RMSE        | 42.8  | 41.2   | 22.9   | 23.5     | 37.8        |
| S061 RMS_Brut    | 65.7  | 65.7   | 65.7   | 65.7     | 65.7        |
| S061 RMS_Methode | 43.4  | 43.4   | 43.4   | 43.4     | 43.4        |
| S061 RMS_ICLABEL | 43.4  | 43.4   | 43.4   | 43.4     | 43.4        |
| S062 RMSE        | 15.1  | 15.0   | 8.0    | 7.8      | 12.8        |
| S062 RMS_Brut    | 28.0  | 28.0   | 28.0   | 28.0     | 28.0        |
| S062 RMS_Methode | 15.2  | 15.2   | 15.2   | 15.2     | 15.2        |
| S062 RMS_ICLABEL | 15.2  | 15.2   | 15.2   | 15.2     | 15.2        |
| S063 RMSE        | 13.8  | 13.5   | 11.0   | 10.9     | 12.6        |
| S063 RMS_Brut    | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S063 RMS_Methode | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S063 RMS_ICLABEL | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S064 RMSE        | 8.6   | 8.5    | 4.9    | 4.8      | 7.1         |
| S064 RMS_Brut    | 13.4  | 13.4   | 13.4   | 13.4     | 13.4        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S064 RMS_Methode | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S064 RMS_ICLABEL | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S065 RMSE        | 23.6  | 23.3   | 16.8   | 17.1     | 20.7        |
| S065 RMS_Brut    | 32.6  | 32.6   | 32.6   | 32.6     | 32.6        |
| S065 RMS_Methode | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S065 RMS_ICLABEL | 23.3  | 23.3   | 23.3   | 23.3     | 23.3        |
| S066 RMSE        | 19.9  | 19.2   | 8.4    | 8.3      | 14.2        |
| S066 RMS_Brut    | 28.9  | 28.9   | 28.9   | 28.9     | 28.9        |
| S066 RMS_Methode | 21.9  | 21.9   | 21.9   | 21.9     | 21.9        |
| S066 RMS_ICLABEL | 21.9  | 21.9   | 21.9   | 21.9     | 21.9        |
| S067 RMSE        | 8.1   | 8.2    | 5.6    | 6.0      | 7.6         |
| S067 RMS_Brut    | 11.5  | 11.5   | 11.5   | 11.5     | 11.5        |
| S067 RMS_Methode | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S067 RMS_ICLABEL | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S068 RMSE        | 25.9  | 25.7   | 18.8   | 19.0     | 22.6        |
| S068 RMS_Brut    | 39.6  | 39.6   | 39.6   | 39.6     | 39.6        |
| S068 RMS_Methode | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S068 RMS_ICLABEL | 23.2  | 23.2   | 23.2   | 23.2     | 23.2        |
| S069 RMSE        | 47.6  | 50.6   | 20.3   | 34.0     | 49.8        |
| S069 RMS_Brut    | 52.8  | 52.8   | 52.8   | 52.8     | 52.8        |
| S069 RMS_Methode | 44.2  | 44.2   | 44.2   | 44.2     | 44.2        |
| S069 RMS_ICLABEL | 44.2  | 44.2   | 44.2   | 44.2     | 44.2        |
| S070 RMSE        | 30.6  | 30.1   | 18.7   | 18.3     | 23.7        |
| S070 RMS_Brut    | 43.6  | 43.6   | 43.6   | 43.6     | 43.6        |
| S070 RMS_Methode | 31.0  | 31.0   | 31.0   | 31.0     | 31.0        |
| S070 RMS_ICLABEL | 31.0  | 31.0   | 31.0   | 31.0     | 31.0        |
| S071 RMSE        | 20.3  | 20.2   | 14.4   | 14.3     | 17.2        |
| S071 RMS_Brut    | 28.7  | 28.7   | 28.7   | 28.7     | 28.7        |
| S071 RMS_Methode | 18.9  | 18.9   | 18.9   | 18.9     | 18.9        |
| S071 RMS_ICLABEL | 18.9  | 18.9   | 18.9   | 18.9     | 18.9        |
| S072 RMSE        | 6.5   | 6.4    | 3.9    | 4.1      | 5.4         |
| S072 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S072 RMS_Methode | 6.1   | 6.1    | 6.1    | 6.1      | 6.1         |
| S072 RMS_ICLABEL | 6.1   | 6.1    | 6.1    | 6.1      | 6.1         |
| S073 RMSE        | 8.0   | 7.9    | 5.5    | 5.5      | 6.8         |
| S073 RMS_Brut    | 12.6  | 12.6   | 12.6   | 12.6     | 12.6        |
| S073 RMS_Methode | 7.4   | 7.4    | 7.4    | 7.4      | 7.4         |
| S073 RMS_ICLABEL | 7.4   | 7.4    | 7.4    | 7.4      | 7.4         |
| S074 RMSE        | 9.9   | 9.4    | 4.2    | 4.0      | 6.2         |
| S074 RMS_Brut    | 11.3  | 11.3   | 11.3   | 11.3     | 11.3        |
| S074 RMS_Methode | 10.6  | 10.6   | 10.6   | 10.6     | 10.6        |
| S074 RMS_ICLABEL | 10.6  | 10.6   | 10.6   | 10.6     | 10.6        |
| S075 RMSE        | 50.5  | 48.9   | 22.6   | 23.9     | 39.1        |
| S075 RMS_Brut    | 68.1  | 68.1   | 68.1   | 68.1     | 68.1        |
| S075 RMS_Methode | 53.5  | 53.5   | 53.5   | 53.5     | 53.5        |
| S075 RMS_ICLABEL | 53.5  | 53.5   | 53.5   | 53.5     | 53.5        |
| S076 RMSE        | 7.6   | 7.3    | 4.7    | 4.8      | 6.5         |
| S076 RMS_Brut    | 10.5  | 10.5   | 10.5   | 10.5     | 10.5        |
| S076 RMS_Methode | 7.5   | 7.5    | 7.5    | 7.5      | 7.5         |
| S076 RMS_ICLABEL | 7.5   | 7.5    | 7.5    | 7.5      | 7.5         |
| S077 RMSE        | 35.3  | 35.3   | 22.1   | 21.9     | 28.8        |
| S077 RMS_Brut    | 55.8  | 55.8   | 55.8   | 55.8     | 55.8        |
| S077 RMS_Methode | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S077 RMS_ICLABEL | 33.2  | 33.2   | 33.2   | 33.2     | 33.2        |
| S078 RMSE        | 17.6  | 17.2   | 8.2    | 8.2      | 13.5        |
| S078 RMS_Brut    | 25.6  | 25.6   | 25.6   | 25.6     | 25.6        |
| S078 RMS_Methode | 18.5  | 18.5   | 18.5   | 18.5     | 18.5        |
| S078 RMS_ICLABEL | 18.5  | 18.5   | 18.5   | 18.5     | 18.5        |
| S079 RMSE        | 41.3  | 40.3   | 19.7   | 25.2     | 31.9        |
| S079 RMS_Brut    | 44.1  | 44.1   | 44.1   | 44.1     | 44.1        |
| S079 RMS_Methode | 42.1  | 42.1   | 42.1   | 42.1     | 42.1        |
| S079 RMS_ICLABEL | 42.1  | 42.1   | 42.1   | 42.1     | 42.1        |
| S080 RMSE        | 22.4  | 22.1   | 11.8   | 12.5     | 18.8        |
| S080 RMS_Brut    | 32.0  | 32.0   | 32.0   | 32.0     | 32.0        |
| S080 RMS_Methode | 22.2  | 22.2   | 22.2   | 22.2     | 22.2        |
| S080 RMS_ICLABEL | 22.2  | 22.2   | 22.2   | 22.2     | 22.2        |
| S081 RMSE        | 19.4  | 19.1   | 10.2   | 10.0     | 14.8        |
| S081 RMS_Brut    | 22.8  | 22.8   | 22.8   | 22.8     | 22.8        |
| S081 RMS_Methode | 17.9  | 17.9   | 17.9   | 17.9     | 17.9        |
| S081 RMS_ICLABEL | 17.9  | 17.9   | 17.9   | 17.9     | 17.9        |
| S082 RMSE        | 17.8  | 17.3   | 11.9   | 11.7     | 14.6        |
| S082 RMS_Brut    | 25.0  | 25.0   | 25.0   | 25.0     | 25.0        |
| S082 RMS_Methode | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S082 RMS_ICLABEL | 18.7  | 18.7   | 18.7   | 18.7     | 18.7        |
| S083 RMSE        | 32.9  | 32.2   | 16.1   | 17.6     | 26.8        |
| S083 RMS_Brut    | 45.6  | 45.6   | 45.6   | 45.6     | 45.6        |
| S083 RMS_Methode | 34.2  | 34.2   | 34.2   | 34.2     | 34.2        |
| S083 RMS_ICLABEL | 34.2  | 34.2   | 34.2   | 34.2     | 34.2        |
| S084 RMSE        | 27.3  | 26.9   | 17.4   | 18.7     | 23.0        |
| S084 RMS_Brut    | 33.0  | 33.0   | 33.0   | 33.0     | 33.0        |
| S084 RMS_Methode | 28.8  | 28.8   | 28.8   | 28.8     | 28.8        |
| S084 RMS_ICLABEL | 28.8  | 28.8   | 28.8   | 28.8     | 28.8        |
| S085 RMSE        | 17.6  | 17.4   | 10.2   | 11.3     | 15.5        |
| S085 RMS_Brut    | 24.8  | 24.8   | 24.8   | 24.8     | 24.8        |
| S085 RMS_Methode | 18.6  | 18.6   | 18.6   | 18.6     | 18.6        |
| S085 RMS_ICLABEL | 18.6  | 18.6   | 18.6   | 18.6     | 18.6        |
| S086 RMSE        | 47.1  | 45.8   | 28.4   | 30.4     | 37.0        |

| Sujet            | DuoCL | GCTNet | ICUNet | ICUNet++ | ICUNet_attn |
|------------------|-------|--------|--------|----------|-------------|
| S086 RMS_Brut    | 59.4  | 59.4   | 59.4   | 59.4     | 59.4        |
| S086 RMS_Methode | 42.0  | 42.0   | 42.0   | 42.0     | 42.0        |
| S086 RMS_ICLABEL | 42.0  | 42.0   | 42.0   | 42.0     | 42.0        |
| S087 RMSE        | 25.3  | 24.5   | 17.6   | 16.8     | 20.2        |
| S087 RMS_Brut    | 36.4  | 36.4   | 36.4   | 36.4     | 36.4        |
| S087 RMS_Methode | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S087 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S088 RMSE        | 51.4  | 57.0   | 20.7   | 31.4     | 56.7        |
| S088 RMS_Brut    | 64.4  | 64.4   | 64.4   | 64.4     | 64.4        |
| S088 RMS_Methode | 51.1  | 51.1   | 51.1   | 51.1     | 51.1        |
| S088 RMS_ICLABEL | 51.1  | 51.1   | 51.1   | 51.1     | 51.1        |
| S089 RMSE        | 5.6   | 5.5    | 2.6    | 2.5      | 4.2         |
| S089 RMS_Brut    | 8.6   | 8.6    | 8.6    | 8.6      | 8.6         |
| S089 RMS_Methode | 5.5   | 5.5    | 5.5    | 5.5      | 5.5         |
| S089 RMS_ICLABEL | 5.5   | 5.5    | 5.5    | 5.5      | 5.5         |
| S090 RMSE        | 22.6  | 22.4   | 18.5   | 18.1     | 21.9        |
| S090 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S090 RMS_Methode | 21.7  | 21.7   | 21.7   | 21.7     | 21.7        |
| S090 RMS_ICLABEL | 21.7  | 21.7   | 21.7   | 21.7     | 21.7        |
| S091 RMSE        | 29.7  | 28.8   | 27.1   | 27.1     | 29.7        |
| S091 RMS_Brut    | 84.1  | 84.1   | 84.1   | 84.1     | 84.1        |
| S091 RMS_Methode | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S091 RMS_ICLABEL | 25.4  | 25.4   | 25.4   | 25.4     | 25.4        |
| S092 RMSE        | 28.5  | 28.0   | 18.0   | 17.7     | 22.1        |
| S092 RMS_Brut    | 40.3  | 40.3   | 40.3   | 40.3     | 40.3        |
| S092 RMS_Methode | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S092 RMS_ICLABEL | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S093 RMSE        | 15.1  | 15.3   | 9.2    | 9.6      | 11.4        |
| S093 RMS_Brut    | 21.8  | 21.8   | 21.8   | 21.8     | 21.8        |
| S093 RMS_Methode | 14.9  | 14.9   | 14.9   | 14.9     | 14.9        |
| S093 RMS_ICLABEL | 14.9  | 14.9   | 14.9   | 14.9     | 14.9        |
| S094 RMSE        | 34.3  | 33.8   | 19.3   | 19.5     | 28.7        |
| S094 RMS_Brut    | 52.4  | 52.4   | 52.4   | 52.4     | 52.4        |
| S094 RMS_Methode | 36.8  | 36.8   | 36.8   | 36.8     | 36.8        |
| S094 RMS_ICLABEL | 36.8  | 36.8   | 36.8   | 36.8     | 36.8        |
| S095 RMSE        | 55.2  | 56.1   | 22.4   | 34.8     | 53.8        |
| S095 RMS_Brut    | 57.8  | 57.8   | 57.8   | 57.8     | 57.8        |
| S095 RMS_Methode | 54.5  | 54.5   | 54.5   | 54.5     | 54.5        |
| S095 RMS_ICLABEL | 54.5  | 54.5   | 54.5   | 54.5     | 54.5        |
| S096 RMSE        | 40.5  | 39.5   | 27.5   | 28.7     | 36.6        |
| S096 RMS_Brut    | 69.0  | 69.0   | 69.0   | 69.0     | 69.0        |
| S096 RMS_Methode | 38.4  | 38.4   | 38.4   | 38.4     | 38.4        |
| S096 RMS_ICLABEL | 38.4  | 38.4   | 38.4   | 38.4     | 38.4        |
| S097 RMSE        | 25.2  | 25.1   | 19.0   | 18.5     | 23.8        |
| S097 RMS_Brut    | 55.3  | 55.3   | 55.3   | 55.3     | 55.3        |
| S097 RMS_Methode | 22.9  | 22.9   | 22.9   | 22.9     | 22.9        |
| S097 RMS_ICLABEL | 22.9  | 22.9   | 22.9   | 22.9     | 22.9        |
| S098 RMSE        | 9.3   | 8.7    | 3.7    | 3.8      | 6.1         |
| S098 RMS_Brut    | 12.8  | 12.8   | 12.8   | 12.8     | 12.8        |
| S098 RMS_Methode | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S098 RMS_ICLABEL | 12.1  | 12.1   | 12.1   | 12.1     | 12.1        |
| S099 RMSE        | 32.5  | 31.4   | 27.2   | 28.3     | 32.0        |
| S099 RMS_Brut    | 54.6  | 54.6   | 54.6   | 54.6     | 54.6        |
| S099 RMS_Methode | 29.9  | 29.9   | 29.9   | 29.9     | 29.9        |
| S099 RMS_ICLABEL | 29.9  | 29.9   | 29.9   | 29.9     | 29.9        |
| S100 RMSE        | 27.4  | 27.0   | 18.5   | 18.5     | 23.6        |
| S100 RMS_Brut    | 45.9  | 45.9   | 45.9   | 45.9     | 45.9        |
| S100 RMS_Methode | 26.2  | 26.2   | 26.2   | 26.2     | 26.2        |
| S100 RMS_ICLABEL | 26.2  | 26.2   | 26.2   | 26.2     | 26.2        |
| S101 RMSE        | 27.2  | 26.9   | 18.0   | 18.2     | 22.2        |
| S101 RMS_Brut    | 32.3  | 32.3   | 32.3   | 32.3     | 32.3        |
| S101 RMS_Methode | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S101 RMS_ICLABEL | 24.9  | 24.9   | 24.9   | 24.9     | 24.9        |
| S102 RMSE        | 7.7   | 7.6    | 4.6    | 4.4      | 5.9         |
| S102 RMS_Brut    | 10.9  | 10.9   | 10.9   | 10.9     | 10.9        |
| S102 RMS_Methode | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S102 RMS_ICLABEL | 7.3   | 7.3    | 7.3    | 7.3      | 7.3         |
| S103 RMSE        | 35.1  | 34.4   | 24.9   | 24.3     | 28.5        |
| S103 RMS_Brut    | 48.8  | 48.8   | 48.8   | 48.8     | 48.8        |
| S103 RMS_Methode | 35.7  | 35.7   | 35.7   | 35.7     | 35.7        |
| S103 RMS_ICLABEL | 35.7  | 35.7   | 35.7   | 35.7     | 35.7        |
| S104 RMSE        | 19.2  | 18.8   | 9.0    | 9.0      | 16.1        |
| S104 RMS_Brut    | 30.9  | 30.9   | 30.9   | 30.9     | 30.9        |
| S104 RMS_Methode | 20.0  | 20.0   | 20.0   | 20.0     | 20.0        |
| S104 RMS_ICLABEL | 20.0  | 20.0   | 20.0   | 20.0     | 20.0        |
| S105 RMSE        | 17.7  | 17.7   | 10.1   | 9.9      | 13.7        |
| S105 RMS_Brut    | 30.1  | 30.1   | 30.1   | 30.1     | 30.1        |
| S105 RMS_Methode | 18.0  | 18.0   | 18.0   | 18.0     | 18.0        |
| S105 RMS_ICLABEL | 18.0  | 18.0   | 18.0   | 18.0     | 18.0        |
| S106 RMSE        | 28.6  | 27.9   | 13.6   | 14.1     | 21.9        |
| S106 RMS_Brut    | 41.8  | 41.8   | 41.8   | 41.8     | 41.8        |
| S106 RMS_Methode | 31.2  | 31.2   | 31.2   | 31.2     | 31.2        |
| S106 RMS_ICLABEL | 31.2  | 31.2   | 31.2   | 31.2     | 31.2        |
| S107 RMSE        | 30.2  | 30.3   | 17.8   | 18.0     | 22.3        |
| S107 RMS_Brut    | 38.7  | 38.7   | 38.7   | 38.7     | 38.7        |
| S107 RMS_Methode | 29.2  | 29.2   | 29.2   | 29.2     | 29.2        |
| S107 RMS_ICLABEL | 29.2  | 29.2   | 29.2   | 29.2     | 29.2        |

| Sujet               | DuoCL | GCTNet | ICUNet      | ICUNet++ | ICUNet_attn |
|---------------------|-------|--------|-------------|----------|-------------|
| S108 RMSE           | 24.2  | 24.3   | 17.7        | 19.3     | 23.1        |
| S108 RMS_Brut       | 51.9  | 51.9   | 51.9        | 51.9     | 51.9        |
| S108 RMS_Methode    | 23.9  | 23.9   | 23.9        | 23.9     | 23.9        |
| S108 RMS_ICLABEL    | 23.9  | 23.9   | 23.9        | 23.9     | 23.9        |
| S109 RMSE           | 40.2  | 39.5   | 38.6        | 41.9     | 46.8        |
| S109 RMS_Brut       | 46.7  | 46.7   | 46.7        | 46.7     | 46.7        |
| S109 RMS_Methode    | 44.4  | 44.4   | 44.4        | 44.4     | 44.4        |
| S109 RMS_ICLABEL    | 44.4  | 44.4   | 44.4        | 44.4     | 44.4        |
| <b>Moyenne RMSE</b> | 23.6  | 23.3   | <b>14.3</b> | 14.9     | 20.2        |
| Moyenne RMS_Brut    | 35.8  | 35.8   | 35.8        | 35.8     | 35.8        |
| Moyenne RMS_Methode | 23.6  | 23.6   | 23.6        | 23.6     | 23.6        |
| Moyenne RMS_ICLABEL | 23.6  | 23.6   | 23.6        | 23.6     | 23.6        |

L'ordre de grandeur des signaux (Nettoyé\_Normalisé\_2) est de 23.6  $\mu$ V (RMS moyen sur les 109 sujets).

TABLE 45 – Récapitulatif — accuracy CSP+LDA sur signal brut, les 109 sujets, selon le nombre de composantes CSP (n\_components), avec le meilleur n\_components par sujet.

| Sujet | 2                  | 4                  | 6                  | 8                  | 10                 | 12                 | 14                 | Meilleur    |
|-------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------|
| S001  | 0.68 ± 0.10        | 0.63 ± 0.21        | 0.69 ± 0.08        | <b>0.71 ± 0.15</b> | 0.71 ± 0.12        | 0.68 ± 0.10        | 0.67 ± 0.16        | 0.71 ± 0.15 |
| S002  | 0.64 ± 0.16        | 0.68 ± 0.18        | 0.66 ± 0.18        | 0.68 ± 0.20        | 0.62 ± 0.22        | <b>0.69 ± 0.18</b> | 0.63 ± 0.21        | 0.69 ± 0.18 |
| S003  | 0.48 ± 0.12        | 0.56 ± 0.16        | 0.57 ± 0.19        | 0.58 ± 0.18        | 0.57 ± 0.14        | <b>0.61 ± 0.09</b> | 0.54 ± 0.14        | 0.61 ± 0.09 |
| S004  | 0.47 ± 0.16        | 0.43 ± 0.16        | 0.44 ± 0.14        | 0.48 ± 0.16        | <b>0.51 ± 0.12</b> | 0.46 ± 0.09        | 0.48 ± 0.18        | 0.51 ± 0.12 |
| S005  | 0.44 ± 0.14        | 0.49 ± 0.14        | 0.54 ± 0.08        | <b>0.56 ± 0.11</b> | 0.52 ± 0.17        | 0.52 ± 0.13        | 0.46 ± 0.12        | 0.56 ± 0.11 |
| S006  | <b>0.56 ± 0.14</b> | 0.50 ± 0.11        | <b>0.56 ± 0.14</b> | 0.53 ± 0.13        | 0.51 ± 0.17        | 0.53 ± 0.16        | 0.52 ± 0.12        | 0.56 ± 0.14 |
| S007  | 0.82 ± 0.19        | <b>0.96 ± 0.07</b> | 0.92 ± 0.09        | 0.94 ± 0.07        | 0.88 ± 0.09        | 0.88 ± 0.09        | 0.83 ± 0.12        | 0.96 ± 0.07 |
| S008  | 0.44 ± 0.07        | 0.48 ± 0.15        | <b>0.54 ± 0.13</b> | 0.51 ± 0.13        | 0.52 ± 0.13        | 0.47 ± 0.11        | 0.49 ± 0.09        | 0.54 ± 0.13 |
| S009  | 0.38 ± 0.11        | <b>0.38 ± 0.19</b> | <b>0.38 ± 0.13</b> | 0.33 ± 0.11        | 0.36 ± 0.16        | 0.28 ± 0.13        | 0.31 ± 0.07        | 0.38 ± 0.19 |
| S010  | 0.56 ± 0.13        | <b>0.58 ± 0.16</b> | 0.50 ± 0.15        | 0.49 ± 0.11        | 0.48 ± 0.14        | 0.46 ± 0.17        | 0.44 ± 0.13        | 0.58 ± 0.16 |
| S011  | <b>0.52 ± 0.19</b> | 0.42 ± 0.14        | 0.44 ± 0.18        | 0.47 ± 0.16        | 0.46 ± 0.13        | 0.43 ± 0.17        | 0.43 ± 0.14        | 0.52 ± 0.19 |
| S012  | 0.40 ± 0.18        | 0.53 ± 0.16        | 0.53 ± 0.18        | 0.51 ± 0.19        | 0.49 ± 0.20        | 0.52 ± 0.20        | <b>0.53 ± 0.16</b> | 0.53 ± 0.16 |
| S013  | 0.47 ± 0.13        | 0.50 ± 0.17        | 0.51 ± 0.15        | 0.53 ± 0.17        | 0.56 ± 0.17        | 0.52 ± 0.15        | <b>0.60 ± 0.16</b> | 0.60 ± 0.16 |
| S014  | <b>0.52 ± 0.15</b> | 0.47 ± 0.12        | 0.44 ± 0.10        | 0.44 ± 0.15        | 0.42 ± 0.11        | 0.48 ± 0.10        | 0.48 ± 0.11        | 0.52 ± 0.15 |
| S015  | 0.64 ± 0.15        | 0.66 ± 0.12        | <b>0.71 ± 0.12</b> | 0.63 ± 0.12        | 0.58 ± 0.10        | 0.62 ± 0.14        | 0.58 ± 0.16        | 0.71 ± 0.12 |
| S016  | <b>0.59 ± 0.18</b> | 0.47 ± 0.15        | 0.47 ± 0.16        | 0.41 ± 0.14        | 0.49 ± 0.11        | 0.43 ± 0.13        | 0.43 ± 0.12        | 0.59 ± 0.18 |
| S017  | 0.46 ± 0.16        | 0.40 ± 0.18        | <b>0.51 ± 0.16</b> | 0.49 ± 0.19        | 0.46 ± 0.16        | 0.49 ± 0.10        | 0.48 ± 0.11        | 0.51 ± 0.16 |
| S018  | 0.41 ± 0.19        | 0.39 ± 0.13        | 0.43 ± 0.19        | 0.50 ± 0.09        | <b>0.59 ± 0.13</b> | 0.57 ± 0.16        | 0.52 ± 0.17        | 0.59 ± 0.13 |
| S019  | <b>0.66 ± 0.14</b> | 0.60 ± 0.07        | 0.54 ± 0.15        | 0.52 ± 0.16        | 0.61 ± 0.17        | 0.59 ± 0.19        | 0.57 ± 0.15        | 0.66 ± 0.14 |
| S020  | 0.50 ± 0.15        | <b>0.62 ± 0.16</b> | 0.58 ± 0.16        | <b>0.62 ± 0.15</b> | 0.60 ± 0.15        | 0.53 ± 0.13        | 0.56 ± 0.12        | 0.62 ± 0.16 |
| S021  | 0.54 ± 0.13        | <b>0.67 ± 0.14</b> | 0.64 ± 0.13        | 0.64 ± 0.15        | 0.62 ± 0.22        | 0.60 ± 0.20        | 0.64 ± 0.18        | 0.67 ± 0.14 |
| S022  | 0.48 ± 0.16        | 0.48 ± 0.16        | <b>0.50 ± 0.14</b> | 0.44 ± 0.10        | 0.46 ± 0.18        | 0.48 ± 0.15        | 0.43 ± 0.12        | 0.50 ± 0.14 |
| S023  | <b>0.62 ± 0.19</b> | 0.54 ± 0.18        | 0.54 ± 0.14        | 0.58 ± 0.13        | 0.51 ± 0.13        | 0.57 ± 0.18        | 0.60 ± 0.11        | 0.62 ± 0.19 |
| S024  | 0.39 ± 0.09        | 0.48 ± 0.10        | 0.56 ± 0.14        | 0.57 ± 0.20        | 0.60 ± 0.17        | <b>0.63 ± 0.12</b> | 0.57 ± 0.13        | 0.63 ± 0.12 |
| S025  | <b>0.56 ± 0.12</b> | 0.51 ± 0.21        | 0.48 ± 0.22        | 0.49 ± 0.21        | 0.49 ± 0.19        | 0.51 ± 0.22        | 0.52 ± 0.21        | 0.56 ± 0.12 |
| S026  | <b>0.73 ± 0.10</b> | 0.69 ± 0.11        | 0.68 ± 0.10        | 0.68 ± 0.14        | 0.68 ± 0.14        | 0.60 ± 0.11        | 0.62 ± 0.15        | 0.73 ± 0.10 |
| S027  | 0.52 ± 0.19        | 0.50 ± 0.13        | 0.54 ± 0.20        | <b>0.61 ± 0.16</b> | 0.57 ± 0.17        | 0.52 ± 0.09        | 0.53 ± 0.12        | 0.61 ± 0.16 |
| S028  | <b>0.59 ± 0.16</b> | 0.44 ± 0.07        | 0.47 ± 0.07        | 0.47 ± 0.13        | 0.49 ± 0.11        | 0.49 ± 0.11        | 0.51 ± 0.16        | 0.59 ± 0.16 |
| S029  | 0.84 ± 0.20        | 0.96 ± 0.05        | 0.96 ± 0.05        | <b>0.97 ± 0.05</b> | <b>0.97 ± 0.05</b> | <b>0.97 ± 0.07</b> | 0.96 ± 0.07        | 0.97 ± 0.05 |
| S030  | 0.52 ± 0.14        | 0.57 ± 0.18        | <b>0.60 ± 0.17</b> | 0.54 ± 0.16        | 0.58 ± 0.15        | 0.53 ± 0.10        | 0.50 ± 0.15        | 0.60 ± 0.17 |
| S031  | 0.56 ± 0.16        | 0.63 ± 0.19        | <b>0.74 ± 0.11</b> | 0.70 ± 0.12        | 0.70 ± 0.12        | 0.67 ± 0.15        | 0.71 ± 0.09        | 0.74 ± 0.11 |
| S032  | 0.44 ± 0.11        | 0.51 ± 0.05        | 0.52 ± 0.12        | 0.62 ± 0.09        | <b>0.63 ± 0.11</b> | 0.63 ± 0.12        | 0.62 ± 0.12        | 0.63 ± 0.11 |
| S033  | <b>0.73 ± 0.16</b> | 0.69 ± 0.12        | 0.53 ± 0.13        | 0.48 ± 0.16        | 0.39 ± 0.19        | 0.40 ± 0.17        | 0.43 ± 0.20        | 0.73 ± 0.16 |
| S034  | <b>0.73 ± 0.11</b> | 0.62 ± 0.07        | 0.61 ± 0.12        | 0.62 ± 0.15        | 0.53 ± 0.17        | 0.54 ± 0.16        | 0.59 ± 0.19        | 0.73 ± 0.11 |
| S035  | <b>0.66 ± 0.17</b> | 0.62 ± 0.17        | 0.54 ± 0.14        | 0.54 ± 0.14        | 0.52 ± 0.10        | 0.60 ± 0.09        | 0.56 ± 0.15        | 0.66 ± 0.17 |
| S036  | <b>0.61 ± 0.18</b> | 0.46 ± 0.14        | 0.40 ± 0.12        | 0.47 ± 0.10        | 0.44 ± 0.15        | 0.43 ± 0.18        | 0.39 ± 0.11        | 0.61 ± 0.18 |
| S037  | <b>0.57 ± 0.16</b> | 0.39 ± 0.18        | 0.42 ± 0.12        | 0.46 ± 0.16        | 0.48 ± 0.19        | 0.48 ± 0.21        | 0.54 ± 0.21        | 0.57 ± 0.16 |
| S038  | <b>0.42 ± 0.16</b> | 0.33 ± 0.10        | 0.40 ± 0.12        | 0.37 ± 0.10        | 0.34 ± 0.16        | 0.33 ± 0.16        | 0.34 ± 0.19        | 0.42 ± 0.16 |
| S039  | 0.49 ± 0.13        | 0.50 ± 0.18        | <b>0.59 ± 0.13</b> | 0.50 ± 0.13        | 0.56 ± 0.14        | 0.57 ± 0.14        | 0.50 ± 0.14        | 0.59 ± 0.13 |
| S040  | 0.38 ± 0.12        | 0.39 ± 0.17        | 0.40 ± 0.11        | 0.40 ± 0.12        | 0.43 ± 0.14        | <b>0.47 ± 0.18</b> | 0.40 ± 0.15        | 0.47 ± 0.18 |
| S041  | <b>0.57 ± 0.13</b> | 0.52 ± 0.15        | 0.51 ± 0.14        | 0.53 ± 0.14        | 0.48 ± 0.17        | 0.53 ± 0.14        | 0.47 ± 0.18        | 0.57 ± 0.13 |
| S042  | <b>0.81 ± 0.09</b> | 0.79 ± 0.10        | 0.78 ± 0.12        | 0.77 ± 0.10        | 0.78 ± 0.14        | <b>0.81 ± 0.15</b> | 0.77 ± 0.14        | 0.81 ± 0.09 |
| S043  | 0.51 ± 0.21        | 0.57 ± 0.14        | 0.54 ± 0.14        | 0.56 ± 0.19        | 0.58 ± 0.19        | 0.54 ± 0.15        | <b>0.61 ± 0.17</b> | 0.61 ± 0.17 |
| S044  | 0.64 ± 0.15        | <b>0.71 ± 0.13</b> | 0.54 ± 0.14        | 0.52 ± 0.15        | 0.47 ± 0.15        | 0.48 ± 0.09        | 0.53 ± 0.10        | 0.71 ± 0.13 |
| S045  | 0.50 ± 0.17        | 0.46 ± 0.16        | 0.44 ± 0.10        | 0.49 ± 0.17        | 0.54 ± 0.13        | 0.54 ± 0.08        | <b>0.60 ± 0.11</b> | 0.60 ± 0.11 |
| S046  | 0.47 ± 0.23        | 0.60 ± 0.25        | <b>0.64 ± 0.21</b> | 0.61 ± 0.16        | 0.60 ± 0.18        | 0.52 ± 0.12        | 0.50 ± 0.17        | 0.64 ± 0.21 |
| S047  | 0.50 ± 0.12        | <b>0.61 ± 0.11</b> | <b>0.61 ± 0.09</b> | 0.54 ± 0.14        | 0.57 ± 0.15        | 0.51 ± 0.16        | 0.52 ± 0.17        | 0.61 ± 0.11 |
| S048  | 0.66 ± 0.12        | <b>0.73 ± 0.16</b> | 0.69 ± 0.11        | 0.68 ± 0.08        | 0.64 ± 0.12        | 0.63 ± 0.16        | 0.62 ± 0.18        | 0.73 ± 0.16 |
| S049  | 0.54 ± 0.15        | 0.83 ± 0.06        | 0.86 ± 0.11        | 0.86 ± 0.09        | 0.90 ± 0.08        | <b>0.92 ± 0.09</b> | 0.86 ± 0.11        | 0.92 ± 0.09 |
| S050  | <b>0.60 ± 0.13</b> | <b>0.60 ± 0.15</b> | 0.51 ± 0.15        | 0.47 ± 0.13        | 0.48 ± 0.12        | 0.52 ± 0.09        | 0.56 ± 0.12        | 0.60 ± 0.13 |
| S051  | 0.37 ± 0.12        | 0.54 ± 0.15        | 0.51 ± 0.12        | 0.50 ± 0.13        | 0.48 ± 0.18        | <b>0.58 ± 0.12</b> | <b>0.58 ± 0.12</b> | 0.58 ± 0.12 |
| S052  | 0.59 ± 0.10        | <b>0.61 ± 0.09</b> | 0.57 ± 0.16        | 0.59 ± 0.11        | 0.53 ± 0.14        | 0.59 ± 0.16        | 0.57 ± 0.19        | 0.61 ± 0.09 |
| S053  | 0.66 ± 0.08        | 0.67 ± 0.09        | 0.72 ± 0.10        | 0.71 ± 0.11        | <b>0.78 ± 0.13</b> | 0.67 ± 0.14        | 0.73 ± 0.13        | 0.78 ± 0.13 |
| S054  | 0.66 ± 0.14        | 0.70 ± 0.11        | 0.79 ± 0.14        | <b>0.81 ± 0.12</b> | 0.72 ± 0.10        | 0.66 ± 0.13        | 0.64 ± 0.10        | 0.81 ± 0.12 |
| S055  | 0.68 ± 0.14        | 0.60 ± 0.17        | 0.64 ± 0.16        | 0.68 ± 0.19        | <b>0.70 ± 0.17</b> | 0.64 ± 0.19        | 0.67 ± 0.16        | 0.70 ± 0.17 |
| S056  | 0.74 ± 0.17        | 0.73 ± 0.15        | 0.71 ± 0.11        | 0.72 ± 0.11        | <b>0.77 ± 0.09</b> | 0.70 ± 0.10        | 0.72 ± 0.13        | 0.77 ± 0.09 |
| S057  | 0.36 ± 0.12        | 0.36 ± 0.18        | 0.36 ± 0.13        | 0.42 ± 0.16        | 0.43 ± 0.16        | 0.39 ± 0.07        | <b>0.46 ± 0.09</b> | 0.46 ± 0.09 |
| S058  | <b>0.59 ± 0.11</b> | 0.53 ± 0.15        | 0.44 ± 0.16        | 0.46 ± 0.18        | 0.53 ± 0.18        | 0.52 ± 0.18        | 0.53 ± 0.17        | 0.59 ± 0.11 |
| S059  | 0.53 ± 0.14        | <b>0.59 ± 0.17</b> | 0.56 ± 0.10        | 0.52 ± 0.19        | 0.49 ± 0.15        | 0.49 ± 0.16        | 0.49 ± 0.15        | 0.59 ± 0.17 |
| S060  | 0.73 ± 0.09        | <b>0.76 ± 0.10</b> | <b>0.76 ± 0.15</b> | 0.62 ± 0.13        | 0.64 ± 0.18        | 0.62 ± 0.15        | 0.66 ± 0.19        | 0.76 ± 0.10 |
| S061  | 0.53 ± 0.15        | 0.57 ± 0.20        | 0.68 ± 0.09        | 0.66 ± 0.12        | <b>0.70 ± 0.13</b> | 0.66 ± 0.16        | 0.68 ± 0.16        | 0.70 ± 0.13 |
| S062  | <b>0.92 ± 0.10</b> | 0.84 ± 0.05        | 0.91 ± 0.07        | 0.87 ± 0.08        | 0.86 ± 0.13        | 0.82 ± 0.07        | 0.86 ± 0.09        | 0.92 ± 0.10 |
| S063  | 0.42 ± 0.11        | 0.37 ± 0.17        | 0.41 ± 0.11        | 0.44 ± 0.13        | 0.49 ± 0.14        | 0.50 ± 0.17        | <b>0.57 ± 0.12</b> | 0.57 ± 0.12 |
| S064  | 0.44 ± 0.09        | <b>0.48 ± 0.11</b> | 0.47 ± 0.18        | 0.41 ± 0.15        | 0.42 ± 0.16        | 0.44 ± 0.17        | 0.44 ± 0.16        | 0.48 ± 0.11 |
| S065  | 0.43 ± 0.13        | 0.42 ± 0.14        | 0.44 ± 0.19        | 0.47 ± 0.18        | 0.51 ± 0.10        | <b>0.54 ± 0.16</b> | 0.48 ± 0.13        | 0.54 ± 0.16 |
| S066  | <b>0.57 ± 0.09</b> | 0.53 ± 0.15        | 0.53 ± 0.13        | 0.49 ± 0.13        | 0.50 ± 0.15        | 0.54 ± 0.16        | 0.50 ± 0.17        | 0.57 ± 0.09 |
| S067  | 0.47 ± 0.16        | 0.49 ± 0.21        | 0.49 ± 0.11        | <b>0.52 ± 0.12</b> | 0.49 ± 0.21        | 0.49 ± 0.14        | 0.46 ± 0.17        | 0.52 ± 0.12 |
| S068  | 0.47 ± 0.12        | 0.53 ± 0.18        | 0.52 ± 0.13        | 0.50 ± 0.10        | 0.50 ± 0.13        | <b>0.58 ± 0.10</b> | 0.49 ± 0.12        | 0.58 ± 0.10 |
| S069  | <b>0.51 ± 0.11</b> | 0.40 ± 0.18        | 0.47 ± 0.13        | 0.44 ± 0.18        | 0.41 ± 0.15        | 0.43 ± 0.15        | 0.43 ± 0.18        | 0.51 ± 0.11 |
| S070  | 0.50 ± 0.26        | <b>0.69 ± 0.13</b> | 0.67 ± 0.18        | 0.60 ± 0.13        | 0.59 ± 0.16        | 0.61 ± 0.15        | 0.58 ± 0.16        | 0.69 ± 0.13 |
| S071  | 0.62 ± 0.19        | <b>0.67 ± 0.10</b> | 0.59 ± 0.11        | 0.59 ± 0.10        | 0.63 ± 0.12        | 0.60 ± 0.14        | 0.63 ± 0.13        | 0.67 ± 0.10 |
| S072  | <b>0.81 ± 0.07</b> | 0.78 ± 0.09        | 0.74 ± 0.09        | 0.71 ± 0.11        | 0.68 ± 0.18        | 0.66 ± 0.16        | 0.68 ± 0.14        | 0.81 ± 0.07 |
| S073  | 0.41 ± 0.10        | 0.37 ± 0.15        | 0.31 ± 0.20        | 0.38 ± 0.25        | <b>0.44 ± 0.20</b> | <b>0.44 ± 0.22</b> | 0.43 ± 0.21        | 0.44 ± 0.20 |
| S074  | 0.59 ± 0.12        | 0.60 ± 0.19        | <b>0.63 ± 0.11</b> | 0.52 ± 0.12        | 0.46 ± 0.08        | 0.51 ± 0.07        | 0.51 ± 0.10        | 0.63 ± 0.11 |
| S075  | 0.54 ± 0.16        | 0.51 ± 0.15        | 0.53 ± 0.11        | <b>0.57 ± 0.09</b> | 0.48 ± 0.14        | 0.49 ± 0.16        | 0.41 ± 0.14        | 0.57 ± 0.09 |
| S076  | 0.51 ± 0.17        | <b>0.54 ± 0.14</b> | 0.52 ± 0.14        | 0.52 ± 0.11        | 0.52 ± 0.17        | 0.51 ± 0.11        | <b>0.54 ± 0.13</b> | 0.54 ± 0.14 |
| S077  | 0.38 ± 0.13        | 0.40 ± 0.15        | 0.43 ± 0.09        | 0.48 ± 0.09        | 0.47 ± 0.08        | 0.49 ± 0.09        | <b>0.49 ± 0.10</b> | 0.49 ± 0.10 |
| S078  | 0.41 ± 0.17        | 0.38 ± 0.12        | 0.31 ± 0.15        | 0.32 ± 0.15        | 0.38 ± 0.12        | 0.41 ± 0.20        | <b>0.46 ± 0.15</b> | 0.46 ± 0.15 |
| S079  | 0.56 ± 0.17        | <b>0.64 ± 0.18</b> | 0.52 ± 0.19        | 0.47 ± 0.18        | 0.41 ± 0.22        | 0.43 ± 0.22        | 0.44 ± 0.16        | 0.64 ± 0.18 |
| S080  | 0.42 ± 0.10        | 0.54 ± 0.14        | 0.54 ± 0.18        | 0.52 ± 0.19        | <b>0.59 ± 0.21</b> | 0.50 ± 0.22        | 0.49 ± 0.17        | 0.59 ± 0.21 |
| S081  | 0.49 ± 0.17        | 0.48 ± 0.15        | 0.46 ± 0.12        | 0.44 ± 0.15        | 0.52 ± 0.22        | 0.51 ± 0.21        | <b>0.54 ± 0.16</b> | 0.54 ± 0.16 |
| S082  | 0.56 ± 0.12        | 0.73 ± 0.12        | 0.66 ± 0.13        | <b>0.74 ± 0.10</b> | 0.67 ± 0.10        | 0.72 ± 0.09        | 0.70 ± 0.07        | 0.74 ± 0.10 |
| S083  | 0.58 ± 0.18        | 0.60 ± 0.10        | 0.58 ± 0.15        | 0.61 ± 0.16        | 0.59 ± 0.11        | <b>0.63 ± 0.16</b> | 0.59 ± 0.17        | 0.63 ± 0.16 |
| S084  | <b>0.68 ± 0.12</b> | 0.59 ± 0.17        | 0.43 ± 0.13        | 0.39 ± 0.16        | 0.33 ± 0.18        | 0.39 ± 0.22        | 0.39 ± 0.15        | 0.68 ± 0.12 |
| S085  | <b>0.84 ± 0.12</b> | 0.72 ± 0.11        | 0.66 ± 0.09        | 0.66 ± 0.12        | 0.68 ± 0.12        | 0.66 ± 0.20        | 0.60 ± 0.20        | 0.84 ± 0.12 |
| S086  | 0.62 ± 0.17        | 0.61 ± 0.19        | 0.64 ± 0.16        | 0.64 ± 0.17        | 0.64 ± 0.16        | 0.53 ± 0.17        | <b>0.68 ± 0.14</b> | 0.68 ± 0.14 |
| S087  | 0.51 ± 0.17        | 0.48 ± 0.10        | <b>0.54 ± 0.10</b> | 0.51 ± 0.14        | 0.47 ± 0.11        | 0.41 ± 0.12        | 0.39 ± 0.13        | 0.54 ± 0.10 |

| Sujet          | 2                  | 4                  | 6                  | 8                  | 10                 | 12                 | 14                 | Meilleur           |
|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| S088           | 0.30 ± 0.11        | 0.38 ± 0.09        | 0.40 ± 0.18        | <b>0.46 ± 0.18</b> | 0.42 ± 0.17        | 0.36 ± 0.07        | 0.39 ± 0.16        | 0.46 ± 0.18        |
| S089           | 0.41 ± 0.12        | 0.41 ± 0.13        | 0.31 ± 0.13        | 0.34 ± 0.14        | 0.40 ± 0.11        | <b>0.42 ± 0.14</b> | 0.41 ± 0.10        | 0.42 ± 0.14        |
| S090           | 0.41 ± 0.12        | 0.44 ± 0.15        | <b>0.48 ± 0.14</b> | 0.40 ± 0.19        | 0.46 ± 0.19        | 0.42 ± 0.18        | 0.44 ± 0.13        | 0.48 ± 0.14        |
| S091           | 0.46 ± 0.12        | 0.53 ± 0.16        | 0.61 ± 0.09        | 0.62 ± 0.15        | 0.60 ± 0.19        | <b>0.63 ± 0.16</b> | 0.60 ± 0.13        | 0.63 ± 0.16        |
| S092           | 0.48 ± 0.10        | <b>0.55 ± 0.12</b> | 0.50 ± 0.09        | 0.44 ± 0.09        | 0.44 ± 0.16        | 0.45 ± 0.08        | 0.44 ± 0.12        | 0.55 ± 0.12        |
| S093           | 0.67 ± 0.20        | <b>0.70 ± 0.07</b> | 0.68 ± 0.15        | 0.64 ± 0.20        | 0.59 ± 0.16        | 0.60 ± 0.22        | 0.62 ± 0.15        | 0.70 ± 0.07        |
| S094           | 0.62 ± 0.09        | 0.72 ± 0.12        | 0.71 ± 0.13        | 0.73 ± 0.12        | 0.71 ± 0.10        | <b>0.76 ± 0.10</b> | 0.74 ± 0.10        | 0.76 ± 0.10        |
| S095           | 0.48 ± 0.10        | <b>0.51 ± 0.12</b> | 0.49 ± 0.14        | 0.42 ± 0.14        | 0.38 ± 0.17        | 0.41 ± 0.15        | 0.41 ± 0.17        | 0.51 ± 0.12        |
| S096           | 0.42 ± 0.16        | <b>0.58 ± 0.17</b> | 0.52 ± 0.13        | 0.50 ± 0.15        | 0.53 ± 0.14        | 0.57 ± 0.18        | 0.52 ± 0.12        | 0.58 ± 0.17        |
| S097           | 0.49 ± 0.16        | 0.51 ± 0.17        | 0.50 ± 0.14        | 0.47 ± 0.14        | 0.50 ± 0.16        | 0.53 ± 0.14        | <b>0.54 ± 0.08</b> | 0.54 ± 0.08        |
| S098           | 0.43 ± 0.13        | 0.58 ± 0.15        | 0.58 ± 0.20        | 0.62 ± 0.14        | <b>0.64 ± 0.12</b> | 0.61 ± 0.17        | 0.60 ± 0.17        | 0.64 ± 0.12        |
| S099           | 0.39 ± 0.16        | <b>0.52 ± 0.17</b> | 0.50 ± 0.14        | 0.50 ± 0.11        | 0.49 ± 0.09        | 0.46 ± 0.12        | 0.50 ± 0.11        | 0.52 ± 0.17        |
| S100           | 0.45 ± 0.16        | 0.49 ± 0.16        | 0.50 ± 0.15        | 0.45 ± 0.16        | 0.44 ± 0.15        | 0.46 ± 0.06        | <b>0.53 ± 0.11</b> | 0.53 ± 0.11        |
| S101           | 0.43 ± 0.16        | 0.40 ± 0.19        | <b>0.47 ± 0.14</b> | 0.40 ± 0.17        | 0.37 ± 0.17        | 0.32 ± 0.10        | 0.32 ± 0.12        | 0.47 ± 0.14        |
| S102           | 0.47 ± 0.15        | 0.51 ± 0.17        | 0.62 ± 0.16        | 0.62 ± 0.13        | <b>0.64 ± 0.13</b> | 0.62 ± 0.11        | 0.52 ± 0.21        | 0.64 ± 0.13        |
| S103           | 0.46 ± 0.10        | <b>0.56 ± 0.09</b> | 0.51 ± 0.13        | 0.52 ± 0.14        | 0.48 ± 0.17        | 0.50 ± 0.12        | 0.52 ± 0.17        | 0.56 ± 0.09        |
| S104           | 0.46 ± 0.13        | 0.53 ± 0.13        | 0.48 ± 0.13        | <b>0.56 ± 0.11</b> | 0.48 ± 0.13        | 0.52 ± 0.15        | 0.46 ± 0.14        | 0.56 ± 0.11        |
| S105           | 0.57 ± 0.15        | 0.47 ± 0.16        | 0.51 ± 0.11        | 0.53 ± 0.12        | 0.54 ± 0.14        | 0.53 ± 0.10        | <b>0.59 ± 0.13</b> | 0.59 ± 0.13        |
| S106           | <b>0.56 ± 0.09</b> | 0.50 ± 0.14        | 0.52 ± 0.19        | 0.49 ± 0.22        | 0.54 ± 0.15        | 0.46 ± 0.16        | 0.47 ± 0.11        | 0.56 ± 0.09        |
| S107           | 0.50 ± 0.18        | 0.46 ± 0.16        | 0.43 ± 0.16        | 0.49 ± 0.12        | <b>0.54 ± 0.09</b> | 0.51 ± 0.09        | 0.48 ± 0.13        | 0.54 ± 0.09        |
| S108           | 0.60 ± 0.11        | 0.56 ± 0.17        | 0.57 ± 0.18        | 0.61 ± 0.22        | 0.57 ± 0.19        | <b>0.62 ± 0.19</b> | 0.61 ± 0.23        | 0.62 ± 0.19        |
| S109           | 0.47 ± 0.11        | <b>0.56 ± 0.12</b> | 0.51 ± 0.05        | 0.46 ± 0.13        | <b>0.56 ± 0.17</b> | 0.48 ± 0.16        | 0.49 ± 0.14        | 0.56 ± 0.12        |
| <b>Moyenne</b> | 0.54 ± 0.12        | <b>0.56 ± 0.12</b> | 0.55 ± 0.12        | 0.55 ± 0.12        | 0.55 ± 0.12        | 0.55 ± 0.12        | 0.54 ± 0.11        | <b>0.61 ± 0.11</b> |

TABLE 46 – Récapitulatif — accuracy CSP+LDA sur ICLabel, les 109 sujets, selon le nombre de composantes CSP (n\_components), avec le meilleur n\_components par sujet.

| Sujet | 2                  | 4                  | 6                  | 8                  | 10                 | 12                 | 14                 | Meilleur    |
|-------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------|
| S001  | <b>0.70 ± 0.15</b> | 0.64 ± 0.23        | 0.69 ± 0.16        | 0.67 ± 0.13        | 0.63 ± 0.11        | 0.61 ± 0.14        | 0.63 ± 0.13        | 0.70 ± 0.15 |
| S002  | 0.73 ± 0.17        | 0.69 ± 0.20        | 0.70 ± 0.19        | <b>0.74 ± 0.17</b> | 0.73 ± 0.14        | 0.70 ± 0.19        | 0.69 ± 0.17        | 0.74 ± 0.17 |
| S003  | 0.41 ± 0.16        | 0.54 ± 0.16        | <b>0.66 ± 0.13</b> | 0.64 ± 0.12        | 0.61 ± 0.15        | 0.62 ± 0.17        | 0.61 ± 0.19        | 0.66 ± 0.13 |
| S004  | 0.50 ± 0.18        | 0.44 ± 0.11        | 0.48 ± 0.12        | <b>0.59 ± 0.07</b> | 0.57 ± 0.13        | 0.54 ± 0.13        | 0.57 ± 0.10        | 0.59 ± 0.07 |
| S005  | 0.46 ± 0.13        | 0.49 ± 0.12        | 0.46 ± 0.14        | 0.50 ± 0.12        | 0.51 ± 0.09        | <b>0.53 ± 0.14</b> | 0.50 ± 0.09        | 0.53 ± 0.14 |
| S006  | 0.50 ± 0.13        | 0.51 ± 0.15        | <b>0.53 ± 0.12</b> | <b>0.53 ± 0.11</b> | 0.52 ± 0.13        | 0.49 ± 0.17        | 0.46 ± 0.14        | 0.53 ± 0.12 |
| S007  | 0.76 ± 0.18        | <b>0.98 ± 0.04</b> | 0.94 ± 0.09        | 0.91 ± 0.12        | 0.93 ± 0.10        | 0.92 ± 0.09        | 0.90 ± 0.09        | 0.98 ± 0.04 |
| S008  | 0.50 ± 0.11        | <b>0.52 ± 0.15</b> | 0.46 ± 0.18        | 0.50 ± 0.21        | 0.42 ± 0.15        | 0.43 ± 0.19        | 0.46 ± 0.15        | 0.52 ± 0.15 |
| S009  | <b>0.58 ± 0.12</b> | 0.56 ± 0.17        | 0.48 ± 0.10        | 0.46 ± 0.12        | 0.43 ± 0.10        | 0.44 ± 0.10        | 0.44 ± 0.09        | 0.58 ± 0.12 |
| S010  | 0.59 ± 0.09        | <b>0.60 ± 0.12</b> | 0.53 ± 0.11        | 0.48 ± 0.14        | 0.52 ± 0.13        | 0.49 ± 0.12        | 0.52 ± 0.17        | 0.60 ± 0.12 |
| S011  | 0.49 ± 0.12        | <b>0.50 ± 0.20</b> | <b>0.50 ± 0.17</b> | 0.49 ± 0.17        | 0.49 ± 0.13        | 0.44 ± 0.15        | <b>0.50 ± 0.18</b> | 0.50 ± 0.20 |
| S012  | 0.41 ± 0.15        | 0.50 ± 0.17        | <b>0.58 ± 0.18</b> | 0.44 ± 0.15        | 0.51 ± 0.14        | 0.52 ± 0.13        | 0.47 ± 0.15        | 0.58 ± 0.18 |
| S013  | <b>0.62 ± 0.14</b> | 0.51 ± 0.18        | 0.48 ± 0.11        | 0.52 ± 0.13        | 0.51 ± 0.13        | 0.50 ± 0.18        | 0.53 ± 0.15        | 0.62 ± 0.14 |
| S014  | 0.52 ± 0.15        | <b>0.54 ± 0.13</b> | <b>0.54 ± 0.10</b> | 0.44 ± 0.19        | 0.41 ± 0.19        | 0.47 ± 0.13        | 0.51 ± 0.17        | 0.54 ± 0.13 |
| S015  | 0.59 ± 0.16        | 0.69 ± 0.16        | 0.64 ± 0.15        | 0.67 ± 0.14        | <b>0.70 ± 0.13</b> | 0.64 ± 0.14        | <b>0.70 ± 0.11</b> | 0.70 ± 0.13 |
| S016  | 0.62 ± 0.16        | <b>0.66 ± 0.15</b> | 0.63 ± 0.19        | 0.58 ± 0.13        | 0.54 ± 0.10        | 0.59 ± 0.13        | 0.48 ± 0.19        | 0.66 ± 0.15 |
| S017  | 0.33 ± 0.09        | 0.34 ± 0.16        | 0.33 ± 0.17        | 0.34 ± 0.17        | <b>0.38 ± 0.19</b> | 0.38 ± 0.17        | 0.37 ± 0.13        | 0.38 ± 0.19 |
| S018  | 0.40 ± 0.12        | 0.36 ± 0.16        | 0.41 ± 0.16        | 0.49 ± 0.13        | 0.48 ± 0.12        | 0.48 ± 0.14        | <b>0.53 ± 0.11</b> | 0.53 ± 0.11 |
| S019  | 0.62 ± 0.11        | <b>0.68 ± 0.08</b> | 0.67 ± 0.12        | 0.60 ± 0.10        | 0.59 ± 0.14        | 0.64 ± 0.15        | 0.63 ± 0.12        | 0.68 ± 0.08 |
| S020  | 0.52 ± 0.10        | 0.66 ± 0.18        | <b>0.67 ± 0.16</b> | 0.66 ± 0.16        | 0.66 ± 0.23        | 0.62 ± 0.19        | 0.60 ± 0.19        | 0.67 ± 0.16 |
| S021  | 0.50 ± 0.18        | 0.61 ± 0.12        | 0.60 ± 0.11        | <b>0.64 ± 0.10</b> | 0.61 ± 0.13        | 0.60 ± 0.16        | 0.54 ± 0.14        | 0.64 ± 0.10 |
| S022  | 0.59 ± 0.14        | <b>0.63 ± 0.17</b> | 0.60 ± 0.14        | 0.56 ± 0.14        | 0.54 ± 0.12        | 0.56 ± 0.14        | 0.59 ± 0.14        | 0.63 ± 0.17 |
| S023  | 0.67 ± 0.16        | 0.66 ± 0.15        | 0.63 ± 0.15        | 0.56 ± 0.16        | 0.56 ± 0.12        | 0.57 ± 0.19        | <b>0.68 ± 0.14</b> | 0.68 ± 0.14 |
| S024  | 0.47 ± 0.18        | 0.52 ± 0.22        | 0.51 ± 0.12        | 0.51 ± 0.10        | 0.56 ± 0.15        | <b>0.56 ± 0.14</b> | 0.53 ± 0.15        | 0.56 ± 0.14 |
| S025  | 0.42 ± 0.13        | 0.42 ± 0.18        | 0.48 ± 0.16        | 0.54 ± 0.20        | <b>0.54 ± 0.15</b> | 0.50 ± 0.19        | 0.48 ± 0.22        | 0.54 ± 0.15 |
| S026  | 0.60 ± 0.21        | <b>0.80 ± 0.11</b> | 0.76 ± 0.13        | 0.71 ± 0.17        | 0.68 ± 0.21        | 0.57 ± 0.26        | 0.59 ± 0.22        | 0.80 ± 0.11 |
| S027  | 0.58 ± 0.16        | <b>0.58 ± 0.16</b> | 0.56 ± 0.14        | 0.43 ± 0.12        | 0.42 ± 0.16        | 0.40 ± 0.12        | 0.46 ± 0.14        | 0.58 ± 0.16 |
| S028  | <b>0.64 ± 0.15</b> | 0.39 ± 0.14        | 0.39 ± 0.07        | 0.39 ± 0.11        | 0.40 ± 0.10        | 0.43 ± 0.12        | 0.47 ± 0.15        | 0.64 ± 0.15 |
| S029  | 0.74 ± 0.22        | 0.97 ± 0.05        | 0.97 ± 0.05        | 0.98 ± 0.04        | 0.97 ± 0.05        | 0.97 ± 0.05        | <b>0.99 ± 0.03</b> | 0.99 ± 0.03 |
| S030  | 0.52 ± 0.16        | 0.56 ± 0.18        | 0.59 ± 0.19        | <b>0.64 ± 0.19</b> | 0.58 ± 0.20        | 0.52 ± 0.20        | 0.49 ± 0.19        | 0.64 ± 0.19 |
| S031  | 0.61 ± 0.12        | 0.64 ± 0.17        | 0.71 ± 0.14        | 0.72 ± 0.15        | <b>0.74 ± 0.09</b> | <b>0.74 ± 0.10</b> | 0.69 ± 0.08        | 0.74 ± 0.09 |
| S032  | 0.53 ± 0.16        | 0.74 ± 0.10        | 0.67 ± 0.15        | 0.68 ± 0.14        | 0.66 ± 0.12        | 0.72 ± 0.10        | <b>0.77 ± 0.12</b> | 0.77 ± 0.12 |
| S033  | 0.63 ± 0.16        | <b>0.67 ± 0.13</b> | 0.58 ± 0.18        | 0.49 ± 0.13        | 0.46 ± 0.17        | 0.42 ± 0.13        | 0.49 ± 0.12        | 0.67 ± 0.13 |
| S034  | <b>0.68 ± 0.13</b> | <b>0.68 ± 0.14</b> | 0.58 ± 0.13        | 0.50 ± 0.17        | 0.52 ± 0.16        | 0.49 ± 0.16        | 0.52 ± 0.17        | 0.68 ± 0.13 |
| S035  | <b>0.68 ± 0.18</b> | 0.62 ± 0.20        | 0.60 ± 0.16        | 0.58 ± 0.17        | 0.61 ± 0.15        | 0.53 ± 0.18        | 0.58 ± 0.13        | 0.68 ± 0.18 |
| S036  | 0.49 ± 0.19        | 0.46 ± 0.09        | 0.47 ± 0.13        | 0.48 ± 0.16        | 0.52 ± 0.13        | <b>0.53 ± 0.17</b> | 0.50 ± 0.17        | 0.53 ± 0.17 |
| S037  | <b>0.48 ± 0.17</b> | 0.39 ± 0.14        | 0.47 ± 0.15        | 0.42 ± 0.10        | 0.47 ± 0.08        | 0.46 ± 0.14        | 0.42 ± 0.08        | 0.48 ± 0.17 |
| S038  | <b>0.42 ± 0.17</b> | 0.36 ± 0.18        | 0.40 ± 0.11        | 0.27 ± 0.12        | 0.34 ± 0.18        | 0.31 ± 0.20        | 0.36 ± 0.18        | 0.42 ± 0.17 |
| S039  | 0.48 ± 0.12        | <b>0.53 ± 0.12</b> | 0.49 ± 0.17        | 0.49 ± 0.15        | 0.49 ± 0.13        | 0.47 ± 0.10        | 0.48 ± 0.09        | 0.53 ± 0.12 |
| S040  | 0.47 ± 0.11        | <b>0.52 ± 0.17</b> | 0.47 ± 0.11        | 0.43 ± 0.08        | 0.46 ± 0.15        | 0.49 ± 0.17        | 0.48 ± 0.14        | 0.52 ± 0.17 |
| S041  | 0.51 ± 0.13        | <b>0.52 ± 0.12</b> | 0.48 ± 0.11        | 0.49 ± 0.09        | <b>0.52 ± 0.12</b> | 0.50 ± 0.13        | 0.47 ± 0.14        | 0.52 ± 0.12 |
| S042  | 0.69 ± 0.17        | <b>0.82 ± 0.12</b> | 0.80 ± 0.11        | 0.80 ± 0.13        | 0.74 ± 0.14        | 0.73 ± 0.14        | 0.72 ± 0.10        | 0.82 ± 0.12 |
| S043  | 0.56 ± 0.16        | 0.58 ± 0.16        | 0.57 ± 0.21        | 0.59 ± 0.23        | 0.57 ± 0.20        | <b>0.60 ± 0.18</b> | 0.56 ± 0.20        | 0.60 ± 0.18 |
| S044  | 0.67 ± 0.18        | <b>0.76 ± 0.12</b> | 0.60 ± 0.12        | 0.63 ± 0.16        | 0.60 ± 0.17        | 0.62 ± 0.15        | 0.54 ± 0.16        | 0.76 ± 0.12 |
| S045  | <b>0.60 ± 0.16</b> | 0.47 ± 0.13        | 0.49 ± 0.14        | 0.53 ± 0.21        | 0.59 ± 0.19        | 0.54 ± 0.12        | 0.48 ± 0.12        | 0.60 ± 0.16 |
| S046  | <b>0.60 ± 0.18</b> | <b>0.60 ± 0.26</b> | <b>0.60 ± 0.24</b> | 0.54 ± 0.25        | 0.53 ± 0.20        | 0.52 ± 0.16        | 0.47 ± 0.13        | 0.60 ± 0.18 |
| S047  | 0.49 ± 0.15        | 0.58 ± 0.10        | 0.58 ± 0.14        | 0.49 ± 0.13        | <b>0.60 ± 0.16</b> | 0.51 ± 0.22        | 0.51 ± 0.15        | 0.60 ± 0.16 |
| S048  | 0.60 ± 0.10        | 0.71 ± 0.14        | <b>0.72 ± 0.14</b> | 0.70 ± 0.13        | 0.67 ± 0.11        | 0.64 ± 0.14        | 0.64 ± 0.13        | 0.72 ± 0.14 |
| S049  | 0.74 ± 0.19        | <b>0.80 ± 0.12</b> | 0.78 ± 0.10        | 0.77 ± 0.09        | 0.78 ± 0.09        | 0.77 ± 0.08        | 0.76 ± 0.11        | 0.80 ± 0.12 |
| S050  | 0.50 ± 0.10        | 0.53 ± 0.13        | 0.51 ± 0.07        | <b>0.57 ± 0.13</b> | 0.54 ± 0.10        | 0.56 ± 0.14        | 0.53 ± 0.16        | 0.57 ± 0.13 |
| S051  | 0.44 ± 0.21        | 0.41 ± 0.17        | 0.42 ± 0.13        | 0.47 ± 0.15        | 0.46 ± 0.13        | <b>0.50 ± 0.14</b> | 0.48 ± 0.12        | 0.50 ± 0.14 |
| S052  | 0.63 ± 0.14        | 0.59 ± 0.12        | 0.61 ± 0.14        | 0.63 ± 0.11        | 0.60 ± 0.14        | 0.61 ± 0.12        | <b>0.64 ± 0.12</b> | 0.64 ± 0.12 |
| S053  | 0.72 ± 0.19        | 0.71 ± 0.14        | 0.72 ± 0.11        | <b>0.73 ± 0.12</b> | 0.72 ± 0.11        | 0.71 ± 0.16        | 0.70 ± 0.15        | 0.73 ± 0.12 |
| S054  | 0.71 ± 0.14        | 0.69 ± 0.17        | 0.78 ± 0.15        | 0.74 ± 0.11        | <b>0.79 ± 0.12</b> | 0.72 ± 0.09        | 0.68 ± 0.12        | 0.79 ± 0.12 |
| S055  | 0.71 ± 0.12        | 0.73 ± 0.11        | 0.72 ± 0.13        | 0.76 ± 0.10        | <b>0.79 ± 0.10</b> | 0.74 ± 0.10        | 0.70 ± 0.12        | 0.79 ± 0.10 |
| S056  | 0.76 ± 0.13        | <b>0.80 ± 0.10</b> | 0.78 ± 0.11        | 0.68 ± 0.13        | 0.63 ± 0.11        | 0.60 ± 0.12        | 0.60 ± 0.14        | 0.80 ± 0.10 |
| S057  | 0.36 ± 0.11        | 0.40 ± 0.21        | 0.37 ± 0.20        | 0.37 ± 0.17        | 0.38 ± 0.17        | 0.38 ± 0.21        | <b>0.41 ± 0.17</b> | 0.41 ± 0.17 |
| S058  | <b>0.74 ± 0.15</b> | 0.53 ± 0.15        | 0.52 ± 0.12        | 0.54 ± 0.13        | 0.52 ± 0.13        | 0.46 ± 0.08        | 0.50 ± 0.13        | 0.74 ± 0.15 |
| S059  | 0.62 ± 0.15        | <b>0.66 ± 0.15</b> | 0.64 ± 0.12        | 0.57 ± 0.09        | 0.53 ± 0.08        | 0.47 ± 0.14        | 0.43 ± 0.13        | 0.66 ± 0.15 |
| S060  | <b>0.77 ± 0.08</b> | 0.72 ± 0.09        | 0.71 ± 0.15        | 0.71 ± 0.11        | 0.68 ± 0.16        | 0.68 ± 0.19        | 0.71 ± 0.18        | 0.77 ± 0.08 |
| S061  | 0.59 ± 0.16        | 0.54 ± 0.19        | 0.70 ± 0.19        | 0.73 ± 0.11        | 0.73 ± 0.16        | <b>0.78 ± 0.13</b> | 0.72 ± 0.16        | 0.78 ± 0.13 |
| S062  | <b>0.93 ± 0.09</b> | 0.92 ± 0.07        | 0.91 ± 0.07        | 0.91 ± 0.04        | 0.90 ± 0.08        | 0.91 ± 0.07        | 0.84 ± 0.11        | 0.93 ± 0.09 |
| S063  | 0.46 ± 0.15        | 0.51 ± 0.10        | 0.58 ± 0.13        | 0.58 ± 0.20        | 0.59 ± 0.17        | 0.57 ± 0.18        | <b>0.62 ± 0.21</b> | 0.62 ± 0.21 |
| S064  | <b>0.58 ± 0.22</b> | 0.50 ± 0.18        | 0.50 ± 0.17        | 0.46 ± 0.15        | 0.47 ± 0.16        | 0.41 ± 0.15        | 0.49 ± 0.19        | 0.58 ± 0.22 |
| S065  | 0.49 ± 0.14        | 0.49 ± 0.19        | 0.51 ± 0.15        | 0.59 ± 0.17        | 0.59 ± 0.19        | 0.59 ± 0.15        | <b>0.63 ± 0.13</b> | 0.63 ± 0.13 |
| S066  | <b>0.60 ± 0.13</b> | 0.57 ± 0.12        | 0.56 ± 0.16        | 0.57 ± 0.18        | 0.57 ± 0.18        | 0.51 ± 0.19        | 0.48 ± 0.14        | 0.60 ± 0.13 |
| S067  | 0.43 ± 0.18        | 0.39 ± 0.17        | 0.44 ± 0.15        | <b>0.47 ± 0.16</b> | 0.33 ± 0.16        | 0.43 ± 0.25        | 0.38 ± 0.19        | 0.47 ± 0.16 |
| S068  | 0.46 ± 0.14        | 0.53 ± 0.16        | <b>0.60 ± 0.15</b> | 0.57 ± 0.12        | 0.59 ± 0.15        | 0.56 ± 0.16        | 0.53 ± 0.11        | 0.60 ± 0.15 |
| S069  | <b>0.53 ± 0.16</b> | 0.51 ± 0.15        | 0.43 ± 0.18        | 0.49 ± 0.15        | 0.46 ± 0.15        | 0.43 ± 0.17        | 0.40 ± 0.12        | 0.53 ± 0.16 |
| S070  | 0.66 ± 0.22        | 0.71 ± 0.22        | <b>0.72 ± 0.19</b> | 0.67 ± 0.13        | 0.69 ± 0.11        | 0.71 ± 0.19        | 0.69 ± 0.16        | 0.72 ± 0.19 |
| S071  | 0.57 ± 0.14        | <b>0.66 ± 0.12</b> | 0.60 ± 0.13        | 0.60 ± 0.11        | 0.61 ± 0.07        | 0.56 ± 0.13        | 0.56 ± 0.17        | 0.66 ± 0.12 |
| S072  | <b>0.81 ± 0.09</b> | 0.79 ± 0.06        | 0.72 ± 0.11        | 0.71 ± 0.09        | 0.72 ± 0.10        | 0.76 ± 0.12        | 0.69 ± 0.11        | 0.81 ± 0.09 |
| S073  | 0.42 ± 0.14        | 0.37 ± 0.11        | <b>0.48 ± 0.17</b> | 0.42 ± 0.13        | 0.42 ± 0.11        | 0.41 ± 0.07        | 0.47 ± 0.11        | 0.48 ± 0.17 |
| S074  | <b>0.63 ± 0.18</b> | 0.60 ± 0.16        | 0.59 ± 0.17        | 0.49 ± 0.20        | 0.52 ± 0.12        | 0.48 ± 0.13        | 0.48 ± 0.17        | 0.63 ± 0.18 |
| S075  | 0.53 ± 0.17        | 0.44 ± 0.20        | 0.56 ± 0.16        | 0.56 ± 0.11        | 0.54 ± 0.10        | 0.53 ± 0.14        | <b>0.59 ± 0.12</b> | 0.59 ± 0.12 |
| S076  | <b>0.53 ± 0.15</b> | 0.52 ± 0.13        | 0.51 ± 0.11        | 0.49 ± 0.10        | 0.38 ± 0.13        | 0.39 ± 0.11        | 0.36 ± 0.13        | 0.53 ± 0.15 |
| S077  | 0.40 ± 0.10        | 0.43 ± 0.15        | 0.41 ± 0.12        | 0.43 ± 0.19        | 0.43 ± 0.13        | <b>0.47 ± 0.14</b> | 0.40 ± 0.16        | 0.47 ± 0.14 |
| S078  | 0.41 ± 0.07        | 0.43 ± 0.13        | 0.48 ± 0.13        | 0.49 ± 0.15        | 0.53 ± 0.18        | <b>0.56 ± 0.15</b> | 0.54 ± 0.14        | 0.56 ± 0.15 |
| S079  | 0.61 ± 0.20        | <b>0.62 ± 0.21</b> | 0.57 ± 0.19        | 0.38 ± 0.16        | 0.43 ± 0.13        | 0.47 ± 0.13        | 0.49 ± 0.13        | 0.62 ± 0.21 |
| S080  | 0.49 ± 0.10        | 0.59 ± 0.09        | <b>0.60 ± 0.14</b> | 0.57 ± 0.14        | 0.57 ± 0.14        | 0.49 ± 0.14        | 0.51 ± 0.14        | 0.60 ± 0.14 |
| S081  | 0.42 ± 0.12        | 0.43 ± 0.13        | 0.51 ± 0.17        | <b>0.54 ± 0.19</b> | 0.53 ± 0.20        | 0.50 ± 0.19        | 0.53 ± 0.14        | 0.54 ± 0.19 |
| S082  | 0.63 ± 0.12        | 0.70 ± 0.15        | 0.70 ± 0.12        | <b>0.73 ± 0.13</b> | 0.70 ± 0.14        | 0.67 ± 0.11        | 0.62 ± 0.07        | 0.73 ± 0.13 |
| S083  | 0.56 ± 0.16        | <b>0.60 ± 0.12</b> | 0.51 ± 0.15        | 0.49 ± 0.12        | 0.49 ± 0.12        | 0.47 ± 0.12        | 0.50 ± 0.20        | 0.60 ± 0.12 |
| S084  | <b>0.60 ± 0.12</b> | 0.58 ± 0.12        | 0.57 ± 0.12        | 0.46 ± 0.12        | 0.43 ± 0.16        | 0.41 ± 0.16        | 0.44 ± 0.13        | 0.60 ± 0.12 |
| S085  | 0.80 ± 0.11        | 0.77 ± 0.14        | <b>0.82 ± 0.10</b> | 0.71 ± 0.09        | 0.62 ± 0.09        | 0.69 ± 0.17        | 0.71 ± 0.12        | 0.82 ± 0.10 |
| S086  | 0.46 ± 0.22        | <b>0.63 ± 0.16</b> | 0.59 ± 0.14        | 0.51 ± 0.17        | 0.49 ± 0.17        | 0.53 ± 0.14        | 0.58 ± 0.08        | 0.63 ± 0.16 |
| S087  | <b>0.53 ± 0.16</b> | 0.48 ± 0.14        | 0.49 ± 0.09        | 0.52 ± 0.10        | 0.48 ± 0.12        | 0.49 ± 0.13        | 0.48 ± 0.17        | 0.53 ± 0.16 |

| Sujet          | 2                  | 4                  | 6                  | 8                  | 10                 | 12                 | 14                 | Meilleur           |
|----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| S088           | 0.36 ± 0.12        | 0.38 ± 0.10        | 0.40 ± 0.12        | 0.39 ± 0.19        | 0.38 ± 0.16        | 0.36 ± 0.15        | <b>0.43 ± 0.05</b> | 0.43 ± 0.05        |
| S089           | 0.41 ± 0.17        | 0.39 ± 0.17        | 0.41 ± 0.14        | <b>0.42 ± 0.14</b> | <b>0.42 ± 0.10</b> | <b>0.42 ± 0.08</b> | <b>0.42 ± 0.12</b> | 0.42 ± 0.14        |
| S090           | 0.49 ± 0.17        | 0.47 ± 0.16        | 0.49 ± 0.18        | 0.48 ± 0.15        | 0.49 ± 0.10        | <b>0.54 ± 0.14</b> | 0.51 ± 0.11        | 0.54 ± 0.14        |
| S091           | <b>0.58 ± 0.18</b> | 0.56 ± 0.20        | 0.54 ± 0.17        | 0.57 ± 0.13        | 0.56 ± 0.16        | 0.47 ± 0.18        | 0.57 ± 0.12        | 0.58 ± 0.18        |
| S092           | 0.43 ± 0.15        | 0.52 ± 0.15        | <b>0.56 ± 0.16</b> | 0.47 ± 0.11        | 0.47 ± 0.13        | 0.48 ± 0.08        | 0.48 ± 0.10        | 0.56 ± 0.16        |
| S093           | 0.69 ± 0.12        | <b>0.71 ± 0.14</b> | 0.67 ± 0.19        | 0.67 ± 0.23        | 0.62 ± 0.19        | 0.63 ± 0.19        | 0.62 ± 0.17        | 0.71 ± 0.14        |
| S094           | 0.67 ± 0.11        | <b>0.80 ± 0.14</b> | 0.73 ± 0.17        | 0.72 ± 0.11        | 0.69 ± 0.11        | 0.78 ± 0.11        | 0.79 ± 0.13        | 0.80 ± 0.14        |
| S095           | 0.47 ± 0.10        | 0.43 ± 0.16        | <b>0.53 ± 0.10</b> | 0.51 ± 0.17        | 0.50 ± 0.16        | 0.47 ± 0.07        | 0.50 ± 0.13        | 0.53 ± 0.10        |
| S096           | 0.47 ± 0.18        | 0.50 ± 0.17        | 0.46 ± 0.18        | 0.49 ± 0.17        | 0.52 ± 0.16        | <b>0.57 ± 0.14</b> | 0.56 ± 0.16        | 0.57 ± 0.14        |
| S097           | 0.48 ± 0.16        | 0.46 ± 0.18        | 0.43 ± 0.15        | <b>0.49 ± 0.18</b> | 0.48 ± 0.16        | 0.44 ± 0.16        | 0.44 ± 0.10        | 0.49 ± 0.18        |
| S098           | 0.51 ± 0.09        | 0.59 ± 0.16        | 0.62 ± 0.16        | 0.58 ± 0.16        | 0.62 ± 0.13        | 0.64 ± 0.15        | <b>0.66 ± 0.14</b> | 0.66 ± 0.14        |
| S099           | <b>0.61 ± 0.17</b> | 0.48 ± 0.14        | 0.41 ± 0.17        | 0.40 ± 0.15        | 0.43 ± 0.13        | 0.46 ± 0.16        | 0.48 ± 0.21        | 0.61 ± 0.17        |
| S100           | 0.56 ± 0.17        | 0.60 ± 0.18        | <b>0.62 ± 0.17</b> | 0.57 ± 0.22        | 0.60 ± 0.17        | 0.59 ± 0.18        | 0.55 ± 0.16        | 0.62 ± 0.17        |
| S101           | <b>0.41 ± 0.17</b> | 0.37 ± 0.10        | 0.36 ± 0.10        | 0.37 ± 0.14        | 0.37 ± 0.16        | 0.34 ± 0.13        | 0.34 ± 0.12        | 0.41 ± 0.17        |
| S102           | 0.53 ± 0.21        | 0.47 ± 0.19        | 0.66 ± 0.18        | 0.67 ± 0.12        | <b>0.71 ± 0.10</b> | <b>0.71 ± 0.10</b> | 0.66 ± 0.15        | 0.71 ± 0.10        |
| S103           | 0.41 ± 0.09        | 0.53 ± 0.10        | 0.50 ± 0.12        | 0.49 ± 0.15        | 0.53 ± 0.13        | 0.54 ± 0.13        | <b>0.54 ± 0.16</b> | 0.54 ± 0.16        |
| S104           | 0.47 ± 0.11        | 0.53 ± 0.07        | 0.46 ± 0.16        | 0.52 ± 0.15        | 0.53 ± 0.17        | <b>0.58 ± 0.14</b> | 0.53 ± 0.18        | 0.58 ± 0.14        |
| S105           | 0.49 ± 0.12        | <b>0.57 ± 0.18</b> | 0.49 ± 0.16        | 0.53 ± 0.13        | 0.52 ± 0.14        | 0.52 ± 0.15        | <b>0.57 ± 0.13</b> | 0.57 ± 0.18        |
| S106           | 0.50 ± 0.13        | 0.54 ± 0.18        | 0.46 ± 0.15        | 0.51 ± 0.19        | 0.56 ± 0.11        | <b>0.59 ± 0.15</b> | 0.47 ± 0.14        | 0.59 ± 0.15        |
| S107           | 0.48 ± 0.20        | 0.50 ± 0.21        | 0.53 ± 0.13        | 0.49 ± 0.10        | 0.53 ± 0.08        | <b>0.58 ± 0.10</b> | 0.56 ± 0.14        | 0.58 ± 0.10        |
| S108           | 0.50 ± 0.10        | 0.53 ± 0.15        | 0.54 ± 0.22        | 0.54 ± 0.19        | 0.58 ± 0.18        | 0.58 ± 0.13        | <b>0.60 ± 0.12</b> | 0.60 ± 0.12        |
| S109           | 0.47 ± 0.15        | 0.51 ± 0.09        | <b>0.52 ± 0.18</b> | 0.51 ± 0.13        | 0.46 ± 0.14        | 0.42 ± 0.15        | 0.47 ± 0.15        | 0.52 ± 0.18        |
| <b>Moyenne</b> | 0.56 ± 0.11        | <b>0.57 ± 0.13</b> | 0.57 ± 0.12        | 0.56 ± 0.12        | 0.56 ± 0.12        | 0.55 ± 0.12        | 0.55 ± 0.11        | <b>0.62 ± 0.12</b> |